CHAPTER

77

ENGINE INDICATING

(LEAP-1B ENGINES)

BOEING

737-7/8/8200/9/10
AIRCRAFT MAINTENANCE MANUAL

CHAPTER 77
ENGINE INDICATING

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A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

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N1 Speed Sensor Installation TASK 77-12-01-400-801-G00			406	PTF ALL
N2 SPEED SENSOR - REMOVAL/INSTALLATION	77-12-02		401	PTF ALL
N2 Speed Sensor Removal TASK 77-12-02-000-801-G00			401	PTF ALL
N2 Speed Sensor Installation TASK 77-12-02-400-801-G00			406	PTF ALL
N1 SPEED SENSOR SHIM - REMOVAL/INSTALLATION	77-12-03		401	PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0127
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ENGINE TEMPERATURE SENSING SYSTEM - INSPECTION/CHECK	77-21-00		601	PTF ALL
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ENGINE TACHOMETER SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Engine tachometer inspection.

TASK 77-12-00-210-801-G00

2. Engine Tachometer Inspection

(Figure 601 and Figure 602)

A. General

(1) This task gives the instructions for the visual inspection of the engine tachometer system (N1 and N2 speed sensor).

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
77-12-01-000-801-G00	N1 Speed Sensor Removal (P/B 401)
77-12-01-400-801-G00	N1 Speed Sensor Installation (P/B 401)
77-12-02-000-801-G00	N2 Speed Sensor Removal (P/B 401)
77-12-02-400-801-G00	N2 Speed Sensor Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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F. Prepare for the Inspection

SUBTASK 77-12-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 77-12-00-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 77-12-00-010-002-G00

(3) Do this task: N1 Speed Sensor Removal, TASK 77-12-01-000-801-G00.

SUBTASK 77-12-00-010-003-G00

(4) Do this task: N2 Speed Sensor Removal, TASK 77-12-02-000-801-G00.

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G. Engine Tachometer Inspection

SUBTASK 77-12-00-212-001-G00

(1) Do the visual inspection of the N1 speed sensor [1] as follows (Figure 601):

(a) Examine the visible surfaces on zones A, B, E, F, FF, G, GG, H, HH, M, T, and S for these conditions:

1) Cracks, dents, scoring/scratches, erosion/peening, corrosion

a) Not permitted.

b) Replace the N1 speed sensor [1].

(b) Deposits

1) Permitted if the identification is readable visually with or without an optical device.

SUBTASK 77-12-00-212-002-G00

(2) Do the visual inspection of the N2 speed sensor [2] as follows:

(a) Examine the surfaces on zones P, J, and the gaskets in zone K and Q for these conditions (Figure 602 (Sheet 3)):

1) Nicks, scoring, scratch

a) Permitted up to 0.002 in. (0.05 mm) maximum depth, 0.01 in. (0.25 mm) maximum bottom of radius and the identification on zone J is readable visually with or without an optical device.

b) Not permitted if more than the limits or the identification is not readable.

<1> Replace the N2 speed sensor [2].

2) Tears on the gaskets

a) Not permitted.

<1> Replace the N2 speed sensor [2].

3) Deposits

a) Permitted if the identification is readable visually with or without an optical device.

(b) Examine the surface on zone F for these conditions (Figure 602 (Sheet 3)):

1) Nicks, scoring, scratch

a) Permitted up to 0.008 in. (0.20 mm) maximum depth and 0.01 in. (0.25 mm) maximum bottom of radius.

b) Not permitted if more than the limits.

<1> Replace the N2 speed sensor [2]..

(c) Examine the connector in zone G, and the screws in zones L, and M for these conditions (Figure 602 (Sheet 3)):

1) Nicks, scoring, scratch

a) Permitted up to 0.004 in. (0.10 mm) maximum depth and 0.01 in. (0.25 mm) maximum bottom of radius.

b) Not permitted if more than the limits.

<1> Replace the N2 speed sensor [2].

(d) Examine the surfaces on zones G, H, and I for these conditions (Figure 602 (Sheet 2)):

1) Cracks, dents, corrosion, erosion, peening, scorings, scratches, nicks, and bent pins

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a) Not permitted.

b) Replace the N2 speed sensor [2].

2) Deposits

a) Permitted if the identification is readable visually with or without an optical device.

(e) Examine the engraved identification in zone F (housing) (Figure 602 (Sheet 2)):

1) The engraved identification is not readable or missing.

a) Not permitted.

b) Replace the N2 speed sensor [2].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 77-12-00-410-002-G00

(1) Do this task: N1 Speed Sensor Installation, TASK 77-12-01-400-801-G00.

SUBTASK 77-12-00-410-003-G00

(2) Do this task: N2 Speed Sensor Installation, TASK 77-12-02-400-801-G00.

SUBTASK 77-12-00-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 77-12-00-860-002-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

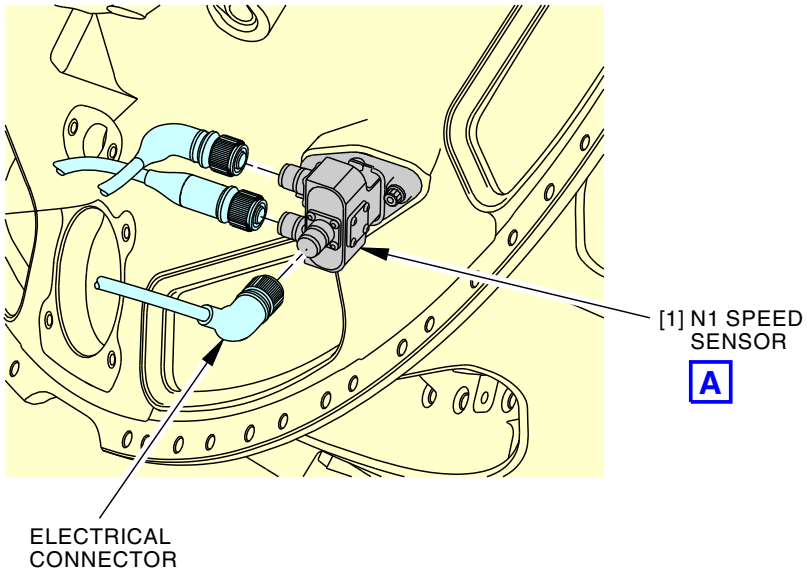
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

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(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

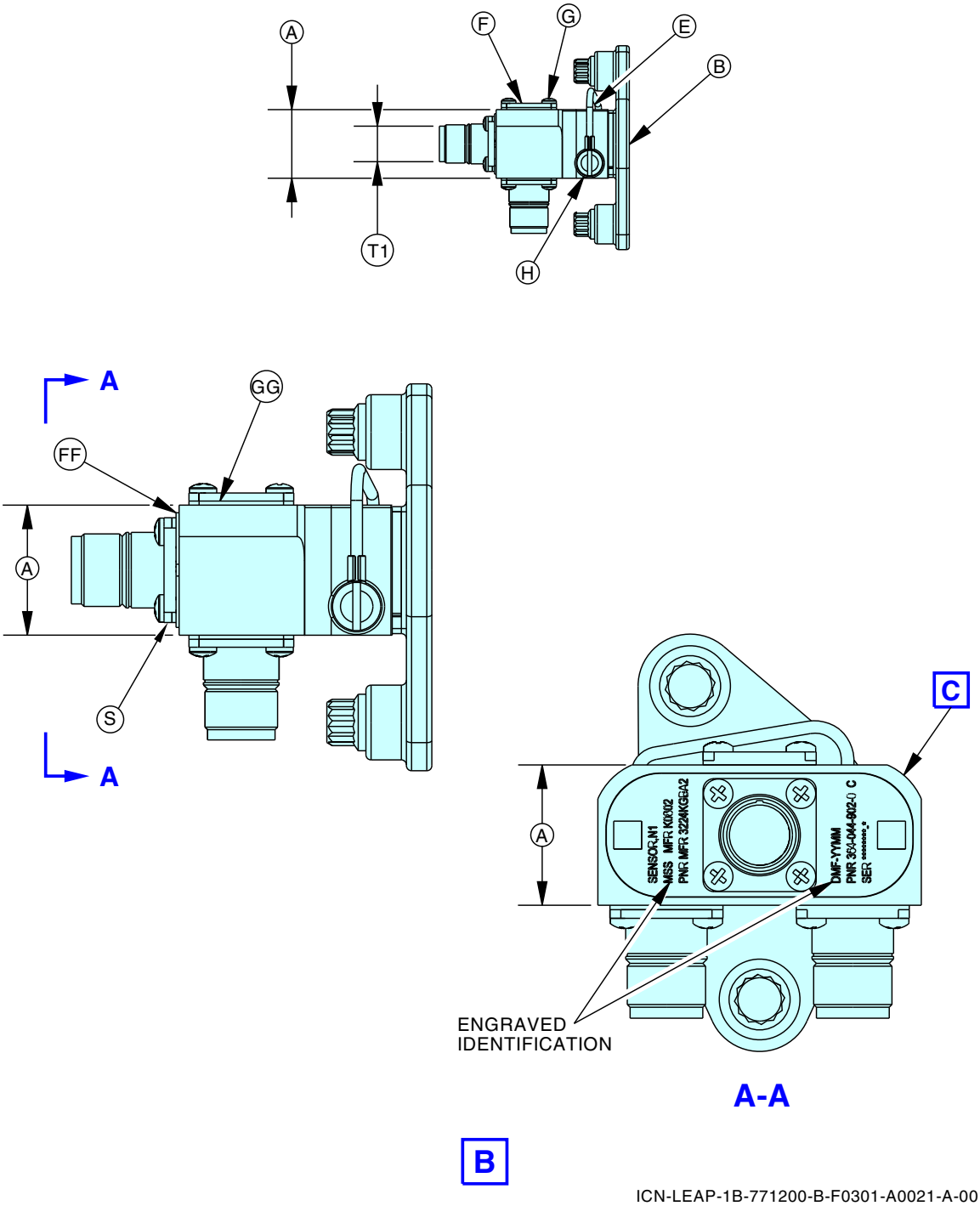
END OF TASK

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Engine N1 Speed Sensor Inspection
Figure 601/77-12-00-990-801-G00 (Sheet 1 of 3)

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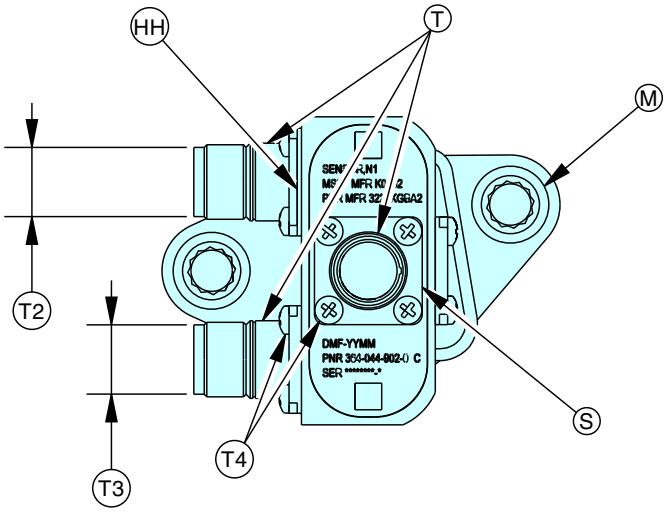


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2572913 S0000618081_V1

Engine N1 Speed Sensor Inspection
Figure 601/77-12-00-990-801-G00 (Sheet 2 of 3)

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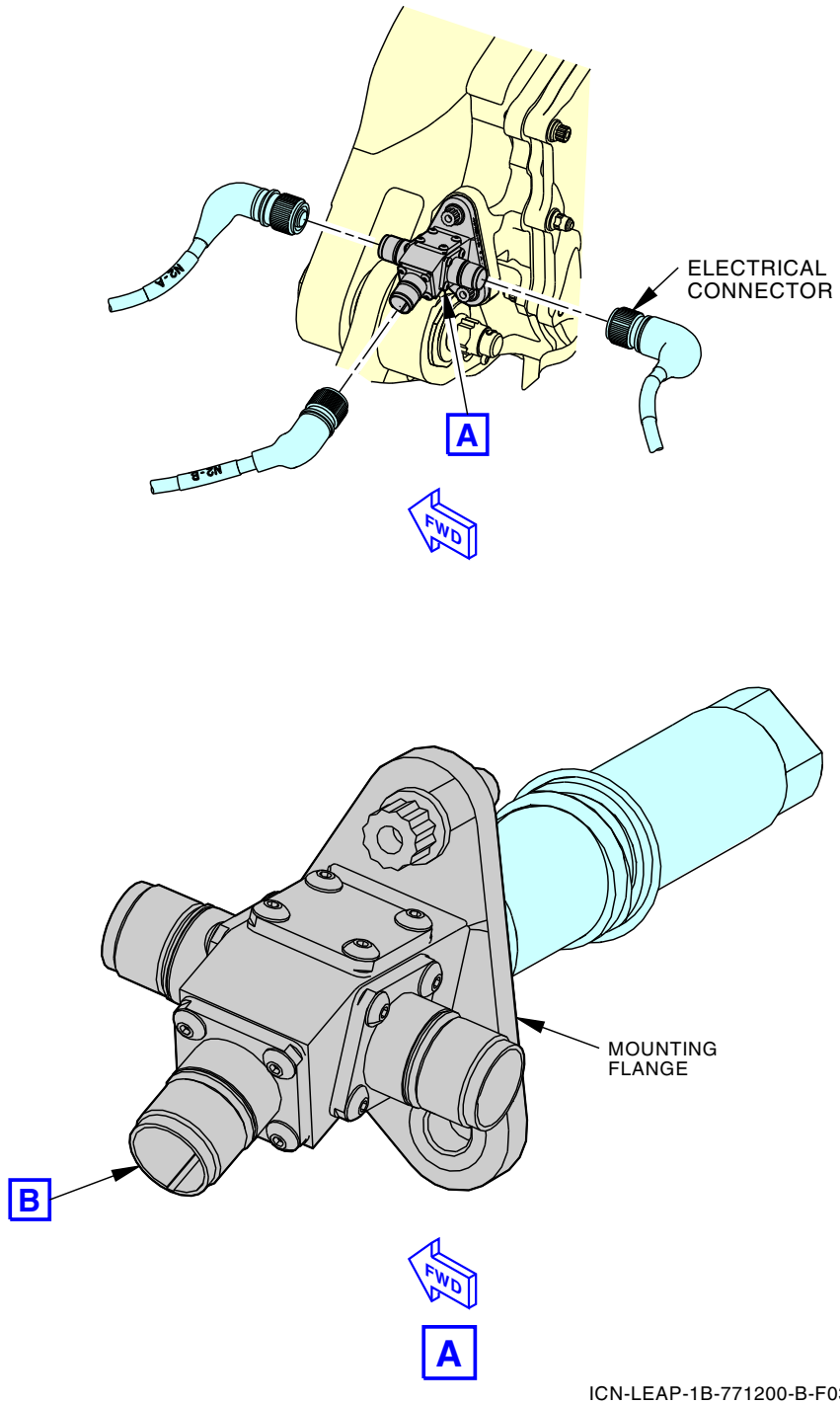
C

ZONE	DESCRIPTION
A	CONNECTOR HOUSING
B	MOUNTING FLANGE
S	CONNECTOR PLATE
E	BONDING STRAP
F	LID
G	LID FIXINGS
H	BONDING STRAP FIXINGS
M	ATTACHMENT FIXINGS
T	CONNECTOR
T1	CONNECTOR B
T2	CONNECTOR A
T3	CONNECTOR C
T4	CONNECTOR FIXINGS
FF	CONNECTOR GASKET
GG	CONNECTOR GASKET
HH	CONNECTOR GASKET

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2573222 S0000618082_V1

Engine N1 Speed Sensor Inspection
Figure 601/77-12-00-990-801-G00 (Sheet 3 of 3)

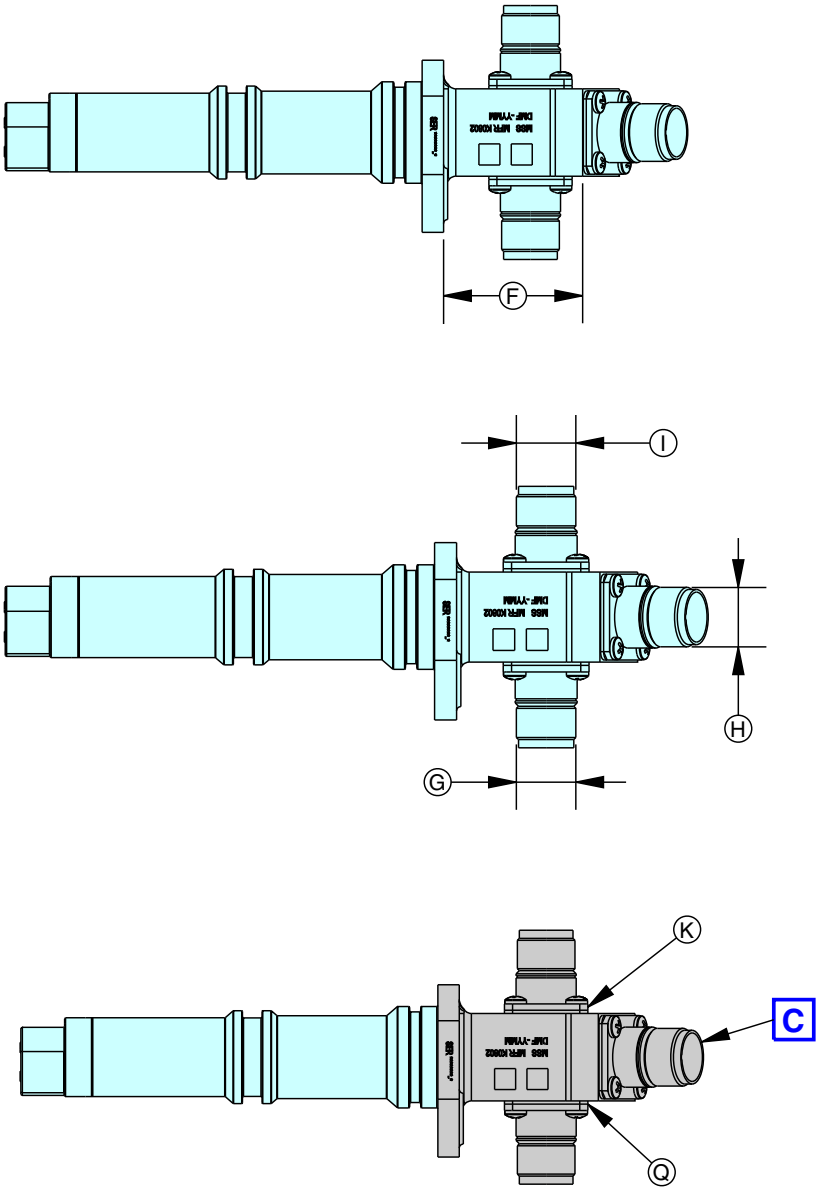
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2573181 S0000618083_V1

Engine N2 Speed Sensor Inspection
Figure 602/77-12-00-990-803-G00 (Sheet 1 of 3)

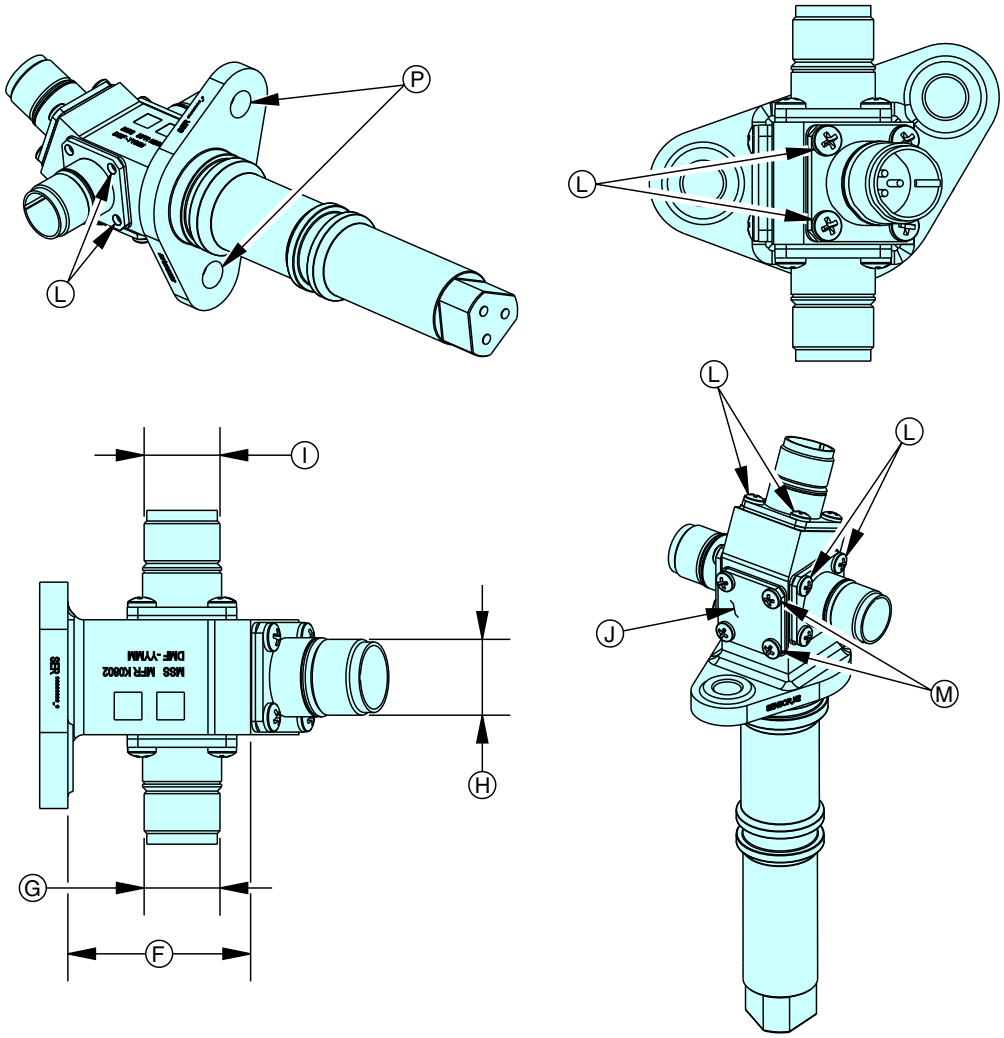
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2573206 S0000618084_V1

Engine N2 Speed Sensor Inspection
Figure 602/77-12-00-990-803-G00 (Sheet 2 of 3)

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C

ZONE	DESCRIPTION
F	HOUSING
G	CONNECTOR - 3
H	CONNECTOR - 1
I	CONNECTOR - 2
J	LID

K	GASKET, CONNECTOR - 2
L	FASTENING SCREWS - CONNECTORS TO HOUSING
M	FASTENING SCREWS - LID TO HOUSING
P	FLANGE MOUNTING HOLES
Q	GASKET, CONNECTOR - 3

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2573209 S0000618085_V1

Engine N2 Speed Sensor Inspection
Figure 602/77-12-00-990-803-G00 (Sheet 3 of 3)

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N1 SPEED SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) N1 speed sensor removal

(2) N1 speed sensor installation.

TASK 77-12-01-000-801-G00

2. N1 Speed Sensor Removal

(Figure 401)

A. General

(1) This task gives the instructions on how to remove the N1 speed sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
77-12-03-000-801-G00	N1 Speed Sensor Shim Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4316	Fiberscope - Flexible Borescope Part #: USEF 2-2.4-1000 Supplier: 7CGG1 Part #: USEF 2-6-1000 Supplier: 7CGG1 Opt Part #: 7110561 Supplier: 32212 Opt Part #: IF2D5 Supplier: 32212 Opt Part #: IF6C5X1-8 Supplier: 32212
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-6547	Thermometer - 0 to 350 deg F Range
STD-10718	Fiberscope (Meets General Electric or CFM Standard Practices Manual Specification)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 77-12-01-010-001-G00

(1) Do these tasks in sequence to safely open the right thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

CAUTION IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(b) Open both fan cowl panels (TASK 71-11-04-010-801-G00).

NOTE: Because the center line of the thrust reversers is off 6:00 o'clock position, both fan cowl panels must be opened to prevent damaging the fan cowl panel if either thrust reverser needs to be opened.

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable right thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

SUBTASK 77-12-01-010-002-G00

(2) Remove the segment shroud 3 (TASK 72-23-04-000-801-G00).

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G. N1 Speed Sensor Removal

SUBTASK 77-12-01-030-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use a teflon-jawed pliers, STD-664, to disconnect the electrical connectors as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the electrical connector [4], the electrical connector [5], and the electrical connector [6] from the N1 speed sensor [1].

(b) Disconnect the electrical connector [4], the electrical connector [5], and the electrical connector [6] from the N1 speed sensor [1].

(c) Install the protective covers on the electrical connectors and the electrical receptacles on the N1 speed sensor [1].

SUBTASK 77-12-01-680-001-G00

(2) Remove the N1 speed sensor [1] as follows:

(a) Put a oil resistant container, STD-203 below the N1 speed sensor [1].

NOTE: A small quantity of oil can drain from the fan hub frame port for the N1 speed sensor.

SUBTASK 77-12-01-020-001-G00

(3) Remove the two bolts [3] that attach the N1 speed sensor [1] to the fan hub frame [2].

NOTE: The N1 speed sensor is 19 inches in length. Be careful when you remove the N1 speed sensor.

SUBTASK 77-12-01-020-002-G00

(4) Remove the N1 speed sensor [1] as follows:

(a) Do not turn the N1 sensor upside down.

(b) Remove the N1 speed sensor [1] from the fan hub frame [2].

(c) Make sure that the shim is located on the top of the N1 speed sensor [1].

SUBTASK 77-12-01-020-003-G00

(5) Remove and discard the preformed packing [7] from the N1 speed sensor [1].

SUBTASK 77-12-01-210-001-G00

(6) Inspect the extremity of the N1 Sensor:

(a) If N1 sensor shim is present, do this task: N1 Speed Sensor Shim Removal, TASK 77-12-03-000-801-G00.

(b) If N1 sensor shim is not present, then continue.

SUBTASK 77-12-01-290-001-G00

(7) Inspect the N1 sensor sleeve:

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(a) Assemble the borescope inspection tools per assembly instructions from the borescope tool manufacturer.

1) Use a fiberscope, STD-10718, or a fiberscope, COM-4316, with video probe.

NOTE: It is permitted to substitute the tools specified in the borescope inspection tasks, provided the tools are approved by CFM.

CAUTION

MAKE SURE THAT THE ENGINE AREA THAT YOU EXAMINE WITH THE BORESCOPE IS LESS THAN 115 DEGREES F (46 DEGREES C). IF THE ENGINE INTERNAL AND EXTERNAL TEMPERATURE IS MORE THAN 115 DEGREES F (46 DEGREES C), DAMAGE TO THE BORESCOPE CAN OCCUR

(b) Make sure that the internal and external temperature of the engine is less than 115°F (46°C) before you insert the borescope probe into the engine.

1) If the engine has been recently run, measure the temperature of the engine with a thermometer, STD-6547.

NOTE: It is recommended to wait 3 to 5 hours after engine shutdown to do the borescope inspection. This will allow the engine to become sufficiently cool.

NOTE: You can dry motor the engine to decrease the engine core temperature.

(c) Use the fiberscope, STD-10718, or the fiberscope, COM-4316, with video probe.

1) Use a forward view optical tip when you inspect the N1 sensor sleeve.

(d) Engage the fiberscope in the guide tube in the 5 o'clock strut of the fan hub.

(e) Check for the presence of N1 speed sensor shim in the guide tube in the 5 o'clock strut of the fan hub.

NOTE: The N1 speed sensor shim can be attached with oil to the bottom of the sensor guide.

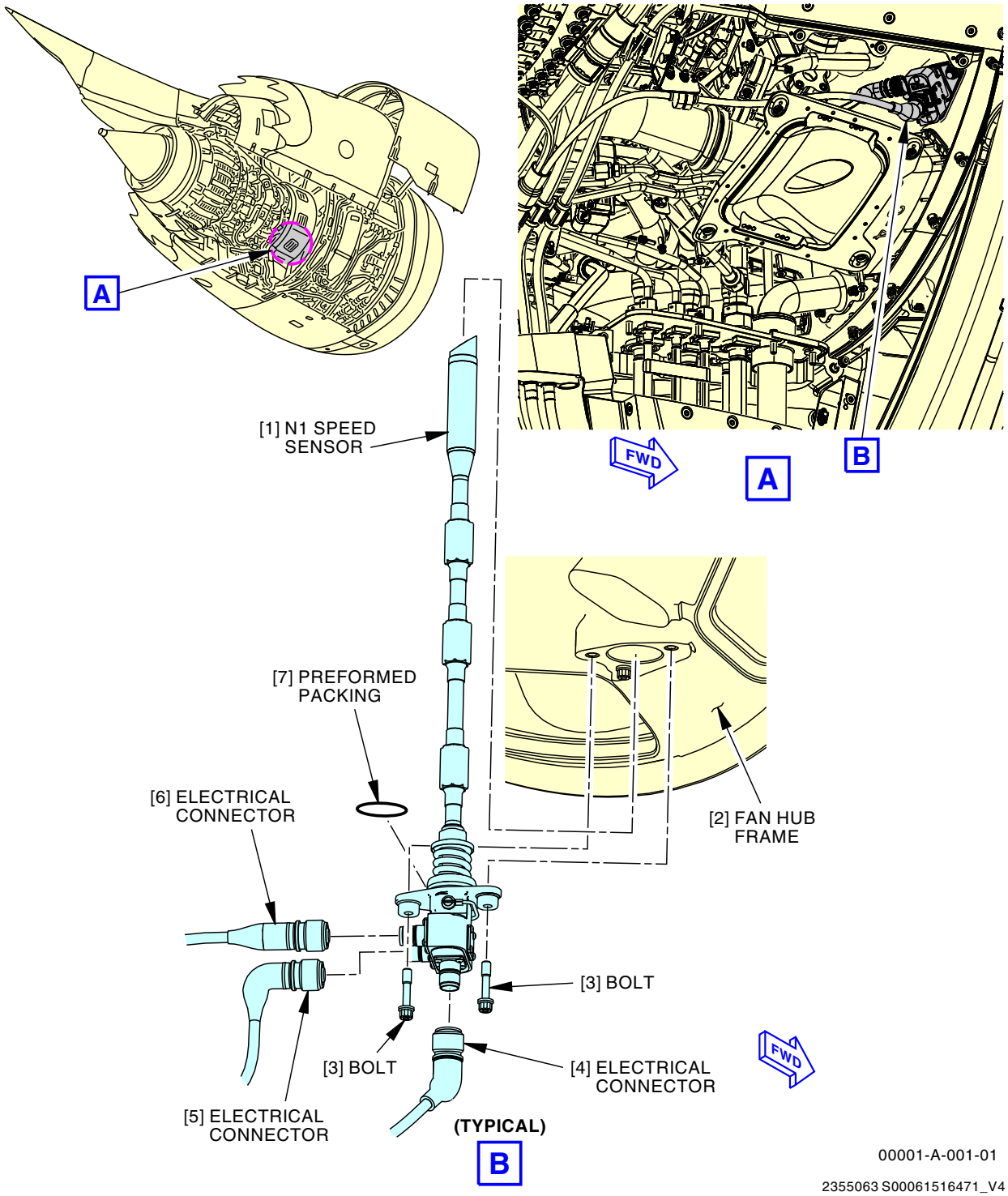
(f) If the N1 speed sensor shim is present, do this task: N1 Speed Sensor Shim Removal, TASK 77-12-03-000-801-G00.

SUBTASK 77-12-01-480-001-G00

(8) Install the protective covers on the sensing elements end of the N1 speed sensor [1] and the opening on the fan hub frame [2].

——— END OF TASK ———

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N1 Speed Sensor Installation
Figure 401/77-12-01-990-801-G00

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TASK 77-12-01-400-801-G00

3. N1 Speed Sensor Installation

(Figure 401)

A. General

(1) This task gives the instructions on how to install the N1 speed sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
77-12-03-400-801-G00	N1 Speed Sensor Shim Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Preformed packing	77-12-01-01-020	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

H. N1 Speed Sensor Installation

SUBTASK 77-12-01-420-001-G00

(1) Install the N1 speed sensor [1] as follows:

(a) Remove the protective covers from the N1 speed sensor [1] and the open port on the fan hub frame [2].

(b) Lubricate a new preformed packing [7] with the engine oil, D00599 [CP2442].

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(c) Install the new preformed packing [7] on the groove on the N1 speed sensor [1].

(d) Lubricate the threads of the bolts [3] with engine oil, D00599 [CP2442].

(e) Make sure that there is no shim at the top of the N1 speed sensor [1] or in the bottom of the N1 speed sensor guide tube.

1) If the shim is installed, make sure that there is no more than one shim.

2) If there is no shim, make sure that you do not install the N1 speed sensor [1] with the shim.

(f) If a N1 speed sensor shim is removed during the N1 speed sensor removal, do this task: N1 Speed Sensor Shim Installation, TASK 77-12-03-400-801-G00.

NOTE: If the visual inspection of the removed shim shows no signs of damage, then it is serviceable and can be used again.

1) Apply vaseline, D00672 [CP5070], on the face of the shim in contact with the N1 speed sensor [1].

(g) Put the N1 speed sensor [1] into the fan hub frame [2].

NOTE: The N1 speed sensor is 19 inches in length. Be careful when you install the N1 speed sensor.

1) Do not move the N1 speed sensor [1] rearward and forward in the guide tube.

a) If you moved the N1 speed sensor [1], do these steps:

<1> Remove the N1 speed sensor [1].

<2> Make sure that the shim is in the correct position.

<3> Put the N1 speed sensor [1] into the fan hub frame [2].

(h) Install the two bolts [3] that attach the N1 speed sensor [1] to the fan hub frame [2].

NOTE: Tighten the bolts [3] one bolt 1/2 turn at a time. This allows even compression of sensor spring.

(i) Tighten the bolts [3] to 109.0 in-lb (12.3 N·m) - 121.0 in-lb (13.7 N·m).

SUBTASK 77-12-01-430-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Connect the electrical connectors as follows (TASK 70-00-01-910-803-G00):

(a) Remove the protective covers from the electrical connector [4], the electrical connector [5], and the electrical connector [6].

(b) Remove the protective covers from the electrical receptacles on the N1 speed sensor [1].

(c) Connect the electrical connector [4], the electrical connector [5], and the electrical connector [6] to the N1 speed sensor [1] .

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(d) Use a teflon-jawed pliers, STD-664, to tighten the electrical connectors to the N1 speed sensor [1].

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I. Put the Airplane Back to Its Usual Condition

SUBTASK 77-12-01-410-001-G00

(1) Install the segment shroud 3 (TASK 72-23-04-400-801-G00).

SUBTASK 77-12-01-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the right thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

(b) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

J. N1 Speed Sensor Installation Test

SUBTASK 77-12-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

N2 SPEED SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) N2 Speed Sensor Removal

(2) N2 Speed Sensor Installation.

TASK 77-12-02-000-801-G00

2. N2 Speed Sensor Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the N2 speed sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 77-12-02-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 77-12-02-010-001-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. N2 Speed Sensor Removal

SUBTASK 77-12-02-020-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the N2 speed sensor [1] as follows:

(a) Put a oil resistant container, STD-203 under the N2 speed sensor [1].

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(b) Disconnect the electrical connectors from the N2 speed sensor [1] as follows (TASK 70-00-01-910-803-G00):

1) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector (W300 harness) [5], the electrical connector (HJ6B harness) [4] and the electrical connector (HJ6A harness) [3] from the N2 speed sensor [1].

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2) Install the protective covers on the electrical connector (W300 harness) [5], the electrical connector (HJ6B harness) [4], the electrical connector (HJ6A harness) [3] and the electrical receptacles on the N2 speed sensor [1].

(c) Remove the two bolts [6] that attach the N2 speed sensor [1] to the accessory gearbox [2].

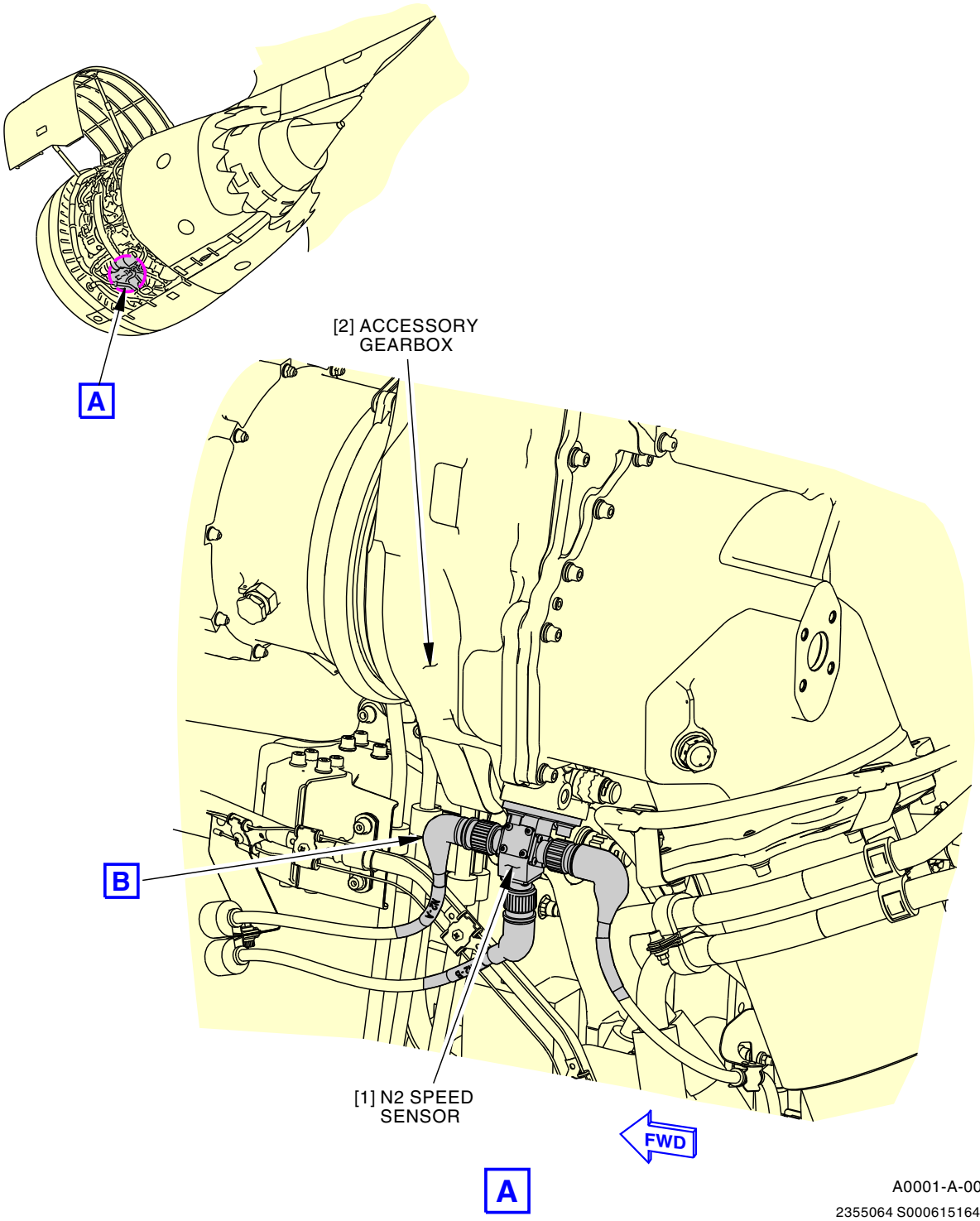
(d) Remove the N2 speed sensor [1] from the AGB.

(e) Remove and discard the preformed packing [7] and the preformed packing [8] from the grooves of the N2 speed sensor [1].

(f) Install the protective cover on the open port for the N2 speed sensor [1].

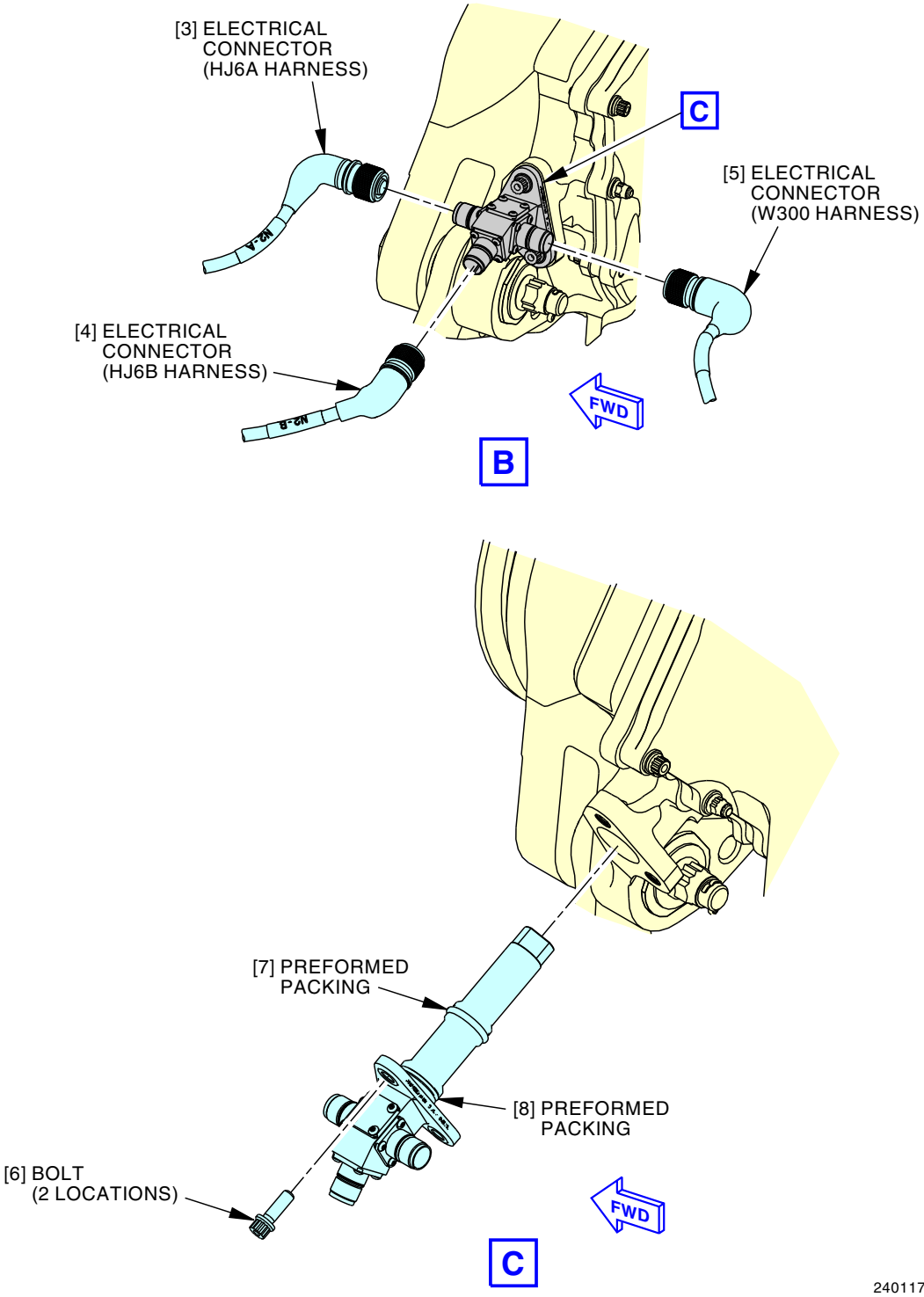
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



N2 Speed Sensor Installation
Figure 401/77-12-02-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



N2 Speed Sensor Installation
Figure 401/77-12-02-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 77-12-02-400-801-G00

3. N2 Speed Sensor Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the N2 speed sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Preformed packing	77-12-02-01-030	PTF ALL
8	Preformed packing	77-12-02-01-040	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Prepare for the Installation

SUBTASK 77-12-02-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 77-12-02-010-002-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

I. N2 Speed Sensor Installation

SUBTASK 77-12-02-420-001-G00

CAUTION DO NOT CAUSE DAMAGE TO THE PACKING WHEN YOU INSTALL THE N2 SPEED SENSOR. A DAMAGED PACKING CAN CAUSE ENGINE OIL LEAKS. OIL LEAKS CAN CAUSE AN IN-FLIGHT SHUTDOWN OF THE ENGINE.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the N2 speed sensor [1] on the accessory gearbox [2]:

(a) Remove the protective cover from the open port for the N2 speed sensor [1] on the accessory gearbox [2].

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the new preformed packing [7], and the new preformed packing [8] with vaseline, D00672 [CP5070].

(d) Lubricate the threads of the bolts [6] with graphite mineral vaseline, D00601 [CP2101]

(e) Install the new preformed packing [7] and new preformed packing [8] into the grooves on the N2 speed sensor [1].

(f) Install the N2 speed sensor [1] on the accessory gearbox with the two bolts [6].

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1) Tighten the bolts [6] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(g) Connect the electrical connectors to the N2 speed sensor [1] as follows (TASK 70-00-01-910-803-G00):

1) Remove the protective covers from the electrical connector (W300 harness) [5], the electrical connector (HJ6B harness) [4], the electrical connector (HJ6A harness) [3], and the electrical receptacles on the N2 speed sensor [1].

2) Use the teflon-jawed pliers, STD-664 to connect the electrical connector (W300 harness) [5], the electrical connector (HJ6B harness) [4] and the electrical connector (HJ6A harness) [3] to the N2 speed sensor [1].

(h) Clean the work area and clear the tools and other items.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 77-12-02-410-001-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 77-12-02-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

K. N2 Speed Sensor Installation Test

SUBTASK 77-12-02-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

N1 SPEED SENSOR SHIM - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) N1 speed sensor shim removal

(2) N1 speed sensor shim installation.

TASK 77-12-03-000-801-G00

2. N1 Speed Sensor Shim Removal

(Figure 401)

A. General

(1) This task gives the instructions on how to remove the N1 speed sensor shim.

B. References

Reference	Title
77-12-01-000-801-G00	N1 Speed Sensor Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4316	Fiberscope - Flexible Borescope Part #: USEF 2-2.4-1000 Supplier: 7CGG1 Part #: USEF 2-6-1000 Supplier: 7CGG1 Opt Part #: 7110561 Supplier: 32212 Opt Part #: IF2D5 Supplier: 32212 Opt Part #: IF6C5X1-8 Supplier: 32212
STD-10718	Fiberscope (Meets General Electric or CFM Standard Practices Manual Specification)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 77-12-03-020-003-G00

(1) Remove the N1 speed sensor [2] (TASK 77-12-01-000-801-G00).

F. N1 Speed Sensor Shim Removal

SUBTASK 77-12-03-020-001-G00

(1) If the N1 speed sensor shim [1] is on the N1 speed sensor [2], remove the N1 speed sensor shim [1] from the N1 speed sensor [2].

SUBTASK 77-12-03-020-002-G00

(2) If the N1 speed sensor shim [1] is in the guide tube in the 5 o'clock strut of the fan hub frame, remove the N1 speed sensor shim [1] from the guide tube, using a stick while watching the area with the fiberscope or flexible borescope.

NOTE: The N1 speed sensor shim can be attached with oil to the bottom of the sensor guide.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Use a fiberscope, STD-10718, or fiberscope, COM-4316, with video probe.

1) Use a forward view optical tip.

SUBTASK 77-12-03-210-001-G00

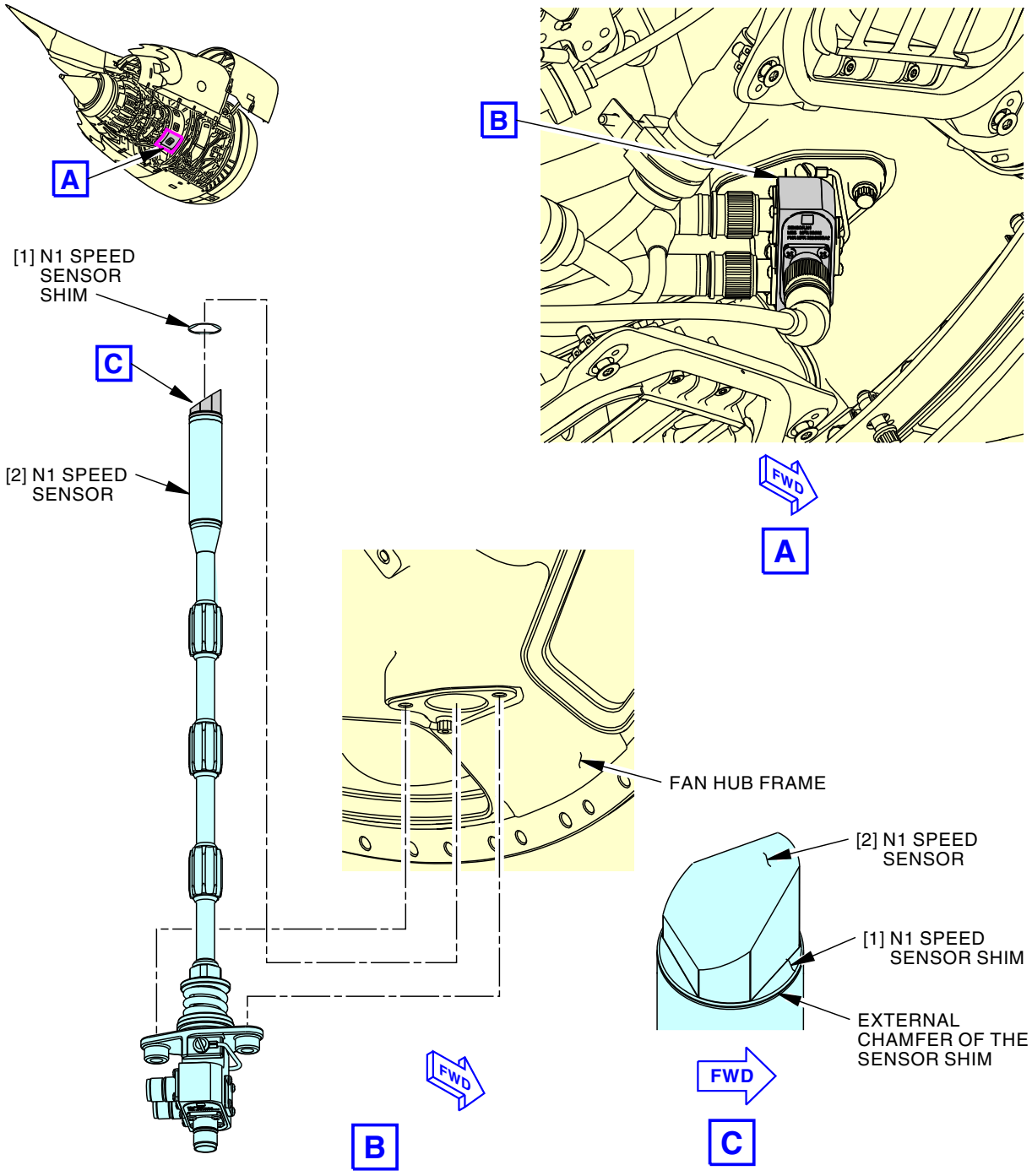
(3) Do a visual inspection of the removed shim.

(a) If indications are found, the shim is not serviceable.

(b) If no indication is found, the shim is serviceable and can be re-used.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



N1 Speed Sensor Shim Installation
Figure 401/77-12-03-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 77-12-03-400-801-G00

3. N1 Speed Sensor Shim Installation

(Figure 401)

A. General

(1) This task gives the instructions on how to install the N1 speed sensor shim.

B. References

Reference	Title
77-12-01-400-801-G00	N1 Speed Sensor Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. N1 Speed Sensor Shim Installation

SUBTASK 77-12-03-860-001-G00

(1) Apply vaseline, D00672 [CP5070], on the face of the N1 speed sensor shim [1] in contact with the N1 speed sensor [2].

SUBTASK 77-12-03-420-001-G00

(2) Install the N1 speed sensor shim [1] on the extremity of the N1 speed sensor [2].

(a) Put the external chamfer of the N1 speed sensor shim [1] to the outside.

SUBTASK 77-12-03-860-002-G00

(3) Make sure that vaseline, D00672 [CP5070], is applied only on the face of the N1 speed sensor shim [1] in contact with the N1 speed sensor [2].

SUBTASK 77-12-03-420-002-G00

(4) Do this task: N1 Speed Sensor Installation, TASK 77-12-01-400-801-G00.

——— END OF TASK ———

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ENGINE TEMPERATURE SENSING SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has two tasks:

(1) Engine temperature sensing system inspection

(2) Exhaust Gas Temperature (EGT) system degradation system test.

TASK 77-21-00-210-801-G00

2. Engine Temperature Sensing System Inspection

(Figure 601)

A. General

(1) This task gives the instructions to do the visual inspection of the engine temperature sensing system.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
77-21-01-000-801-G00	Exhaust Gas Temperature Sensors Removal (P/B 401)
77-21-01-400-801-G00	Exhaust Gas Temperature Sensors Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-26216	Fixture - Go No Go, T48 Sensor Part #: 956L6700 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G50163 [C10-217]	Tape - Reinforced Silicone Stretchtape - RL 6000SA	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 77-21-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 77-21-00-860-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Engine Temperature Sensing System Inspection

SUBTASK 77-21-00-211-001-G00

(1) Do a visual inspection of the EGT sensors [1] for these conditions:

(a) Loose EGT sensor [1]

1) Not permitted.

2) If the bolts are loose, tighten the bolts [2] (TASK 77-21-01-400-801-G00).

(b) Air leakage

1) Not permitted.

2) Replace the gasket [3] on the EGT sensor [1] (TASK 77-21-01-400-801-G00).

(c) Chaffing on the lead [6] or cable [5]

1) Not permitted, replace the lead [6] or cable [5].

2) Continue-In-Service (CIS) limit for chaffing of the leads is 10 days with these conditions:

a) Chaffing is limited to the EGT sensor cable [5] or the lead [6] connected to only one EGT sensor [1].

b) Exhaust Gas Temperature (EGT) system maintenance messages are limited to only one EGT sensor [1].

<1> Do the EGT system degradation system test (TASK 77-21-00-730-801-G00).

<2> Make sure that you find the maintenance messages only for one EGT sensor [1].

c) Disconnect the lead [6] and the cable [5] from the EGT sensor [1] to prevent EGT fluctuations as follows:

<1> Disconnect all connections to the EGT sensor [1].

<2> Put RL 6000SA tape, G50163 [C10-217] on the lead [6].

<3> Fold the lead [6] back on the EGT sensor cable [5].

<4> Separate the lead [6] to prevent an electrical short.

<5> Attach the folded lead [6] to the EGT sensor cable [5] with safetywire or RL 6000SA tape, G50163 [C10-217].

(d) Loose or bad terminal studs [4]

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

3) Continue-In-Service (CIS) limit for loose or bad terminal studs is 10 days with these conditions:

a) Loose or bad terminal studs [4] should be only on one EGT sensor [1].

b) EGT system maintenance messages are limited to only one EGT sensor [1].

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<1> Do the EGT system degradation system test (TASK 77-21-00-730-801-G00)

<2> Make sure that you find the maintenance messages only for one EGT sensor [1].

c) Disconnect the lead [6] and the cable [5] from the EGT sensor [1] to prevent EGT fluctuations as follows:

<1> Disconnect all connections to the EGT sensor [1].

<2> Put RL 6000SA tape, G50163 [C10-217] on the lead [6].

<3> Fold the lead [6] back on the EGT sensor cable [5].

<4> Separate the lead [6] to prevent an electrical short.

<5> Attach the folded lead [6] to the EGT sensor cable [5] with safetywire or RL 6000SA tape, G50163 [C10-217].

(e) Do a visual inspection of the sensor stem of the EGT sensor [1] (area F, Figure 601) for these conditions:

1) Creeping

a) Maximum serviceable limit is 0.0393 in. (0.9982 mm).

<1> Put the EGT sensor in the fixture, SPL-26216 (Figure 601 (Sheet 5)):

NOTE: The fixture, SPL-26216 has a 0.0393 in. (0.9982 mm) depth groove.

<a> If the groove has the EGT sensor stem on it, the creeping is within serviceable limit.

<b> If you can fully see the groove, the creeping is out of serviceable limit, replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

<c> You can use the metal ruler to measure the creeping.

SUBTASK 77-21-00-211-002-G00

(2) Do a visual inspection of the flange [7] on the EGT sensors [1] for these conditions:

(a) Cracks, dents, tear and erosion/peening

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(b) Nicks

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(c) Scoring/Scratches

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(d) Deposit

1) Serviceable

(e) Engraved identification cannot be read with visual and optical device

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1) Not Serviceable

(f) Engraved identification absence

1) Not Serviceable

(g) Scores and shock marks on engraved identification

1) Not Serviceable

SUBTASK 77-21-00-211-003-G00

(3) Do a visual inspection of the housing [8] on the EGT sensors [1] and for these conditions:

(a) Cracks, dents, tear, and erosion/peening

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(b) Nicks

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(c) Scoring/Scratches

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(d) Deposit

1) Serviceable

SUBTASK 77-21-00-211-004-G00

(4) Do a visual inspection of the collar [9] on the EGT sensors [1] for these conditions:

(a) Cracks, dents, tear, oxidation and erosion/peening

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(b) Nicks

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(c) Scoring/Scratches

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(d) Deposit

1) Serviceable

SUBTASK 77-21-00-211-005-G00

(5) Do the visual inspection of the terminal housing [10] on the EGT sensors [1] for these conditions:

(a) Cracks, dents, tear, oxidation and erosion/peening

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1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(b) Nicks

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(c) Scoring/Scratches

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

(d) Discoloration

1) Serviceable

NOTE: A light deposit of discoloration is permitted.

(e) Deposit

1) Serviceable

SUBTASK 77-21-00-210-001-G00

(6) Do a visual inspection of the fill plug [13] and the lid [14] on the EGT sensors [1] for these conditions:

(a) Cracks, nicks, dents, tear, oxidation, scoring/scratches and erosion/peening

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

SUBTASK 77-21-00-210-002-G00

(7) Do a visual inspection of the KP terminal stud assembly [15] and the KN terminal stud assembly [16] on the EGT sensors [1] for these conditions:

(a) Cracks, nicks, dents, tear, oxidation, scoring/scratches, erosion/peening and deposits

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

SUBTASK 77-21-00-211-006-G00

(8) Do a visual inspection of the weld joint and braze joint [11], weld joint and braze joint [12], weld joint and braze joint [17], weld joint and braze joint [18], weld joint and braze joint [19] and weld joint and braze joint [20] for these conditions:

(a) Cracks, nicks, dents, tear, oxidation, scoring/scratches, erosion/peening and deposits.

1) Not permitted.

2) Replace the EGT sensor [1] (TASK 77-21-01-000-801-G00 and TASK 77-21-01-400-801-G00).

SUBTASK 77-21-00-160-001-G00

(9) Clean the work area and clear the tools and other items.

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I. Put the Airplane Back to Its Usual Condition

SUBTASK 77-21-00-860-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 77-21-00-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Engine Temperature Sensing System Tests

SUBTASK 77-21-00-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

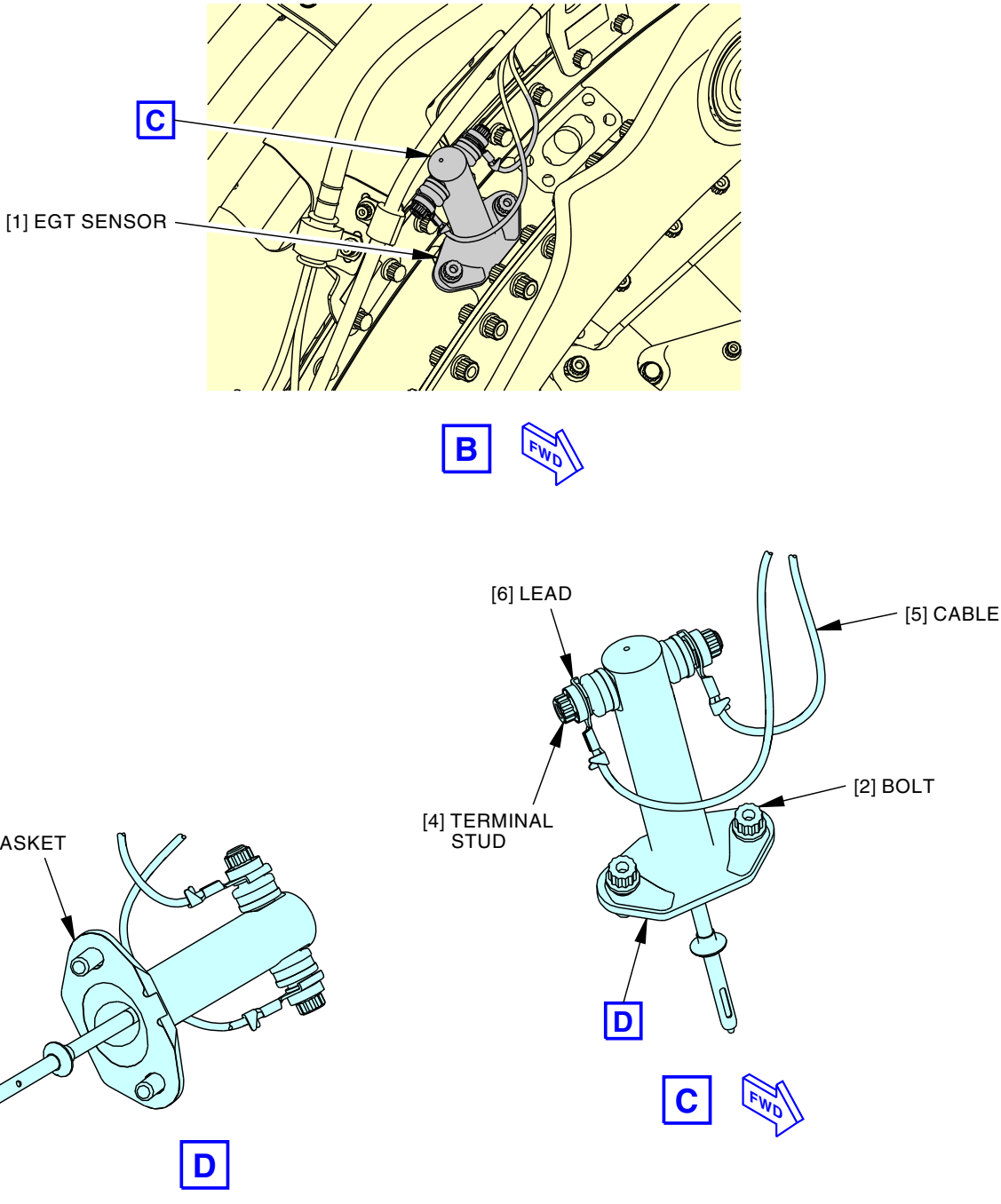
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355065 S00061516482_V1

Engine Temperature Sensing System Inspection
Figure 601/77-21-00-990-801-G00 (Sheet 1 of 5)

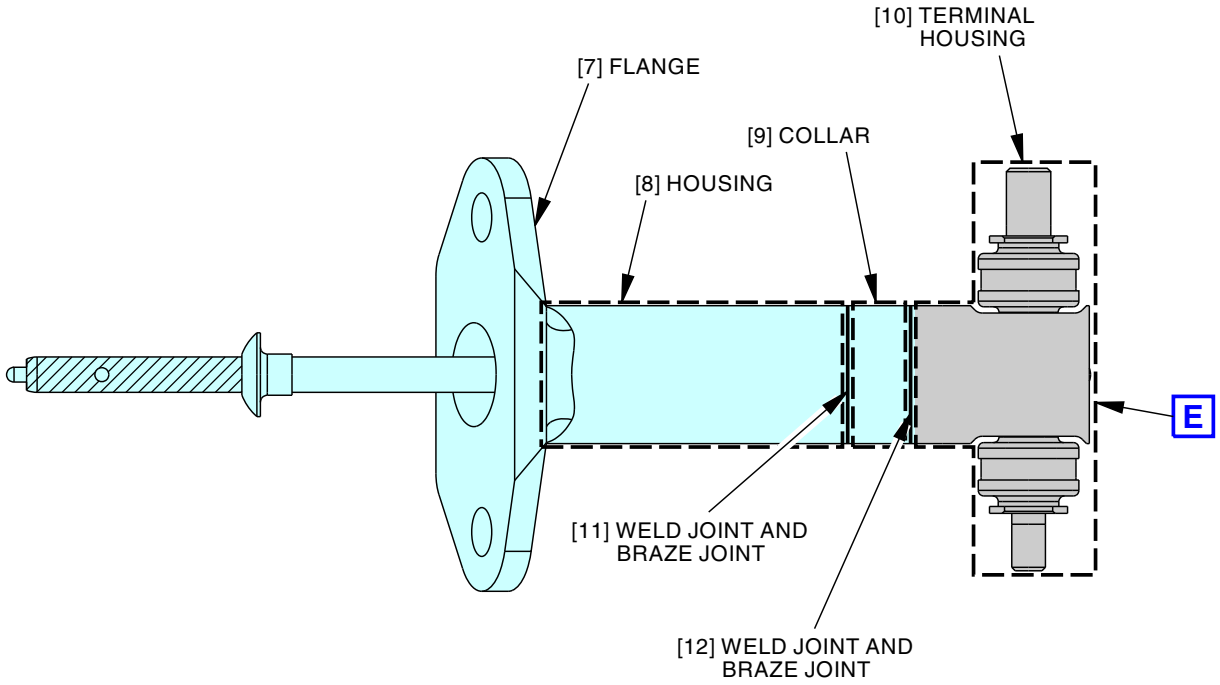
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2457749 S0000572048_V2

Engine Temperature Sensing System Inspection
Figure 601/77-21-00-990-801-G00 (Sheet 2 of 5)

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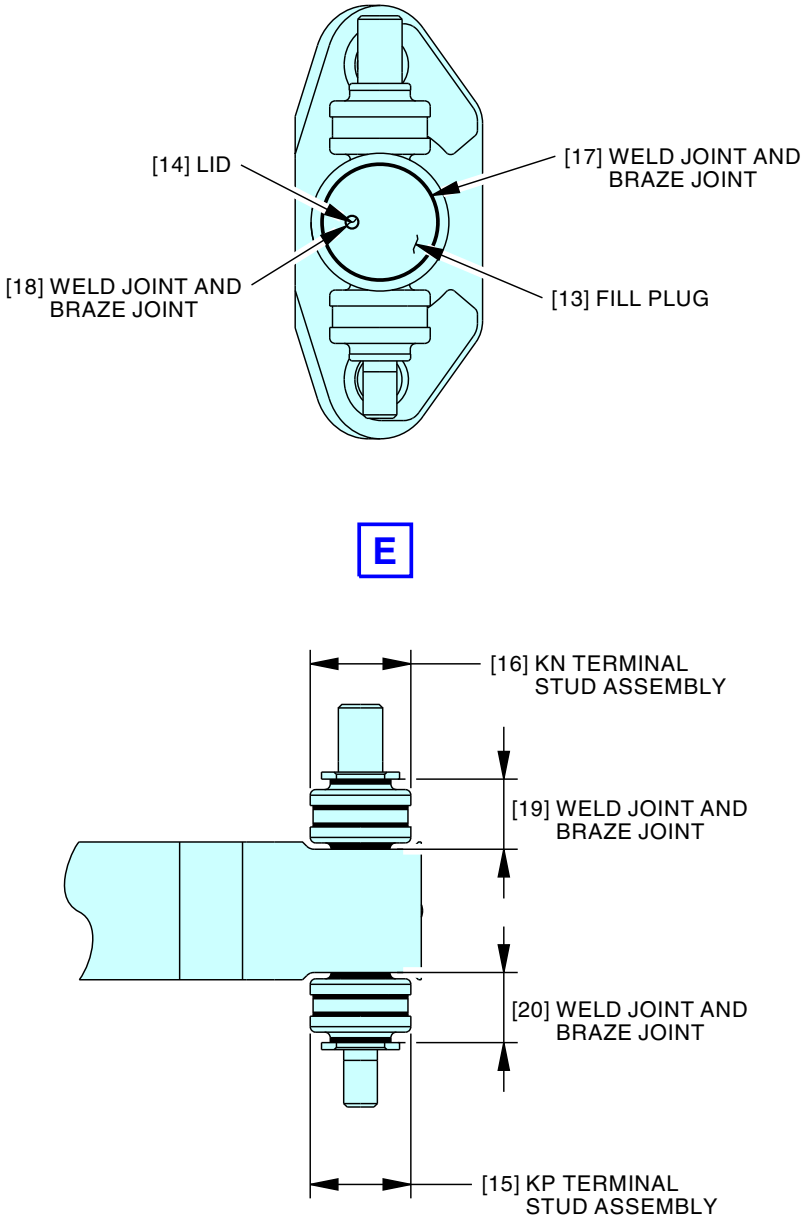
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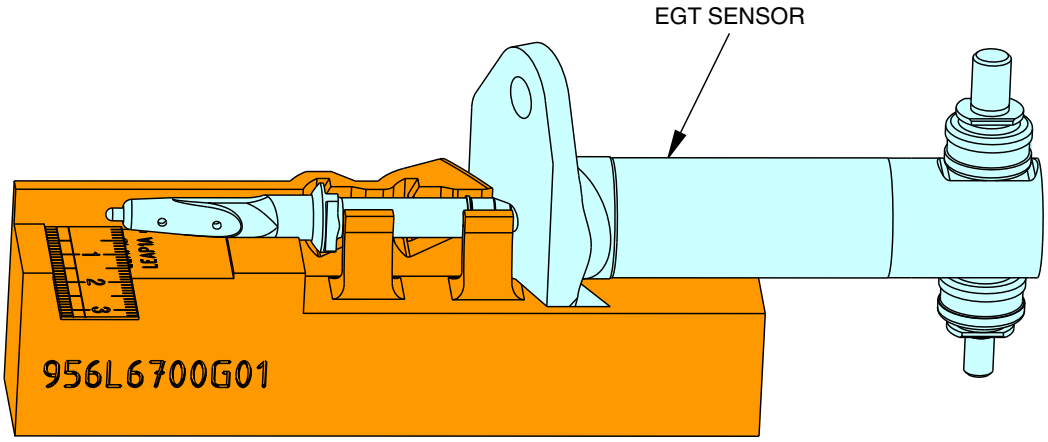
Engine Temperature Sensing System Inspection
Figure 601/77-21-00-990-801-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Temperature Sensing System Inspection
Figure 601/77-21-00-990-801-G00 (Sheet 4 of 5)

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3122329 S0000856908_V1

Engine Temperature Sensing System Inspection
Figure 601/77-21-00-990-801-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 77-21-00-730-801-G00

3. EGT System Degradation System Test

A. General

(1) This task gives the instructions on how to do the Exhaust Gas Temperature (EGT) system degradation test.

B. References

Reference	Title
46-13-02-700-806	Fault History Check (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. EGT System Degradation Test (Preferred)

SUBTASK 77-21-00-200-001-G00

(1) Do the EGT system degradation test as follows:

(a) Set the EEC MAINT POWER switch on the MISC SYSTEM CTRLS to TEST (TASK 73-21-00-800-801-G00)

(b) Wait 30 seconds before you start the test.

(c) Use the MAX display system to find the EGT system fault messages (TASK 46-13-02-700-806).

(d) Select the criteria below on the MAX display system:

1) ONBD MAINT

2) EXTENDED MAINT

3) FAULT HISTORY

4) 71-80 Engine 1 or 2

(e) Look for the EGT system degradation maintenance messages that show in the Table 601 that follows:

1) Look for these maintenance messages in the last 3 flight legs (LEG 0, LEG–1 and LEG-2).

2) If you find one or more of the EGT system degradation maintenance messages, do the corrective action in the fault isolation manual (FIM) for the message(s) that you find.

Table 601/77-21-00-993-803-G00

Maintenance Message Number	Maintenance Message
77–4130X*[1]	EGT Probe 1 Sensor Fault
77–4162X	EGT Probe 2 Sensor Fault
77–4164X	EGT Probe 3 Sensor Fault
77–4166X	EGT Probe 4 Sensor Fault
77–4168X	EGT Probe 5 Sensor Fault

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Table 601/77-21-00-993-803-G00 (Continued)

Maintenance Message Number	Maintenance Message
77–4170X	EGT Probe 6 Sensor Fault
77–4172X	EGT Probe 7 Sensor Fault
77–4174X	EGT Probe 8 Sensor Fault

*[1] X = Engine Position 1 or 2

E. EGT System Degradation Test (Alternate)

SUBTASK 77-21-00-200-002-G00

(1) Do this task: Engine Temperature Sensing System Inspection, TASK 77-21-00-210-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EXHAUST GAS TEMPERATURE (EGT) SENSORS - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Exhaust Gas Temperature (EGT) sensors removal

(2) Exhaust Gas Temperature (EGT) sensors installation.

TASK 77-21-01-000-801-G00

2. Exhaust Gas Temperature Sensors Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to remove the exhaust gas temperature sensor.

(2) The exhaust gas temperature sensor is referred to as the EGT sensor.

(3) There are eight EGT sensors which are installed on the turbine center frame. The EGT sensors are interchangeable.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14947	Protector - SACOC Part #: 956A8681G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 77-21-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 77-21-01-860-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 77-21-01-860-003-G00

(3) For the left engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 77-21-01-860-004-G00

(4) For the right engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 77-21-01-010-001-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 77-21-01-480-001-G00

(6) Install the protector, SPL-14947 on the oil coolers.

G. Exhaust Gas Temperature Sensors Removal

SUBTASK 77-21-01-020-001-G00

CAUTION MAKE SURE THAT YOU INSTALL THE SACOC PROTECTOR TO PREVENT DAMAGE TO THE OIL COOLER.

(1) For the EGT sensor [1] at location 7, remove the EGT sensor [1] as follows:

NOTE: The EGT sensor at location 7 is installed at the 1:00 o'clock position aft looking forward on the turbine center frame.

(a) Remove the bolt [7] that attaches the loop clamp [8] and the EGT harness [3] to the bracket [6].

(b) Tie the EGT harness [3] out of the way with tie wraps.

CAUTION DO NOT REMOVE THE CAPTIVE NUTS FROM THE LEADS. DAMAGE CAN OCCUR TO THE LEADS OR THE CAPTIVE NUTS.

(c) Loosen the captive nuts on the lead [2A] and the lead [2B].

(d) Remove the lead [2A] and the lead [2B] from the EGT sensor [1].

(e) Remove and discard the safety cable or lockwire from the two bolts [5].

(f) Remove the two bolts [5] that attach the EGT sensor [1] to the mounting pad of the turbine center frame.

(g) Remove the EGT sensor [1].

(h) Remove and discard the gasket [4] from the EGT sensor [1].

(i) Install the protective cover on the open port for the EGT sensor [1] on the turbine center frame.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 77-21-01-010-002-G00

CAUTION

MAKE SURE THAT YOU INSTALL THE SACOC PROTECTOR TO PREVENT DAMAGE TO THE OIL COOLER.

(2) For the EGT sensor [1] at locations 1 to 6, remove the EGT sensor [1] as follows:

NOTE: The EGT sensors at locations 1 to 6 are installed anti-clockwise at the 10:00 to 3:00 o'clock positions aft looking forward on the turbine center frame.

CAUTION

DO NOT REMOVE THE CAPTIVE NUTS FROM THE LEADS. DAMAGE CAN OCCUR TO THE LEADS OR THE CAPTIVE NUTS.

(a) Loosen the captive nuts on the lead [2A] and the lead [2B].

(b) Remove the lead [2A] and the lead [2B] from the EGT sensor [1].

(c) Remove and discard the safety cable or lockwire from the two bolts [5].

(d) Remove the two bolts [5] that attach the EGT sensor [1] to the mounting pad of the turbine center frame.

(e) Remove the EGT sensor [1].

(f) Remove and discard the gasket [4] from the EGT sensor [1].

(g) Install the protective cover on the open port for the EGT sensor [1] on the turbine center frame.

SUBTASK 77-21-01-010-003-G00

CAUTION

MAKE SURE THAT YOU INSTALL THE SACOC PROTECTOR TO PREVENT DAMAGE TO THE OIL COOLER.

(3) For the EGT sensor [1] at location 8, remove the EGT sensor [1] as follows:

NOTE: The EGT sensor at location 8 is installed at the 12:00 o'clock position aft looking forward on the turbine center frame.

CAUTION

DO NOT REMOVE THE CAPTIVE NUTS FROM THE LEADS. DAMAGE CAN OCCUR TO THE LEADS OR THE CAPTIVE NUTS.

(a) Loosen the captive nuts on the lead [2A] and the lead [2B].

(b) Remove the lead [2A] and the lead [2B] from the EGT sensor [1].

(c) Remove and discard the safety cable or lockwire from the two bolts [5].

(d) Remove the two bolts [5] that attach the EGT sensor [1] to the mounting pad of the turbine center frame.

NOTE: You can only access the right side bolt from the right side of the engine.

(e) Remove the EGT sensor [1].

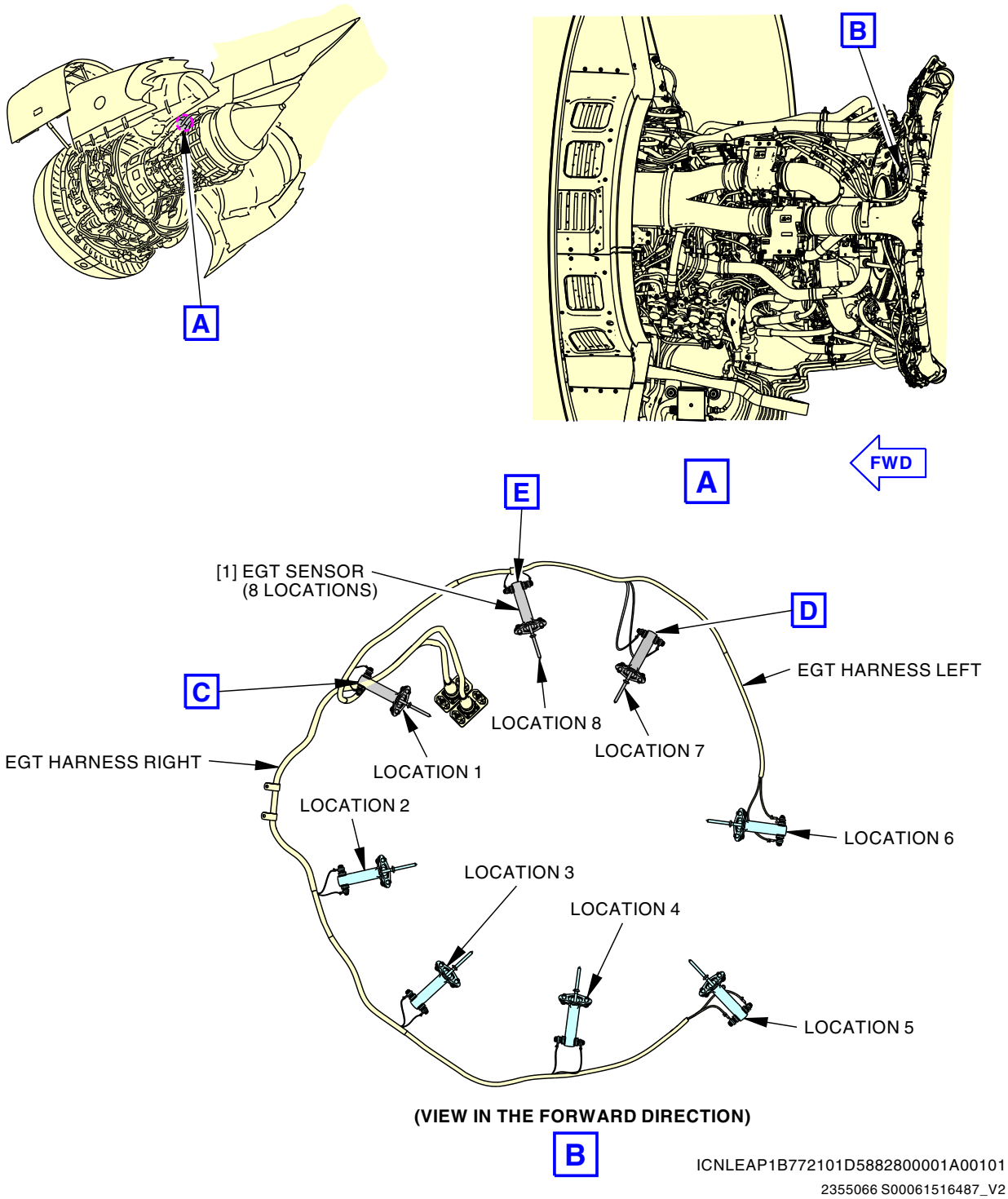
(f) Remove and discard the gasket [4] from the EGT sensor [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Install the protective cover on the open port for the EGT sensor [1] on the turbine center frame.

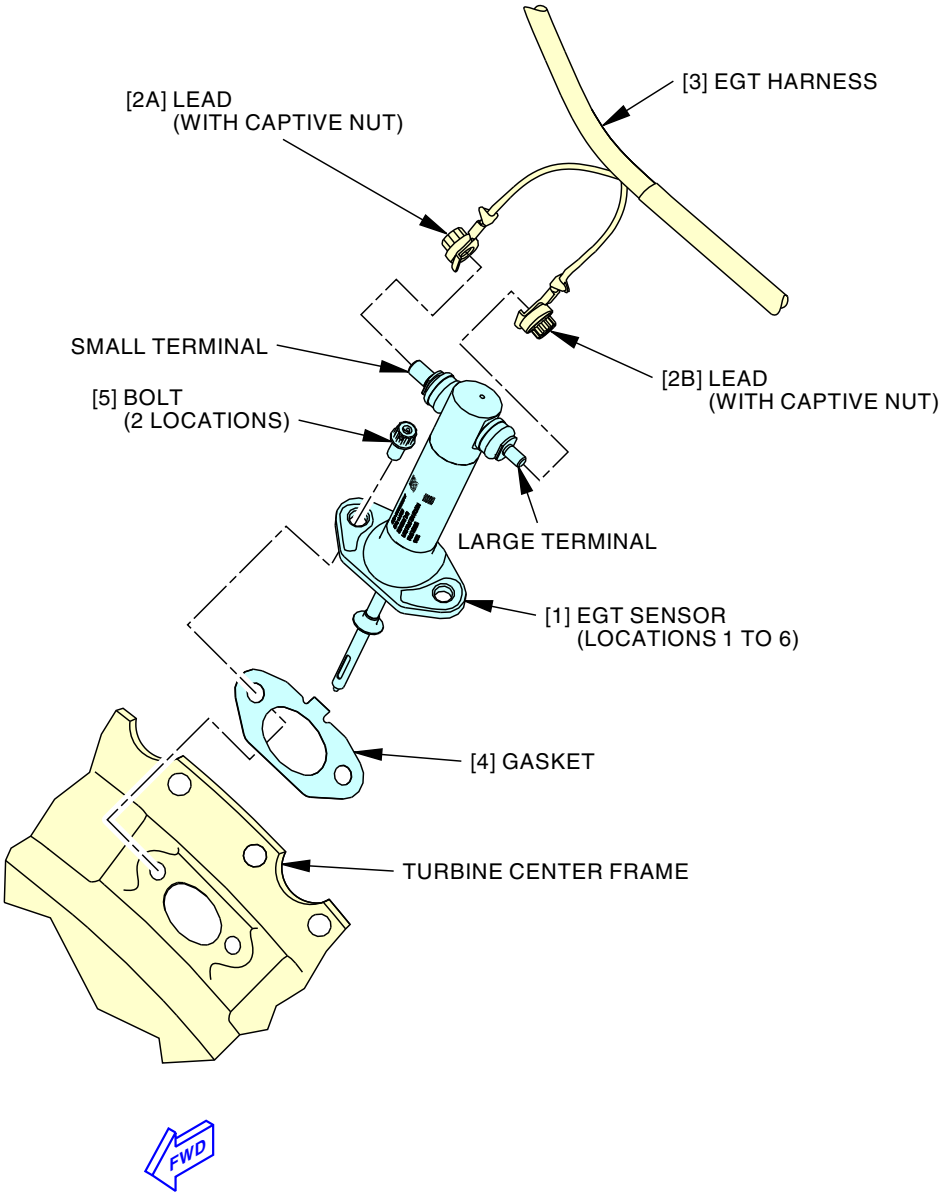
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exhaust Gas Temperature Sensors Installation
Figure 401/77-21-01-990-801-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EGT SENSORS LOCATIONS 1 TO 6

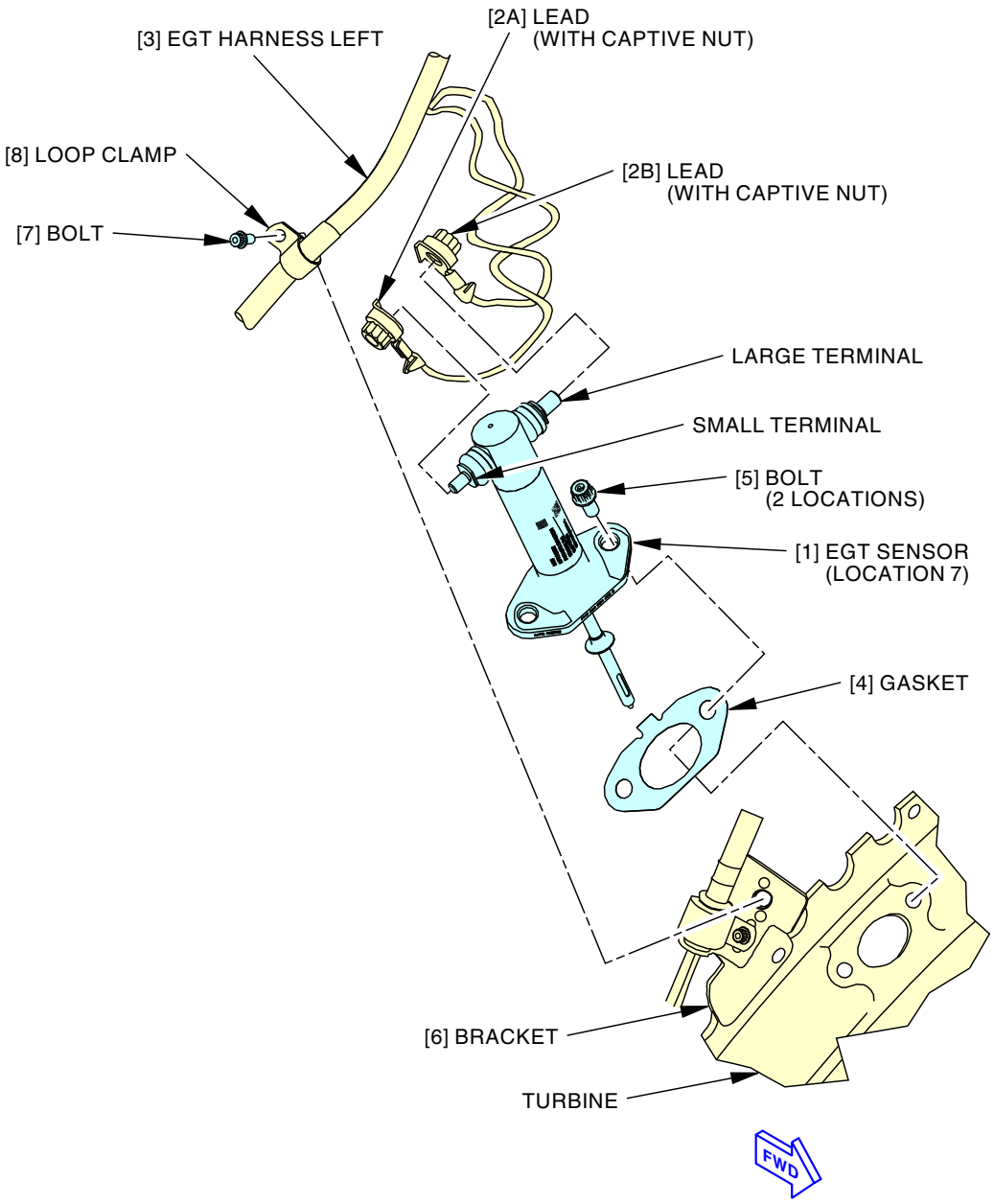
C

ICNLEAP1B772101D5882800002A00101

2355067 S00061516488_V3

Exhaust Gas Temperature Sensors Installation
Figure 401/77-21-01-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EGT SENSOR LOCATION 7

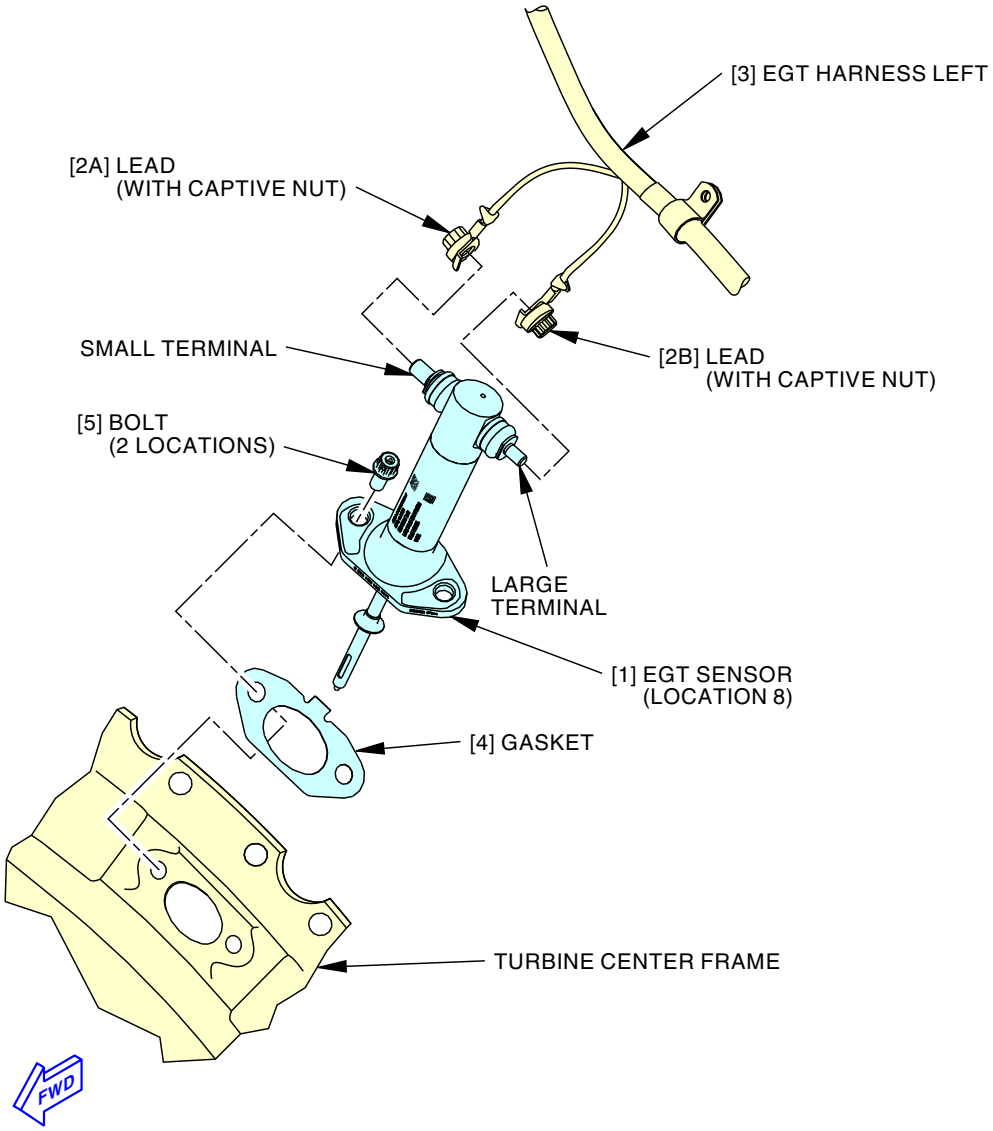
D

ICNLEAP1B772101D5882800003A00101

2402884 S0000554880_V3

Exhaust Gas Temperature Sensors Installation
Figure 401/77-21-01-990-801-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EGT SENSOR LOCATION 8

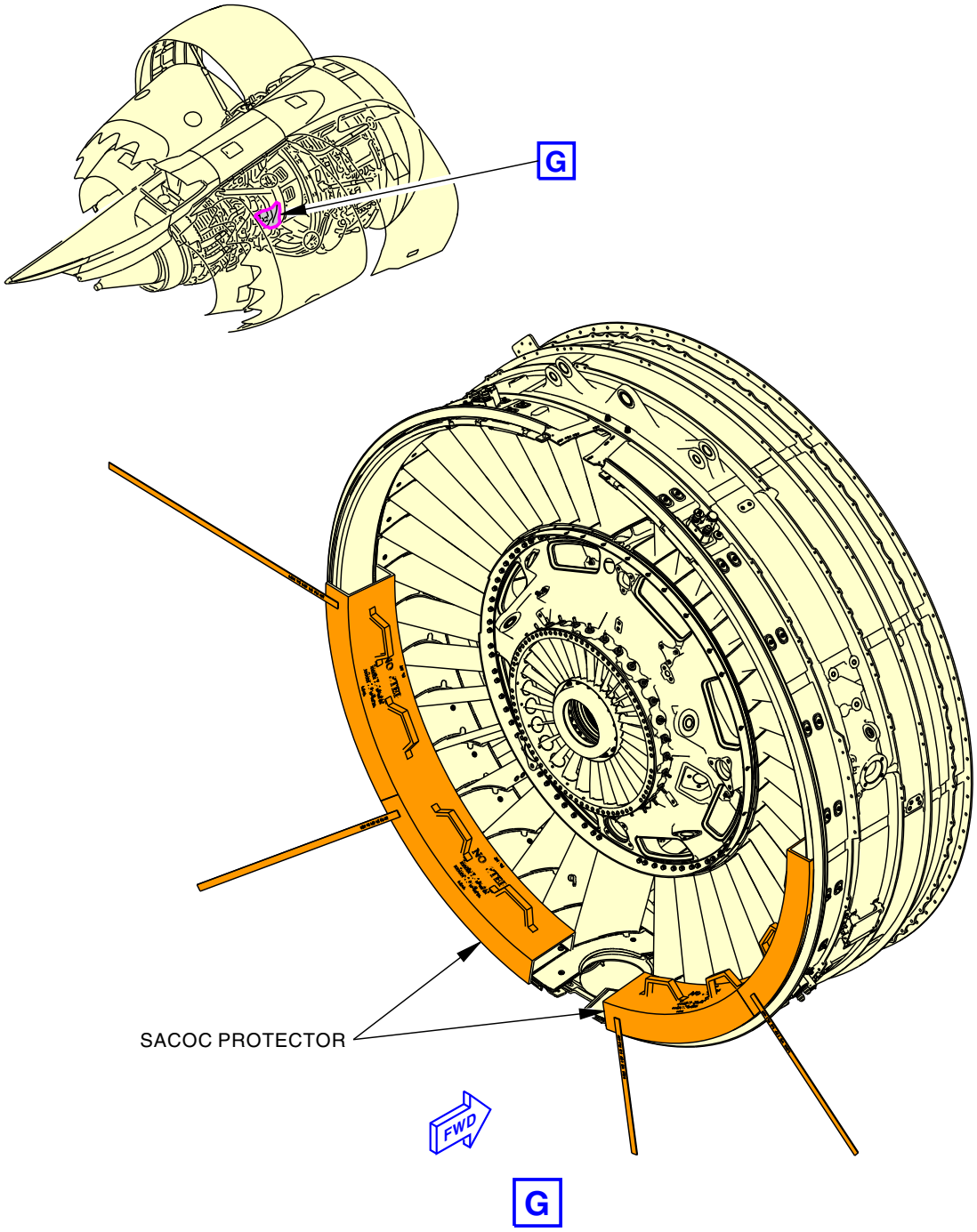
E

ICNLEAP1B772101D5882800004A00101

2402885 S0000554881_V3

Exhaust Gas Temperature Sensors Installation
Figure 401/77-21-01-990-801-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SACOC Protector Installation
Figure 402/77-21-01-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 77-21-01-400-801-G00

3. Exhaust Gas Temperature Sensors Installation

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to install the exhaust gas temperature sensor.

(2) The exhaust gas temperature sensor is referred to as the EGT sensor.

(3) There are eight EGT sensors which are installed on the turbine center frame. The EGT sensors are interchangeable.

B. References

Reference	Title
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14947	Protector - SACOC Part #: 956A8681G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
G02495 [CP8002]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS5689
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A
G51435 [CP2635]	Wax - Stick	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	EGT sensor	77-21-01-01-020	PTF ALL
4	Gasket	77-21-01-01-030	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Exhaust Gas Temperature Sensors Installation

SUBTASK 77-21-01-420-001-G00

(1) For the EGT sensor [1] at location 7, install the EGT sensor [1] as follows:

NOTE: The EGT sensor at location 7 is installed at the 1:00 o'clock position aft looking forward on the turbine center frame.

(a) Remove the protective cover from the open port for the EGT sensor [1] on the turbine center frame.

(b) Put a new gasket [4] on the mounting pad flange.

NOTE: Make sure that the tab on the gasket is in the correct position. The gasket must have full sealing around the EGT sensor to prevent leaks in the probe hole.

(c) Install the EGT sensor [1] on the mounting pad.

(d) Lubricate the threads of the two bolts [5] with graphite mineral vaseline, D00601 [CP2101].

(e) Install the two bolts [5].

1) Tighten the two bolts [5] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

2) Install the safety wire, G02495 [CP8002] or safety cable, G50065 [CP8006]on the bolts [5] (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

CAUTION

MAKE SURE THAT THE EGT HARNESS LEADS ARE IN THE CORRECT POSITION BEFORE YOU TIGHTEN THE CAPTIVE NUTS. IF THE LEADS ARE NOT IN THE CORRECT POSITION, YOU CAN CAUSE DAMAGE TO EGT HARNESS LEADS AND THE EGT PROBE.

CAUTION

DO NOT APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS. IF YOU APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS, YOU CAN CAUSE DAMAGE TO THE TERMINALS ON THE EGT PROBE.

(f) Install the lead [2A] on the EGT sensor [1] as follows:

NOTE: The EGT probe has a large terminal and a small terminal. Make sure that you install the captive nuts on the correct terminals.

1) Attach the lead [2A] to the EGT sensor [1] smaller terminal.

2) Hold the wiring harness lead [2A] to keep it in its position.

3) Tighten the captive nut to 25.2 in-lb (2.85 N·m) - 27.9 in-lb (3.15 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE THAT THE EGT HARNESS LEADS ARE IN THE CORRECT POSITION BEFORE YOU TIGHTEN THE CAPTIVE NUTS. IF THE LEADS ARE NOT IN THE CORRECT POSITION, YOU CAN CAUSE DAMAGE TO EGT HARNESS LEADS AND THE EGT PROBE.

CAUTION DO NOT APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS. IF YOU APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS, YOU CAN CAUSE DAMAGE TO THE TERMINALS ON THE EGT PROBE.

(g) Install the lead [2B] to the EGT sensor [1] as follows:

NOTE: The EGT probe has a large terminal and a small terminal. Make sure that you install the captive nuts on the correct terminals.

NOTE: Wrap the lead [2B] around the lead [2A] three to five times to prevent the leads from contacting the wire harness or adjacent hardware. Do not wrap too tight to prevent damage to the leads.

1) Attach the lead [2B] to the EGT sensor [1] larger terminal.
2) Hold the wiring harness lead [2B] to keep it in its position.
3) Tighten the captive nut to 34.0 in-lb (3.8 N·m) - 37.0 in-lb (4.2 N·m).

(h) Remove the tie wraps from the EGT harness [3].

(i) Apply engine oil, D00599 [CP2442] on the threads of the bolt [7].

(j) Install the bolt [7] that attaches the loop clamp [8] and the EGT harness [3] to the bracket [6].

1) Tighten the bolt [7] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

SUBTASK 77-21-01-410-001-G00

(2) For the EGT sensor [1] at locations 1 to 6, install the EGT sensor [1] as follows:

NOTE: The EGT sensor at locations 1 to 6 are installed anti-clockwise at the 10:00 to 3:00 o'clock positions aft looking forward on the turbine center frame.

(a) Remove the protective cover from the open port for the EGT sensor [1] on the turbine center frame.

(b) Put a new gasket [4] on the mounting pad flange.

NOTE: The gasket must have full sealing around the EGT sensor to prevent leaks in the probe hole.

(c) Install the EGT sensor [1] on the mounting pad.

(d) Lubricate the threads of the two bolts [5] with graphite mineral vaseline, D00601 [CP2101].

(e) Install the two bolts [5].

1) Tighten the bolts [5] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).
2) Install the safety wire, G02495 [CP8002] or safety cable, G50065 [CP8006] on the bolts [5] (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

CAUTION MAKE SURE THAT THE EGT HARNESS LEADS ARE IN THE CORRECT POSITION BEFORE YOU TIGHTEN THE CAPTIVE NUTS. IF THE LEADS ARE NOT IN THE CORRECT POSITION, YOU CAN CAUSE DAMAGE TO EGT HARNESS LEADS AND THE EGT PROBE.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION DO NOT APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS. IF YOU APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS, YOU CAN CAUSE DAMAGE TO THE TERMINALS ON THE EGT PROBE.

(f) Install the lead [2A] on the EGT sensor [1] as follows:

NOTE: The EGT probe has a large terminal and a small terminal. Make sure that you install the captive nuts on the correct terminals.

1) Attach the lead [2A] to the EGT sensor [1] smaller terminal.
2) Hold the wiring harness lead [2A] to keep it in its position.
3) Tighten the captive nut to 25.2 in-lb (2.85 N·m) - 27.9 in-lb (3.15 N·m).

CAUTION MAKE SURE THAT THE EGT HARNESS LEADS ARE IN THE CORRECT POSITION BEFORE YOU TIGHTEN THE CAPTIVE NUTS. IF THE LEADS ARE NOT IN THE CORRECT POSITION, YOU CAN CAUSE DAMAGE TO EGT HARNESS LEADS AND THE EGT PROBE.

CAUTION DO NOT APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS. IF YOU APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS, YOU CAN CAUSE DAMAGE TO THE TERMINALS ON THE EGT PROBE.

(g) Install the lead [2B] to the EGT sensor [1] as follows:

NOTE: The EGT probe has a large terminal and a small terminal. Make sure that you install the captive nuts on the correct terminals.

NOTE: Wrap the lead [2B] around the lead [2A] three to five times to prevent the leads from contacting the wire harness or adjacent hardware. Do not wrap too tight to prevent damage to the leads.

1) Attach the lead [2B] to the EGT sensor [1] larger terminal.
2) Hold the wiring harness lead [2B] to keep it in its position.
3) Tighten the captive nut to 34.0 in-lb (3.8 N·m) - 37.0 in-lb (4.2 N·m).

SUBTASK 77-21-01-410-002-G00

(3) For the EGT sensor [1] at location 8, install the EGT sensor [1] as follows:

NOTE: The EGT sensor at location 8 is installed at the 12:00 o'clock position aft looking forward on the turbine center frame.

(a) Remove the protective cover from the open port for the EGT sensor [1] on the turbine center frame.

(b) Put a new gasket [4] on the mounting pad flange as follows:

NOTE: The gasket must have full sealing around the EGT sensor to prevent leaks in the probe hole.

1) Use stick wax, G51435 [CP2635] to hold the gasket [4] in place on the mount pad flange.

(c) Install the EGT sensor [1] on the location 8 mounting pad.

(d) Lubricate the threads of the two bolts [5] with graphite mineral vaseline, D00601 [CP2101].

(e) Install the two bolts [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Tighten the bolts [5] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

2) Install the safety wire, G02495 [CP8002] or safety cable, G50065 [CP8006] on the bolts [5] (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

CAUTION MAKE SURE THAT THE EGT HARNESS LEADS ARE IN THE CORRECT POSITION BEFORE YOU TIGHTEN THE CAPTIVE NUTS. IF THE LEADS ARE NOT IN THE CORRECT POSITION, YOU CAN CAUSE DAMAGE TO EGT HARNESS LEADS AND THE EGT PROBE.

CAUTION DO NOT APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS. IF YOU APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS, YOU CAN CAUSE DAMAGE TO THE TERMINALS ON THE EGT PROBE.

(f) Install the lead [2A] on the EGT sensor [1] as follows:

NOTE: The EGT probe has a large terminal and a small terminal. Make sure that you install the captive nuts on the correct terminals.

1) Attach the lead [2A] to the EGT sensor [1] smaller terminal.

2) Hold the wiring harness lead [2A] to keep it in its position.

3) Tighten the captive nut to 25.2 in-lb (2.85 N·m) - 27.9 in-lb (3.15 N·m).

CAUTION MAKE SURE THAT THE EGT HARNESS LEADS ARE IN THE CORRECT POSITION BEFORE YOU TIGHTEN THE CAPTIVE NUTS. IF THE LEADS ARE NOT IN THE CORRECT POSITION, YOU CAN CAUSE DAMAGE TO EGT HARNESS LEADS AND THE EGT PROBE.

CAUTION DO NOT APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS. IF YOU APPLY TOO MUCH TORQUE TO THE CAPTIVE NUTS, YOU CAN CAUSE DAMAGE TO THE TERMINALS ON THE EGT PROBE.

(g) Install the lead [2B] to the EGT sensor [1] as follows:

NOTE: The EGT probe has a large terminal and a small terminal. Make sure that you install the captive nuts on the correct terminals.

NOTE: Wrap the lead [2B] around the lead [2A] three to five times to prevent the leads from contacting the wire harness or adjacent hardware. Do not wrap too tight to prevent damage to the leads.

1) Attach the lead [2B] to the EGT sensor [1] larger terminal.

2) Hold the wiring harness lead [2B] to keep it in its position.

3) Tighten the captive nut to 34 in-lb (3.8 N·m) - 37 in-lb (4.2 N·m).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 77-21-01-080-002-G00

(1) Remove the protector, SPL-14947 from the engine.

SUBTASK 77-21-01-860-005-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 77-21-01-860-006-G00

(3) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 77-21-01-860-007-G00

(4) For the left engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 77-21-01-860-008-G00

(5) For the right engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 77-21-01-410-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(6) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

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J. Exhaust Gas Temperature Sensors Installation Test

SUBTASK 77-21-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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EXHAUST GAS TEMPERATURE SENSOR HARNESS 1 - REMOVAL/INSTALLATION

1. General

A. This procedure contains the following tasks:

(1) A removal of the exhaust gas temperature sensor harness 1

(2) An installation of the exhaust gas temperature sensor harness 1.

TASK 77-21-04-000-801-G00

2. Exhaust Gas Temperature Sensor Harness 1 Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the Exhaust Gas Temperature (EGT) harness 1.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 77-21-04-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 77-21-04-865-001-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 77-21-04-010-001-G00

(3) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Exhaust Gas Temperature Sensor Harness 1 Removal

SUBTASK 77-21-04-010-002-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use a teflon-jawed pliers, STD-664, to disconnect these electrical connectors (TASK 70-00-01-910-803-G00):

(a) Disconnect the electrical connector [7] from the EGT harness 1 [1].

(b) Disconnect the electrical connector [5] from the TCF accelerometer cable [2].

(c) Disconnect the electrical connector [6] from the HPTACC valve [3].

(d) Install the protective covers on the electrical receptacles on the EGT harness 1 [1], TCF accelerometer cable [2], HPTACC valve [3], and electrical connectors.

SUBTASK 77-21-04-020-001-G00

(2) Remove the EGT harness 1 [1] as follows:

(a) Remove the two bolts [8] that attach the EGT harness 1 [1] to the support bracket [9].

1) Remove the EGT harness 1 [1] from the support bracket [9].

(b) Put the protective covers on the TCF accelerometer cable [2] and EGT harness 1 [1] receptacles.

CAUTION DO NOT REMOVE THE CAPTIVE NUTS FROM THE LEADS. DAMAGE CAN OCCUR TO THE LEADS OR THE CAPTIVE NUTS.

(c) Disconnect the EGT harness 1 lead [50] and lead [51] from the four EGT sensors [33] as follows:

1) Loosen the captive nuts.

2) Remove the lead [50] and lead [51] from the four EGT sensors [33].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Disconnect the EGT harness 1 [1] loop clamps from the engine brackets and the LPTACC air duct [34] as follow:

1) Remove the bolt [11] that attaches the loop clamp [10] and EGT harness 1 [1] to the support bracket [12].

2) Remove the bolt [15] that attaches the loop clamp [14] and EGT harness 1 [1] to the support bracket [13].

3) Remove the bolt [18] that attaches the loop clamp [17] and EGT harness 1 [1] to the support bracket [16].

PTF ALL; AIRPLANES WITH PRE LEAP-1B SB 72-0270

4) Remove the bolt [19] that attaches the loop clamp [20] and EGT harness 1 [1] to the support bracket [21].

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0270

5) Remove the bolt [20] that attaches the loop clamp [19] and EGT harness 1 [1] to the LPTACC air duct [34].

PTF ALL

6) Remove the bolt [32] that attaches the loop clamp [31] and EGT harness 1 [1] to the LPTACC air duct [34].

7) Remove the bolt [30] that attaches the loop clamp [29] and EGT harness 1 [1] to the LPTACC air duct [34].

PTF ALL; AIRPLANES WITH PRE LEAP-1B SB 72-0270

8) Remove the bolt [28] that attaches the loop clamp [27] and EGT harness 1 [1] to the LPTACC air duct [34].

9) Remove the bolt [26] that attaches the loop clamp [25] and EGT harness 1 [1] to the LPTACC air duct [34].

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0270

10) Remove the bolt [28] that attaches the loop clamp [27] and EGT harness 1 [1] to the support bracket [53].

11) Remove the bolt [26] that attaches the loop clamp [25] and EGT harness 1 [1] to the support bracket [52].

PTF ALL

12) Remove the bolt [23] that attaches the loop clamp [22] and EGT harness 1 [1] to the support bracket [24].

13) Remove the bolt [48] that attaches the loop clamp [49] and EGT harness 1 [1] to the support bracket [47].

14) Remove the bolt [45] that attaches the loop clamp [46] and EGT harness 1 [1] to the support bracket [44].

PTF ALL; AIRPLANES WITH PRE LEAP-1B SB 72-0270

15) Remove the bolt [36] that attaches the loop clamp [35] and EGT harness 1 [1] to the support bracket [37].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0270

16) Remove the two bolts [36] that attach the two loop clamps [35] and EGT harness 1 [1] to the support bracket [37].

PTF ALL

17) Remove the bolt [39] that attaches the loop clamp [38] and EGT harness 1 [1] to the support bracket [40].

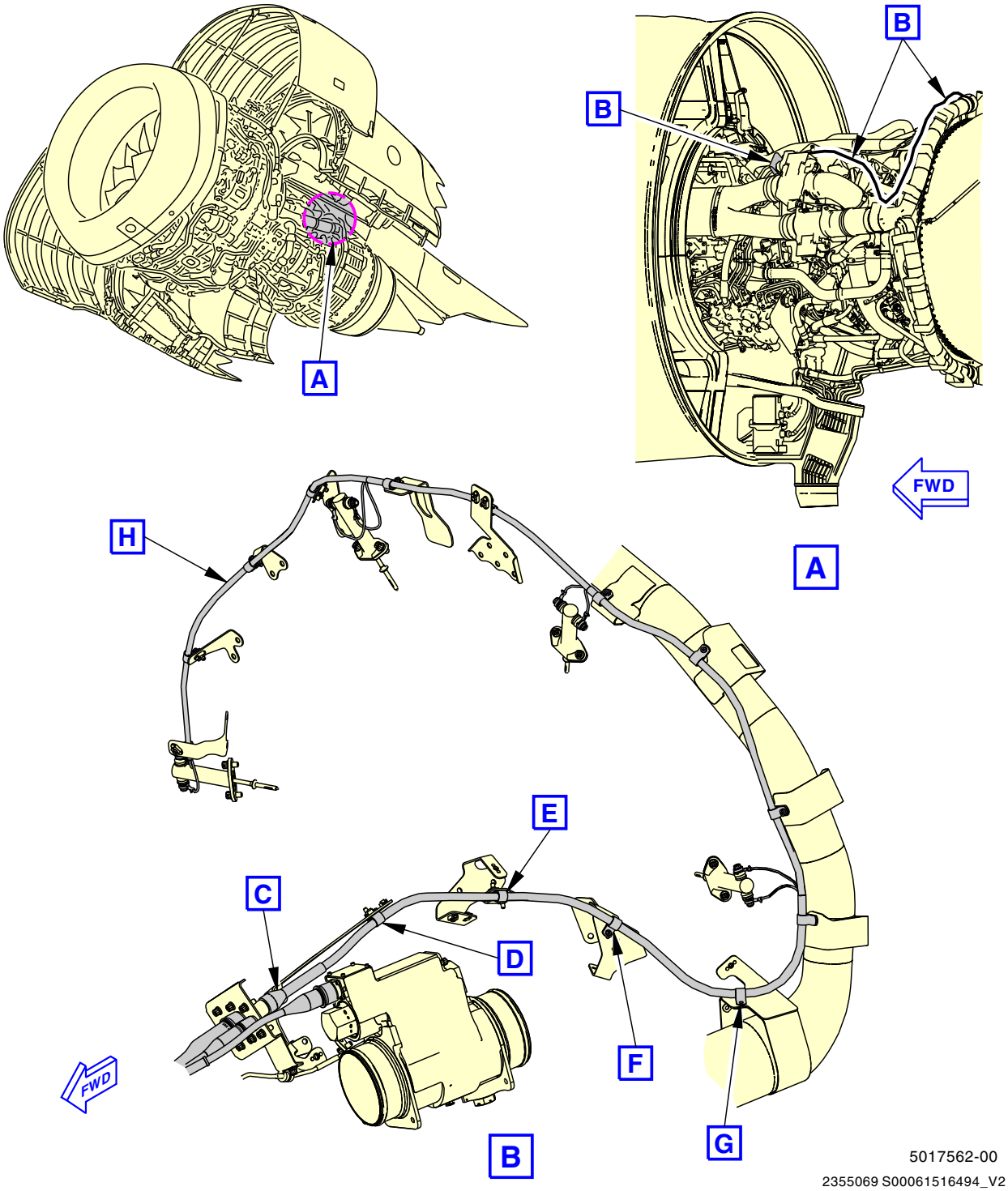
18) Remove the bolt [42] that attaches the loop clamp [41] and EGT harness 1 [1] to the support bracket [43].

(e) Remove the EGT harness 1 [1] from the engine.

(f) Put the protective covers on the four EGT sensors [33] terminals and EGT electrical harness 1 lead [50] and lead [51].

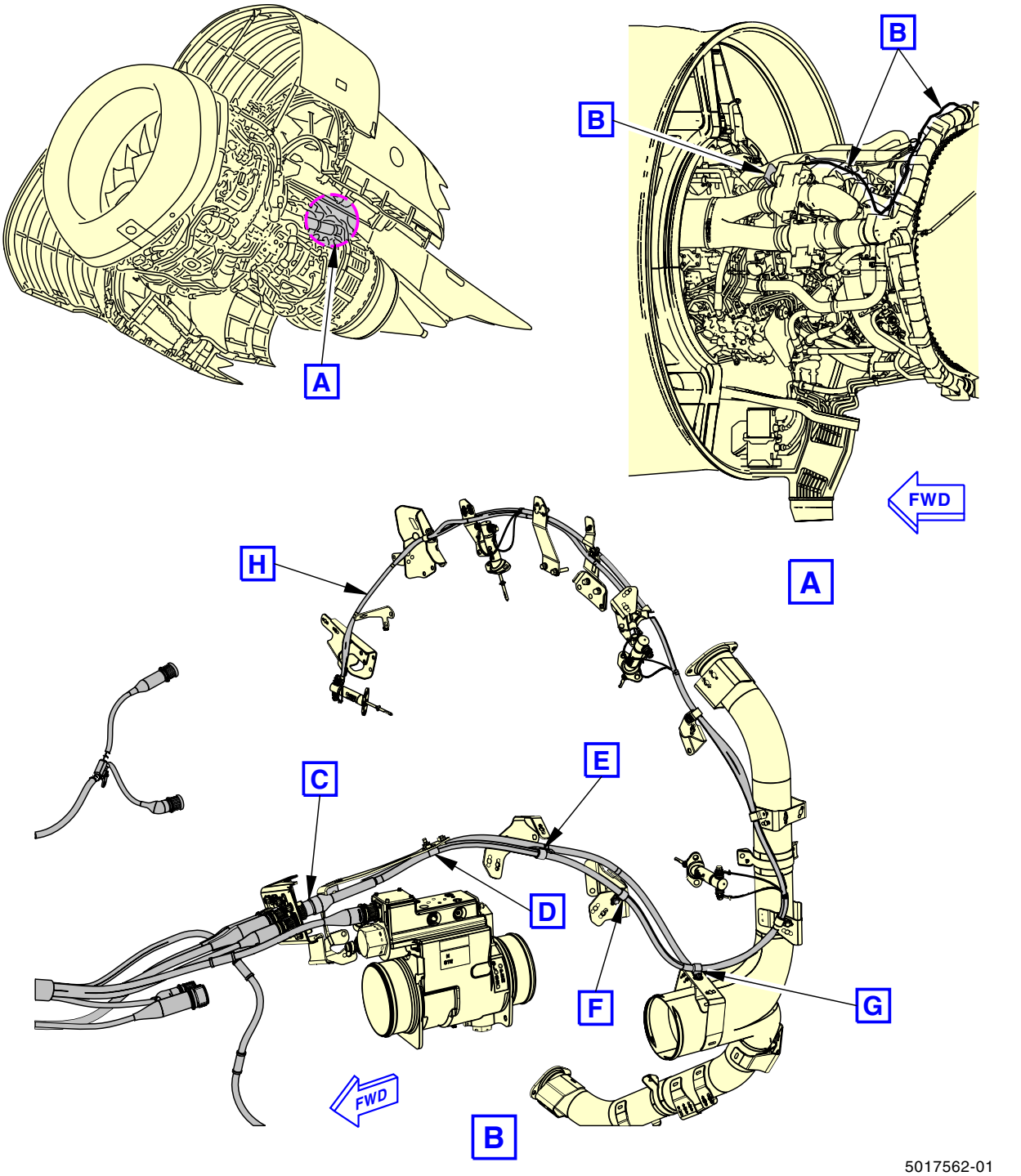
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



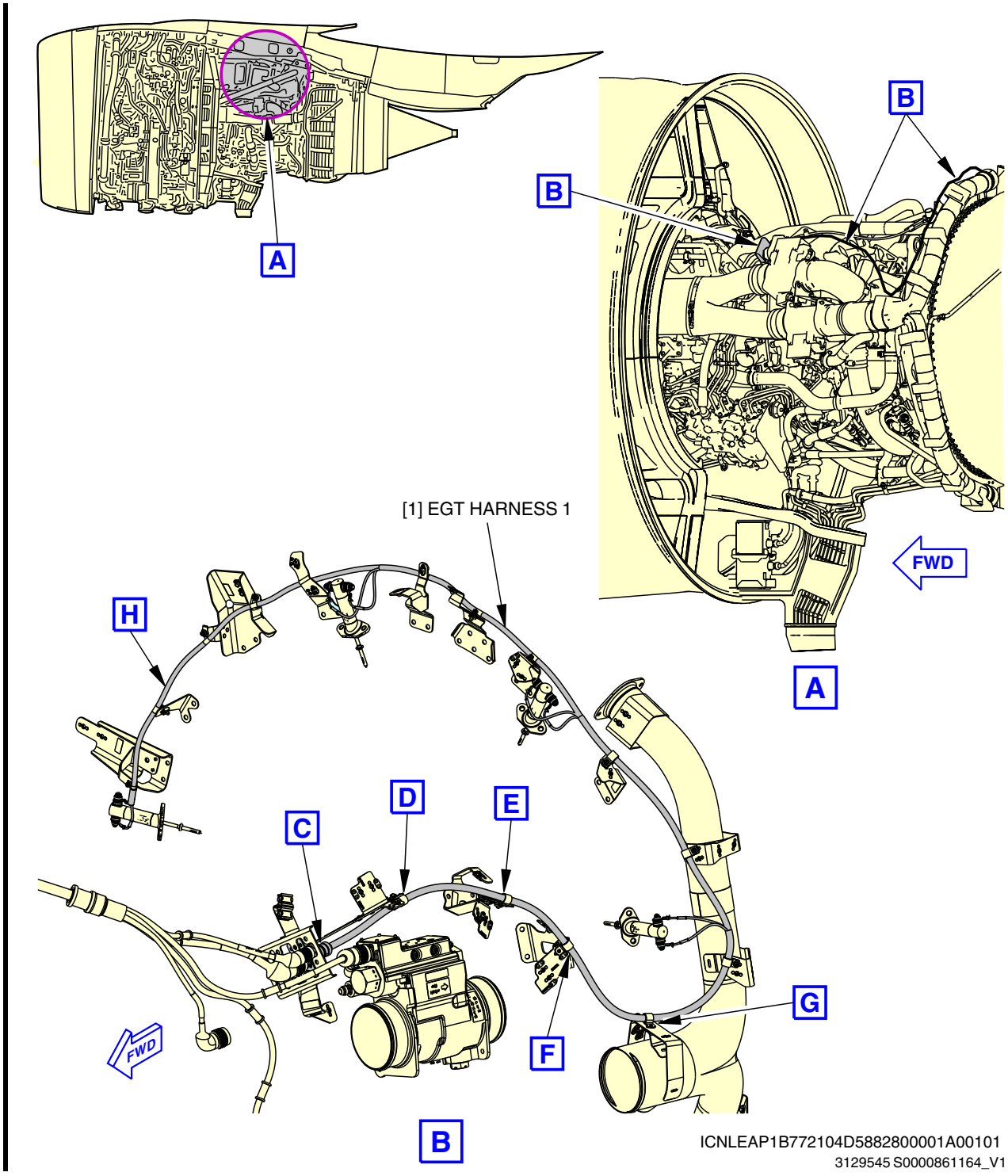
Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 1 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



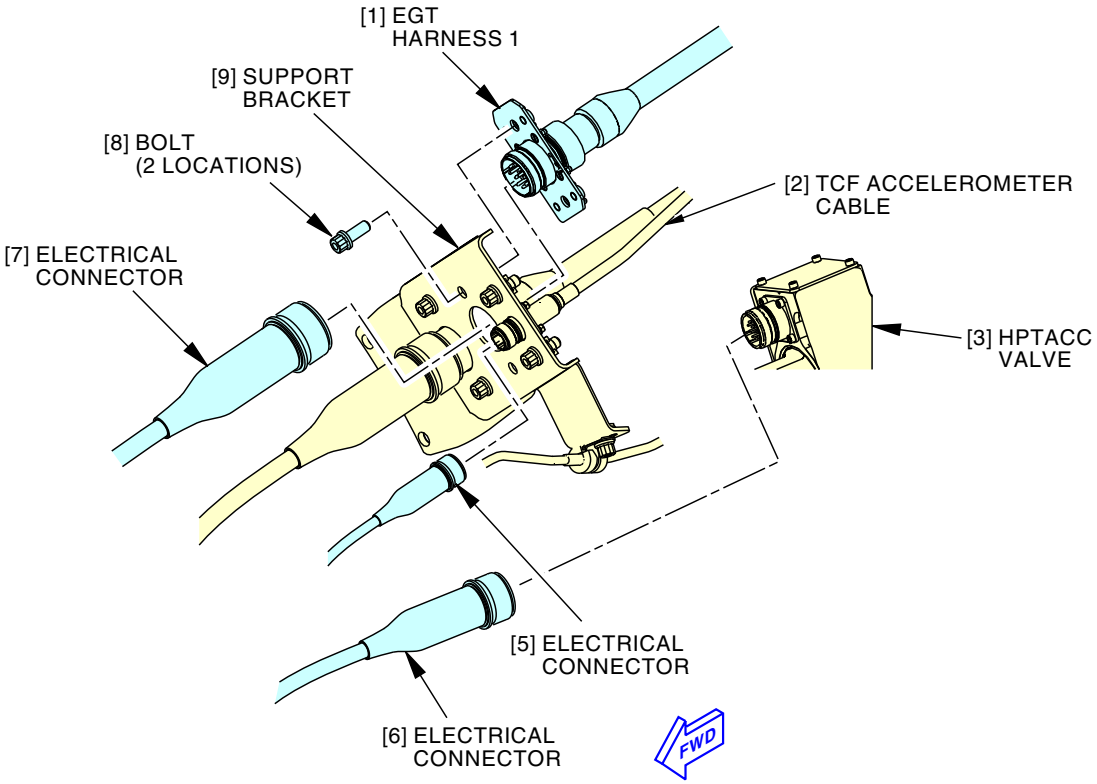
Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 2 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



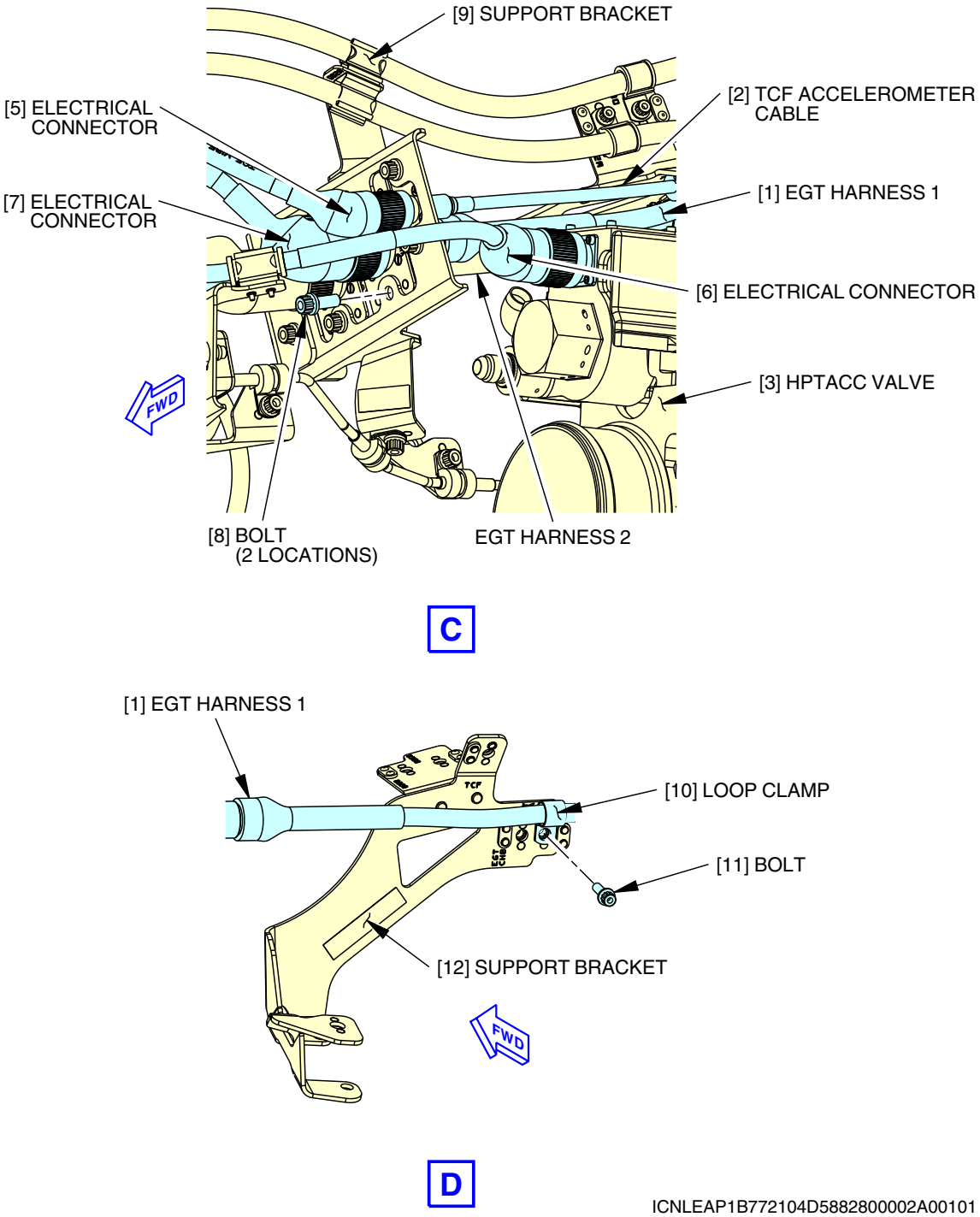
Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 3 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



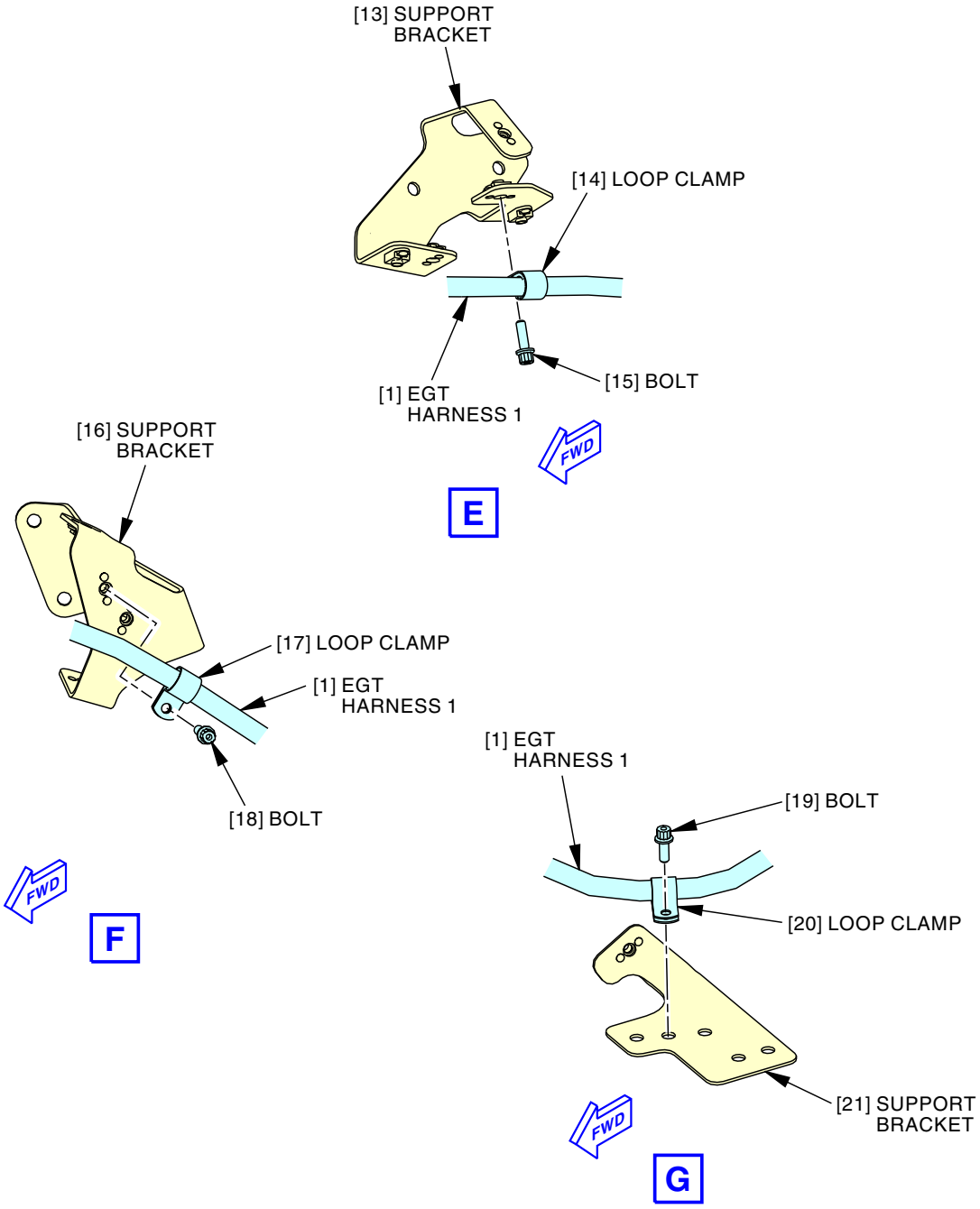
Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 4 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 5 of 11)

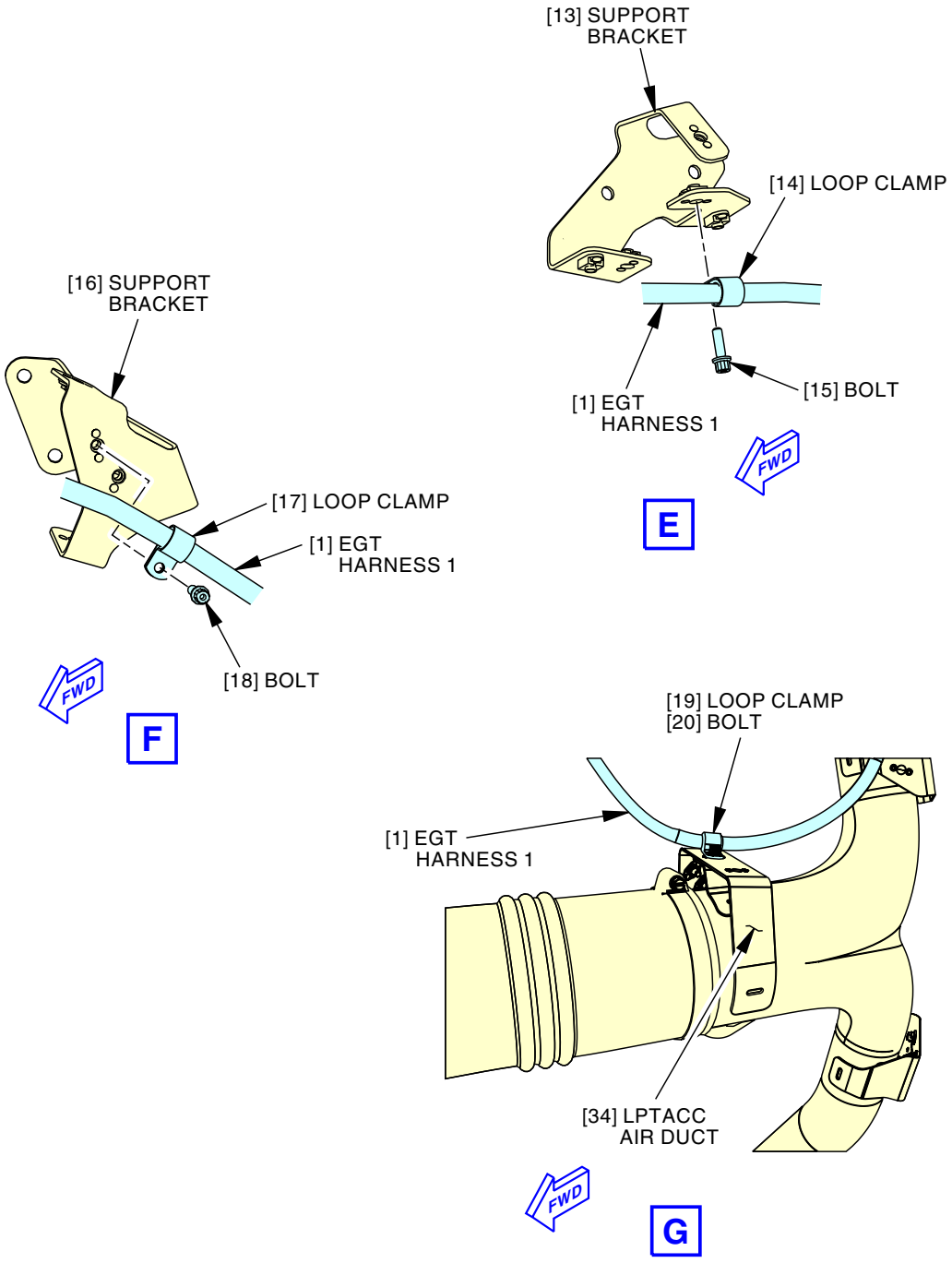
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5017566-00
2355071 S00061516496_V2

Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 6 of 11)

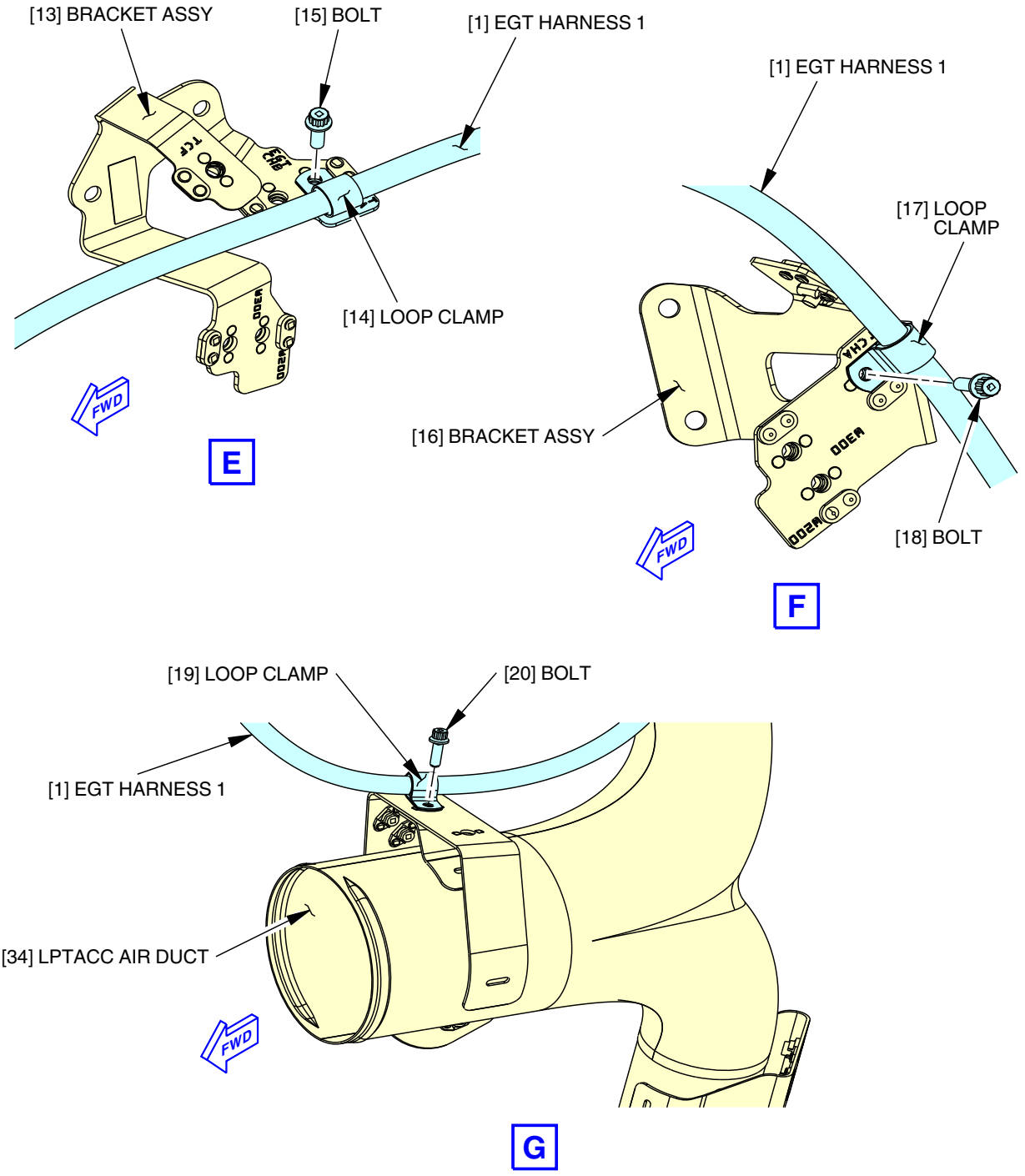
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5017566-00
6014451-00
2933875 S0000710587_V1

Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 7 of 11)

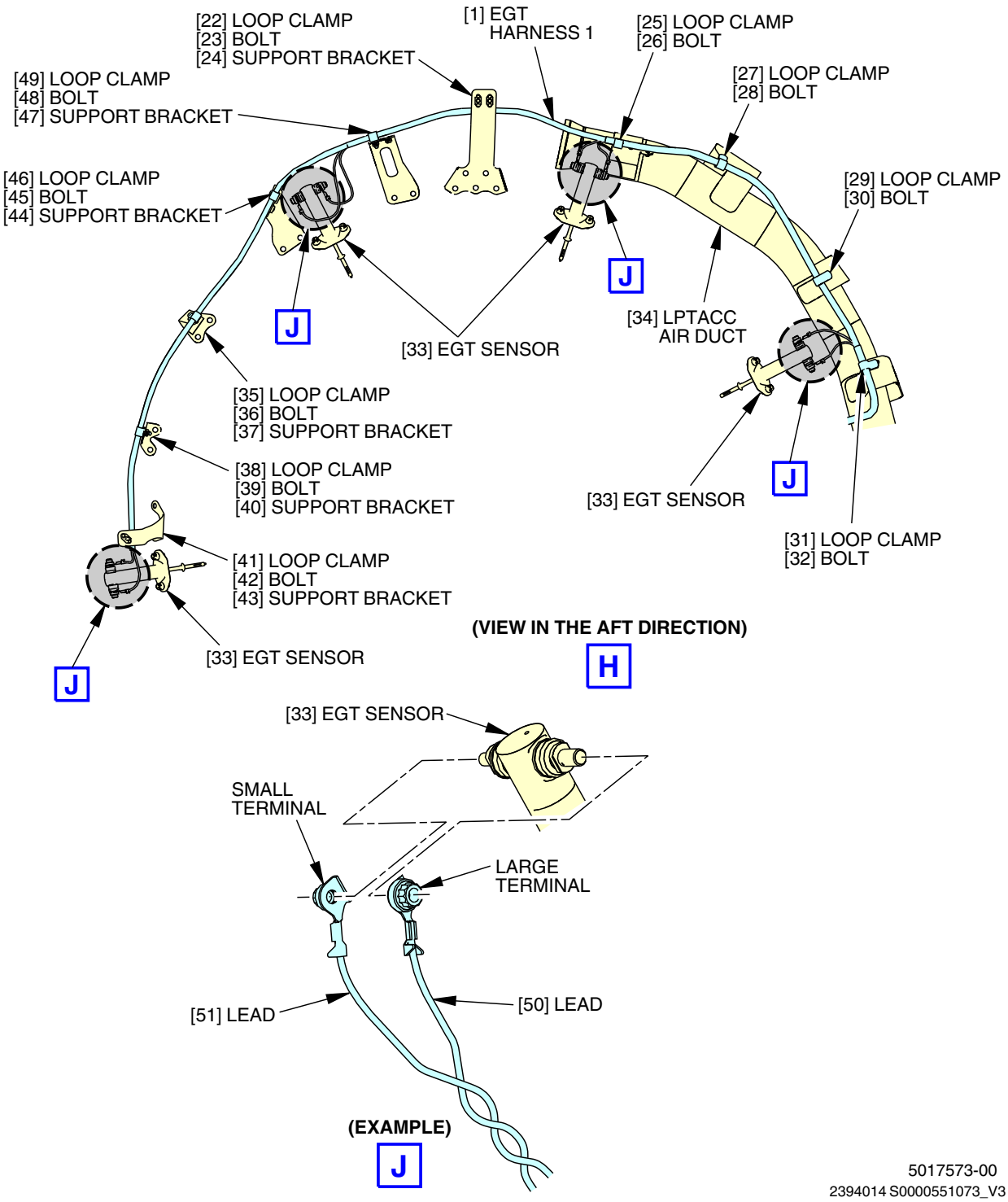
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B772104D5882800003A00101
3129557 S0000861167_V1

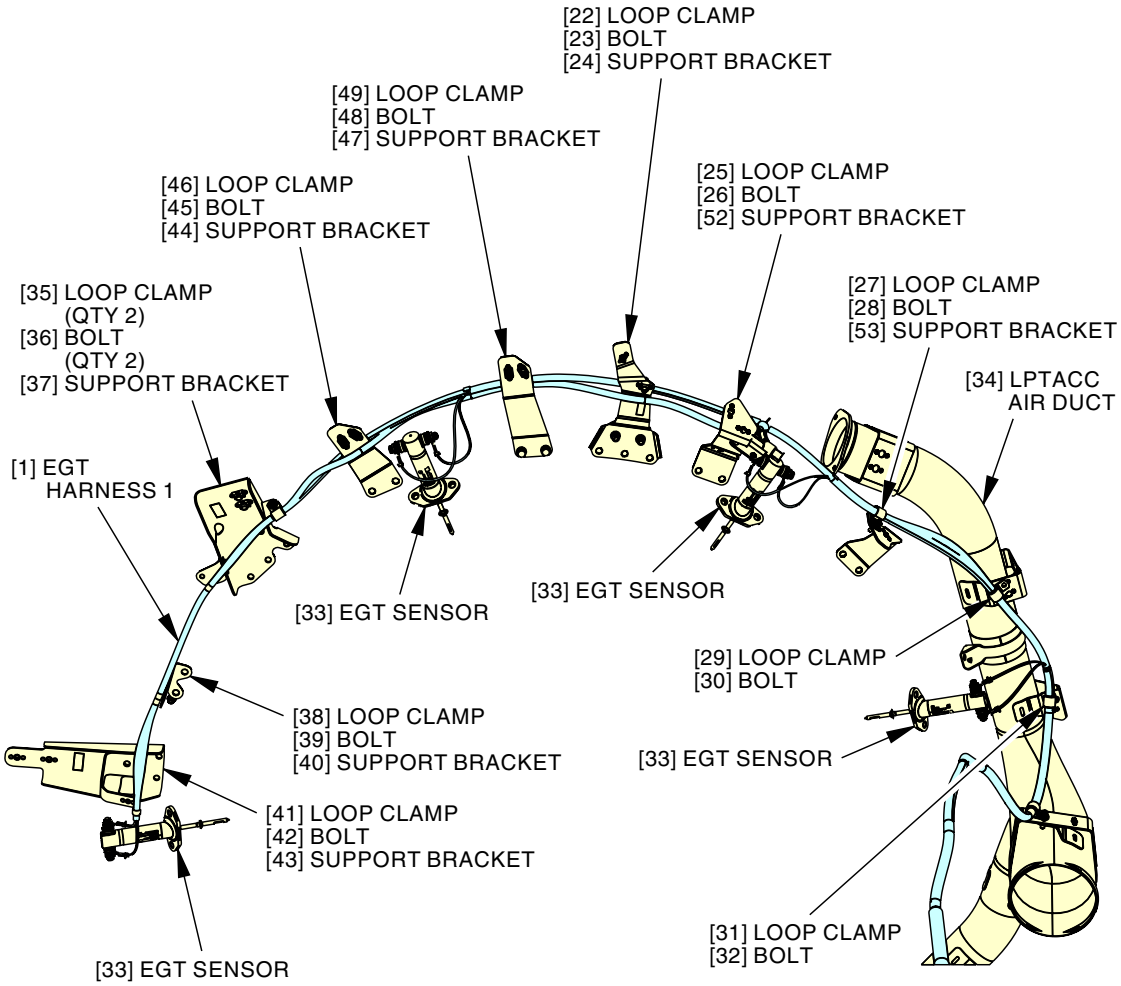
Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 8 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 9 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



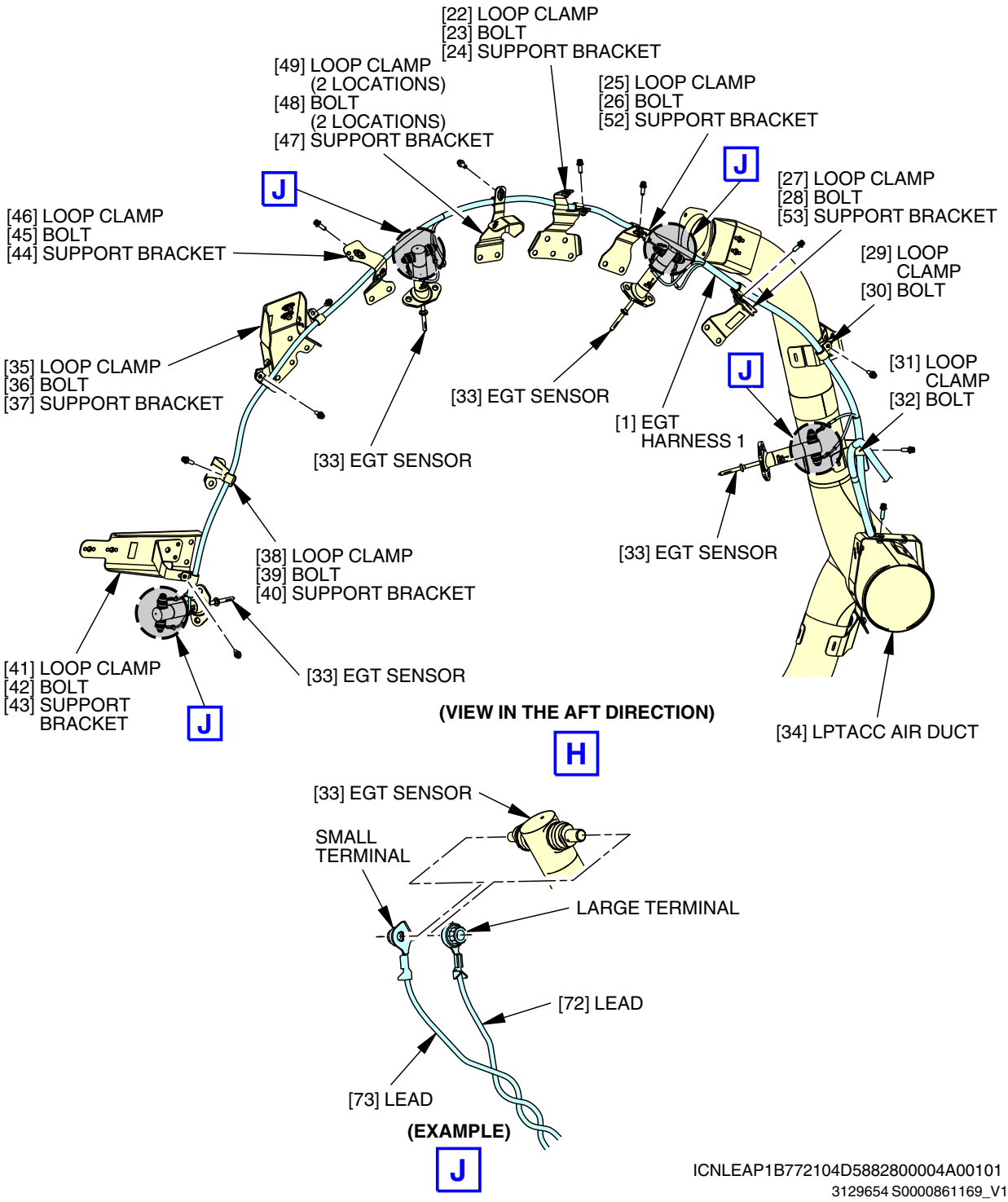
(VIEW IN THE AFT DIRECTION)

H

6014452-00
2933890 S0000710589_V1

Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 10 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exhaust Gas Temperature Sensor Harness 1 Installation
Figure 401/77-21-04-990-801-G00 (Sheet 11 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 77-21-04-400-801-G00

3. Exhaust Gas Temperature Sensor Harness 1 Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the Exhaust Gas Temperature (EGT) harness 1.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Exhaust Gas Temperature Sensor Harness 1 Installation

SUBTASK 77-21-04-420-001-G00

(1) Install the EGT harness 1 [1] as follows:

(a) Remove the protective covers from the four EGT sensors [33] terminals and EGT electrical harness 1 lead [50] and lead [51].

(b) Put the EGT harness 1 [1] in position on the brackets and LPTACC air duct [34].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

MAKE SURE THAT THE POSITION OF THE HARNESS IS CORRECT. IF YOU DO NOT OBEY, DAMAGE TO THE HARNESS CAN OCCUR WHEN IT RUBS THE ADJACENT COMPONENTS.

(c) Loosely install the EGT harness 1 [1] and loop clamps on the engine brackets and LPTACC air duct [34] as follows:

CAUTION

MAKE SURE THAT YOU USE THE BOLTS IDENTIFIED IN THE PARTS LIST. THESE BOLTS WILL HAVE THE CORRECT LENGTH. BOLTS THAT ARE TOO LONG CAN CAUSE DAMAGE.

1) Lubricate the threads of these bolts with the engine oil, D00599 [CP2442]:

PTF ALL; AIRPLANES WITH PRE LEAP-1B SB 72-0270

a) The bolt [11], bolt [15], bolt [18], bolt [19], and bolt [23].

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0270

b) The bolt [11], bolt [15], bolt [18], bolt [20], and bolt [23].

PTF ALL

c) The bolt [26], bolt [28], bolt [30], bolt [32], and bolt [36].

d) The bolt [39], bolt [42], bolt [45], and bolt [48].

2) Install the bolt [42] that attaches the loop clamp [41] and EGT harness 1 [1] to the support bracket [43].

3) Install the bolt [39] that attaches the loop clamp [38] and EGT harness 1 [1] to the support bracket [40].

PTF ALL; AIRPLANES WITH PRE LEAP-1B SB 72-0270

4) Install the bolt [36] that attaches the loop clamp [35] and EGT harness 1 [1] to the support bracket [37].

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0270

5) Install the two bolts [36] that attach the two loop clamps [35] and EGT harness 1 [1] to the support bracket [37].

PTF ALL

6) Install the bolt [45] that attaches the loop clamp [46] and EGT harness 1 [1] to the support bracket [44].

7) Install the bolt [48] that attaches the loop clamp [49] and EGT harness 1 [1] to the support bracket [47].

8) Install the bolt [23] that attaches the loop clamp [22] and EGT harness 1 [1] to the support bracket [24].

PTF ALL; AIRPLANES WITH PRE LEAP-1B SB 72-0270

9) Install the bolt [26] that attaches the loop clamp [25] and EGT harness 1 [1] to the LPTACC air duct [34].

10) Install the bolt [28] that attaches the loop clamp [27] and EGT harness 1 [1] to the LPTACC air duct [34].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0270

11) Install the bolt [26] that attaches the loop clamp [25] and EGT harness 1 [1] to the support bracket [52].

12) Install the bolt [28] that attaches the loop clamp [27] and EGT harness 1 [1] to the support bracket [53].

PTF ALL

13) Install the bolt [30] that attaches the loop clamp [29] and EGT harness 1 [1] to the LPTACC air duct [34].

14) Install the bolt [32] that attaches the loop clamp [31] and EGT harness 1 [1] to the LPTACC air duct [34].

PTF ALL; AIRPLANES WITH PRE LEAP-1B SB 72-0270

15) Install the bolt [19] that attaches the loop clamp [20] and EGT harness 1 [1] to the support bracket [21].

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0270

16) Install the bolt [20] that attaches the loop clamp [19] and EGT harness 1 [1] to the LPTACC air duct [34].

PTF ALL

17) Install the bolt [18] that attaches the loop clamp [17] and EGT harness 1 [1] to the support bracket [16].

18) Install the bolt [15] that attaches the loop clamp [14] and EGT harness 1 [1] to the support bracket [13].

19) Install the bolt [11] that attaches the loop clamp [10] and EGT harness 1 [1] to the support bracket [12].

CAUTION

MAKE SURE THAT THE EGT HARNESS LEADS ARE IN THE CORRECT POSITION BEFORE YOU TIGHTEN THE CAPTIVE NUTS. IF THE LEADS ARE NOT IN THE CORRECT POSITION, YOU CAN CAUSE DAMAGE TO EGT HARNESS LEADS AND THE EGT PROBE.

CAUTION

DO NOT TIGHTEN THE CONNECTORS TOO TIGHT. TOO MUCH TORQUE ON THE CONNECTORS CAN CAUSE DAMAGE TO THE EQUIPMENT.

(d) Connect the EGT harness 1 leads [50] and leads [51] to the four EGT sensors [33] as follows:

NOTE: The EGT sensor has two terminals and two connector nuts. One terminal has a smaller diameter than the other terminal. The smaller connector nut is installed on the smaller terminal. The larger connector nut is installed on the larger terminal.

1) Install the lead [50] on the larger terminal of the EGT sensor [33].

2) Install the lead [51] on the smaller terminal of the EGT sensor [33].

3) If there is any overlength of the lead [50] and lead [51] connected to the EGT sensor [33], wind the longest branch around the smallest one.

4) Tighten each of the lead [50] nut to 29.5 in-lb (3.33 N·m) - 32.5 in-lb (3.67 N·m).

5) Tighten each of the lead [51] nut to 16.2 in-lb (1.83 N·m) - 17.8 in-lb (2.01 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Lubricate the threads of the two bolts [8] with engine oil, D00599 [CP2442].

(f) Remove the protective covers from the TCF accelerometer cable [2] and EGT harness 1 [1].

(g) Put the EGT harness 1 [1] to the support bracket [9].

(h) Install the two bolts [8] that attach the EGT harness 1 [1] to the support bracket [9].

(i) Tighten the two bolts [8] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

PTF ALL; AIRPLANES WITH PRE LEAP-1B SB 72-0270

(j) Tighten the bolt [11], bolt [15], bolt [18], and bolt [19] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0270

(k) Tighten the bolt [11], bolt [15], bolt [18], and bolt [20] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

PTF ALL

(l) Tighten the bolt [23], bolt [26], and bolt [28] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

(m) Tighten the bolt [30], bolt [32], bolt [36], and bolt [39] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

(n) Tighten the bolt [42], bolt [45], and bolt [48] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(o) Connect the electrical connectors as follows:

1) Remove the protective covers from the electrical receptacles on the EGT harness 1 [1], TCF accelerometer cable [2], and HPTACC valve [3].

2) Remove the protective covers from the electrical connector [7], electrical connector [5], and electrical connector [6].

3) Connect the electrical connector [7] to the EGT harness 1 [1].

4) Connect the electrical connector [5] to the TCF accelerometer cable [2].

5) Connect the electrical connector [6] to the HPTACC valve [3].

6) Use a teflon-jawed pliers, STD-664, to tighten the electrical connectors (TASK 70-00-01-910-803-G00).

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 77-21-04-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 77-21-04-865-002-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 77-21-04-860-002-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Exhaust Gas Temperature Sensor Harness 1 Installation Test

SUBTASK 77-21-04-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EXHAUST GAS TEMPERATURE SENSOR HARNESS 2 - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Exhaust gas temperature sensor harness 2 removal

(2) Exhaust gas temperature sensor harness 2 installation.

TASK 77-21-05-000-801-G00

2. Exhaust Gas Temperature Sensor Harness 2 Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the Exhaust Gas Temperature (EGT) sensor harness 2.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 77-21-05-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 77-21-05-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 77-21-05-010-001-G00

(3) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Exhaust Gas Temperature Sensor Harness 2 Removal

SUBTASK 77-21-05-010-002-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use a teflon-jawed pliers, STD-664, to disconnect these electrical connectors (TASK 70-00-01-910-803-G00):

(a) Disconnect the electrical connector [7] from the EGT harness 1 [1].

(b) Disconnect the electrical connector [5] from the TCF accelerometer cable [2].

(c) Disconnect the electrical connector [62] from the EGT harness 2 [61].

(d) Disconnect the electrical connector [3] from the HPTACC valve [4].

(e) Install protective covers on the electrical receptacles on the EGT harness 1 [1], TCF accelerometer cable [2], and on the electrical connectors.

SUBTASK 77-21-05-020-001-G00

(2) Remove the EGT harness 2 [61] as follows:

(a) Remove the two bolts [63] that attach the EGT harness 2 [61] to the support bracket [9].

(b) Remove the EGT harness 2 [61] from the support bracket [9].

(c) Put a protective cover on the EGT harness 2 [61] receptacle.

CAUTION DO NOT REMOVE THE CAPTIVE NUTS FROM THE LEADS. DAMAGE CAN OCCUR TO THE LEADS OR THE CAPTIVE NUTS.

(d) Disconnect the EGT harness 2 lead [72] and lead [73] from each of the four EGT sensors [33] as follows:

1) Loosen the captive nuts.

2) Remove the leads [72] and leads [73] from the four EGT sensors [33].

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(e) Disconnect the EGT harness 2 loop clamps from the engine brackets, LPTACC air duct [34], and oil supply manifold [92] as follows:

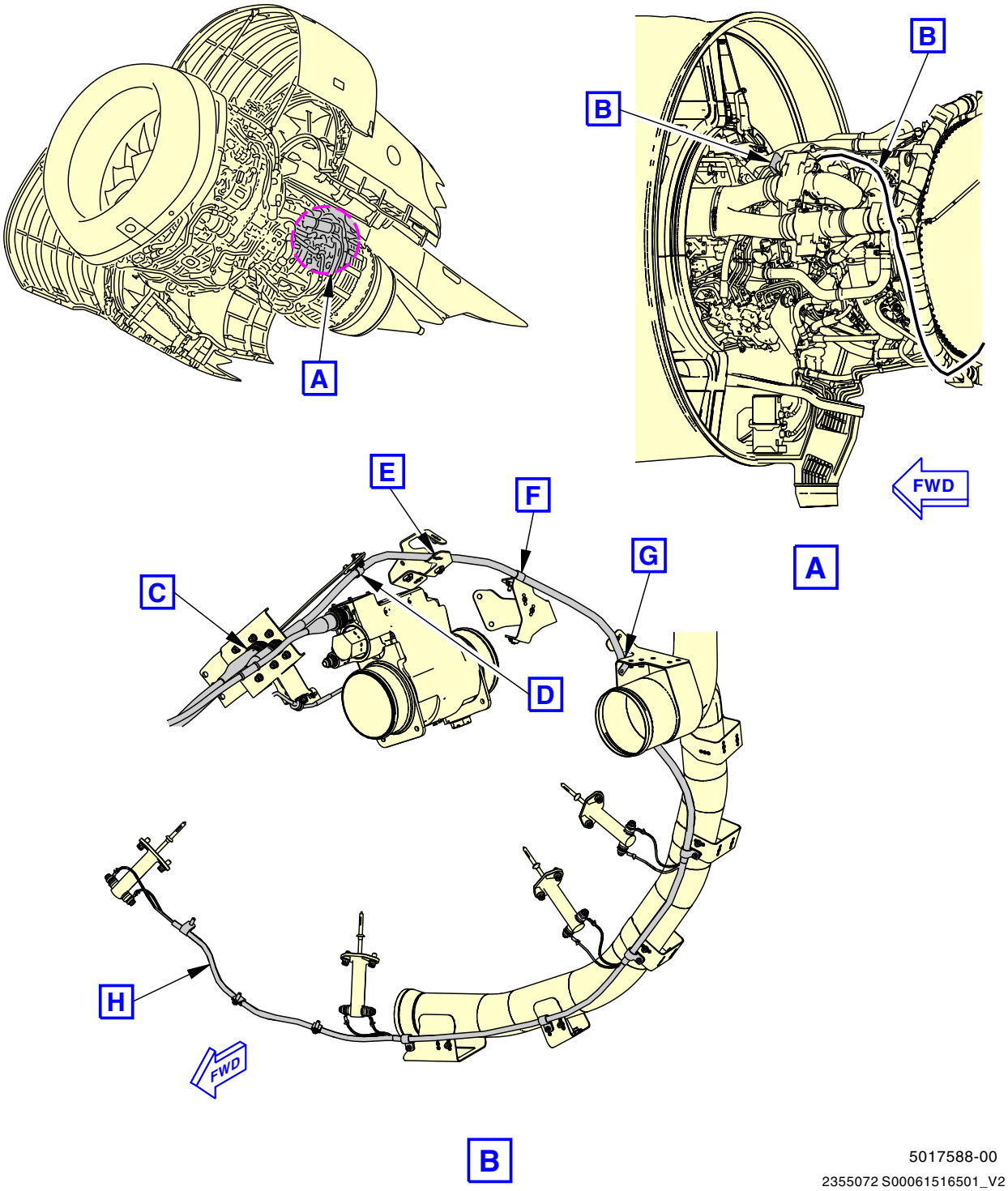
1) Remove the bolt [65] that attaches the loop clamp [64] and EGT harness 2 [61] to the support bracket [12].
2) Remove the bolt [66] that attaches the loop clamp [67] and EGT harness 2 [61] to the support bracket [13].
3) Remove the bolt [68] that attaches the loop clamp [69] and EGT harness 2 [61] to the support bracket [16].
4) Remove the bolt [71] that attaches the loop clamp [70] and EGT harness 2 [61] to LPTACC air duct [34].
5) Remove the bolt [75] that attaches the loop clamp [74] and EGT harness 2 [61] to the support bracket [76].
6) Remove the bolt [78] that attaches the loop clamp [77] and EGT harness 2 [61] to the LPTACC air duct [34].
7) Remove the bolt [80] that attaches the loop clamp [79] and EGT harness 2 [61] to the support bracket [95].
8) Remove the bolt [82] that attaches the loop clamp [81] and EGT harness 2 [61] to the support bracket [94].
9) Remove the bolt [83] that attaches the loop clamp [84] and EGT harness 2 [61] to the support bracket [93].
10) Remove the bolt [85] that attaches the loop clamp [86] and EGT harness 2 [61] to the oil supply manifold [92].
11) Remove the bolt [87] that attaches the loop clamp [88] and EGT harness 2 [61] to the oil supply manifold [92].
12) Remove the bolt [90] that attaches the loop clamp [91] and EGT harness 2 [61] to the support bracket [89].

(f) Remove the EGT harness 2 [61] from the engine.

(g) Put protective covers on the EGT sensors [33] terminals, EGT harness 2 lead [72], and lead [73].

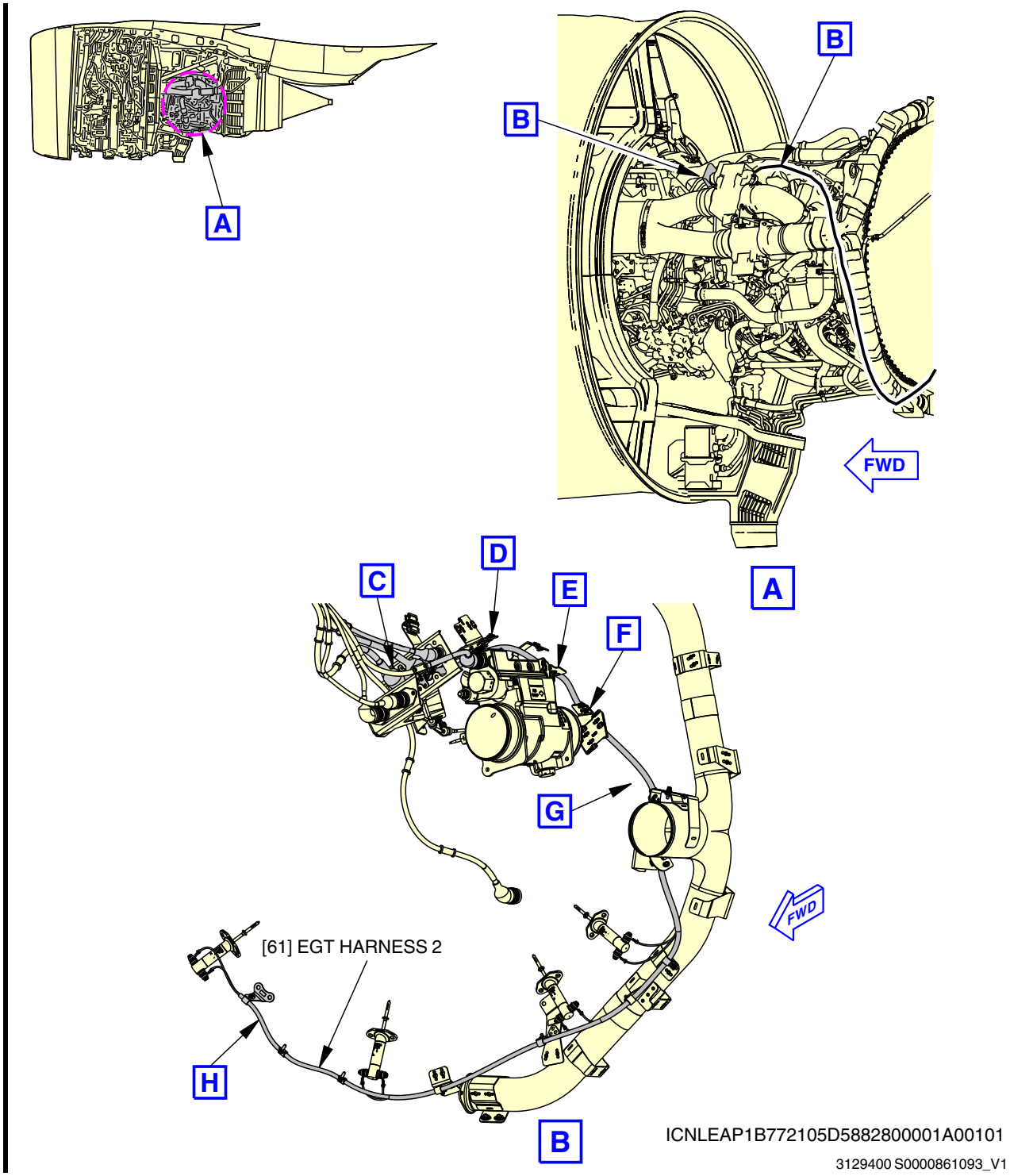
——— END OF TASK ———

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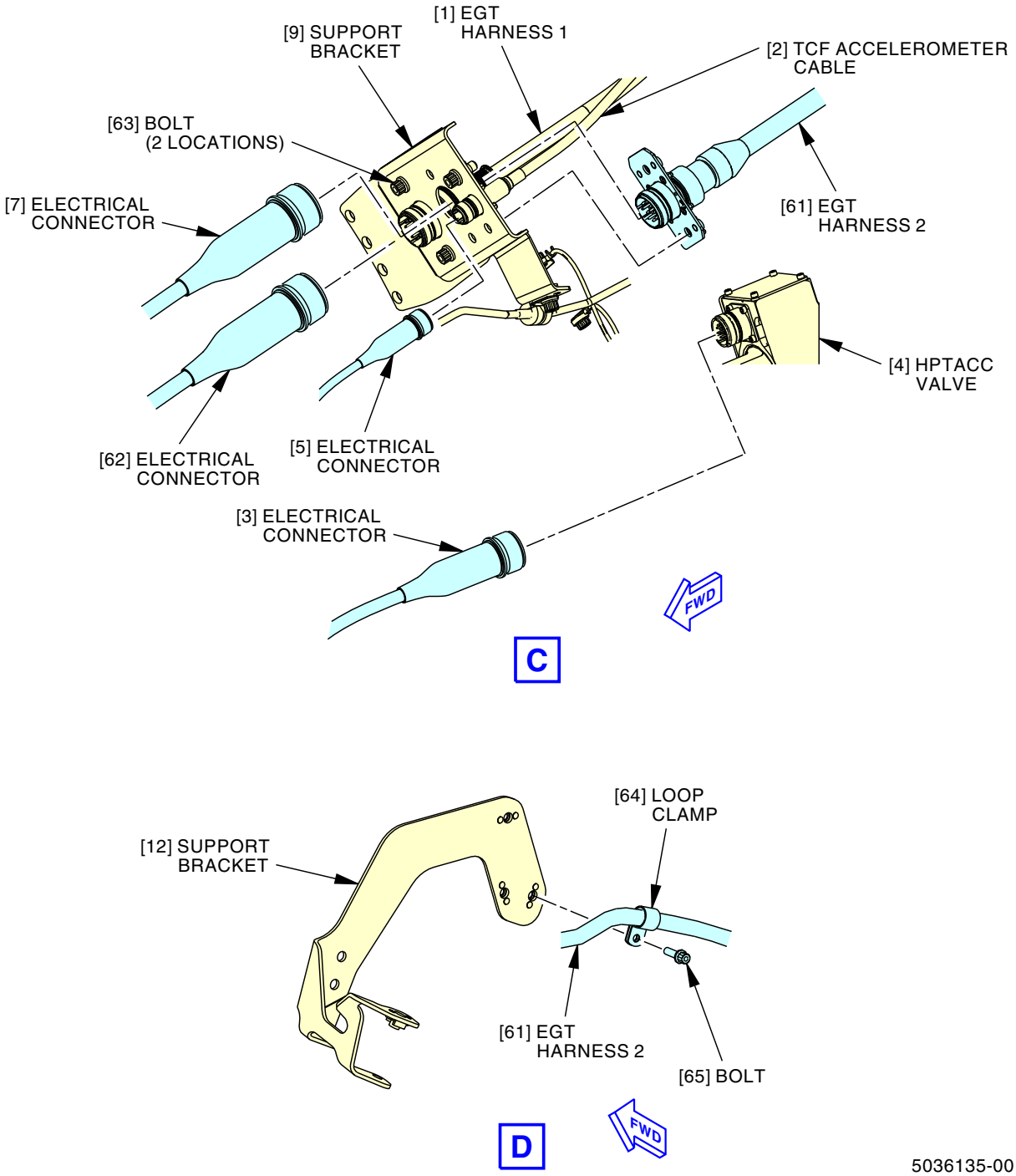
Exhaust Gas Temperature Sensor Harness 2 Installation
Figure 401/77-21-05-990-801-G00 (Sheet 1 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



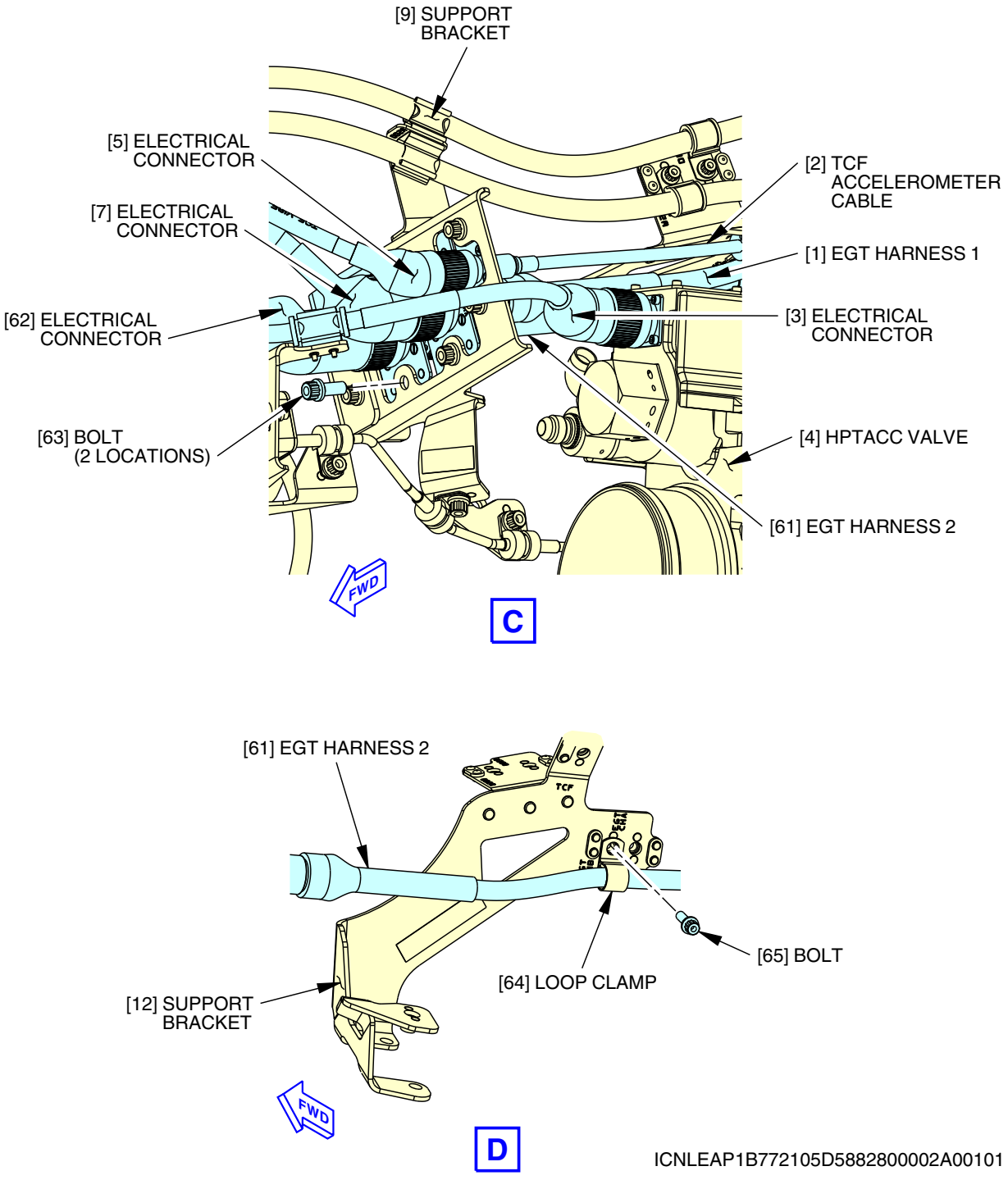
Exhaust Gas Temperature Sensor Harness 2 Installation
Figure 401/77-21-05-990-801-G00 (Sheet 2 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



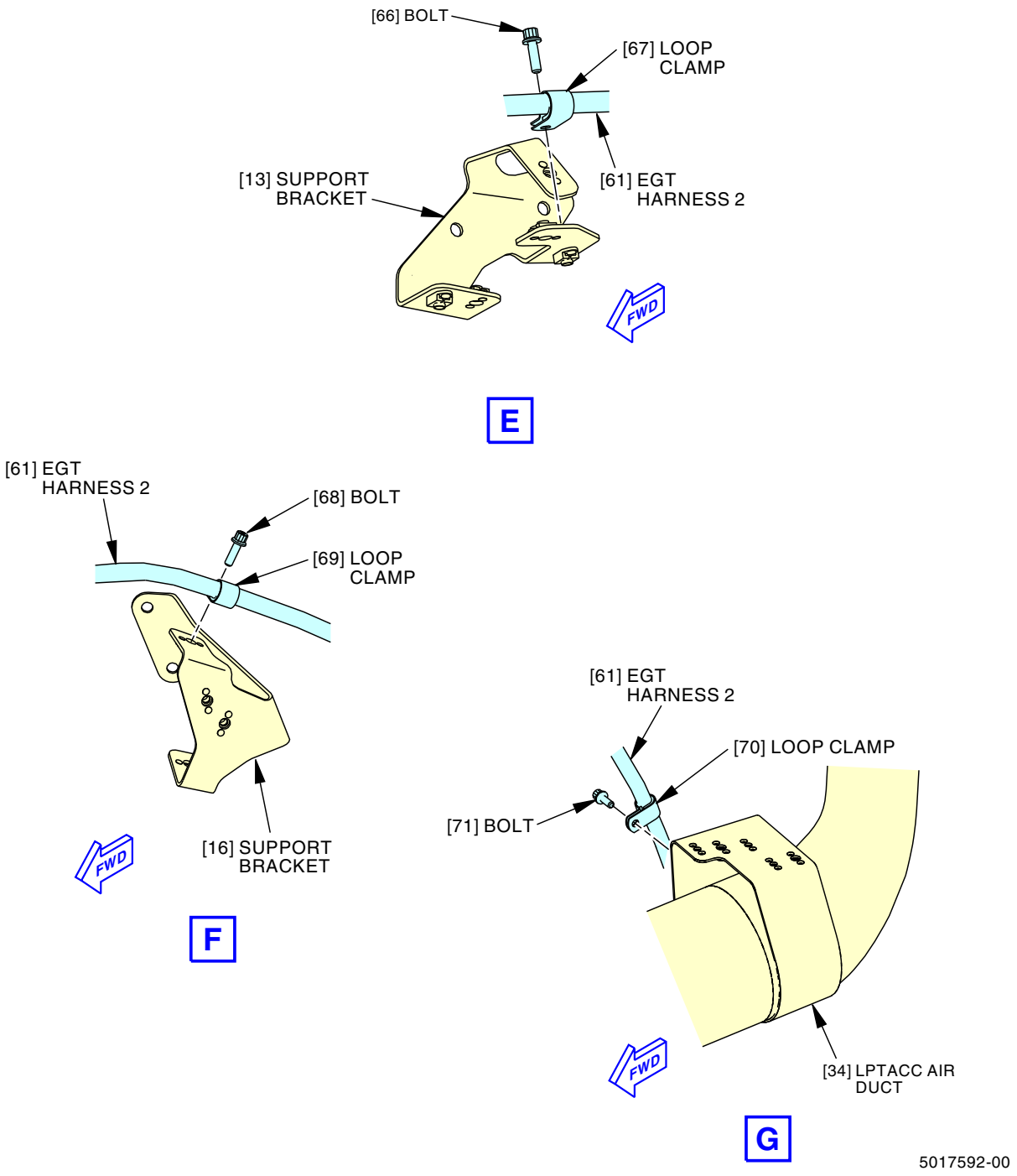
Exhaust Gas Temperature Sensor Harness 2 Installation
Figure 401/77-21-05-990-801-G00 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



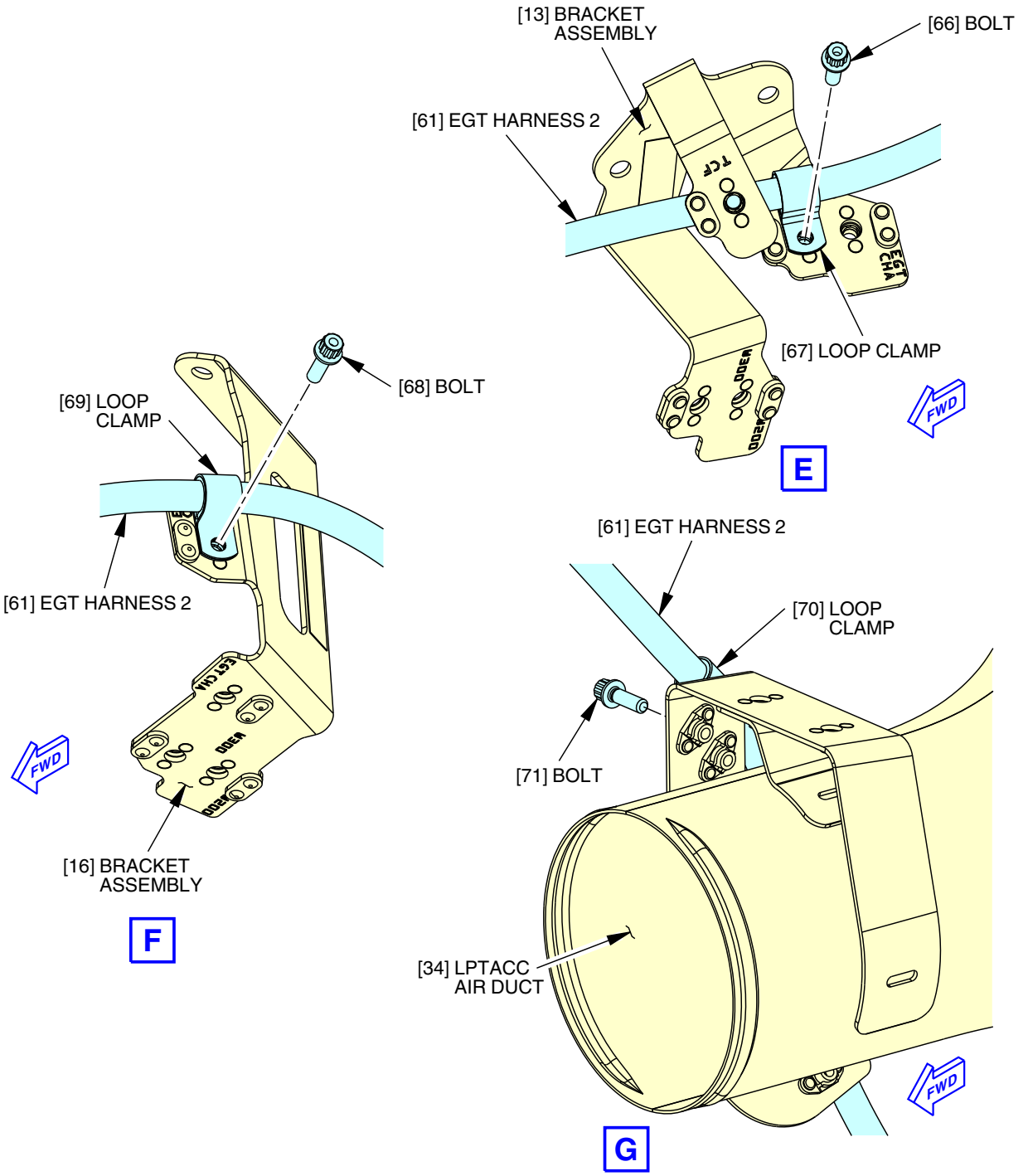
Exhaust Gas Temperature Sensor Harness 2 Installation
Figure 401/77-21-05-990-801-G00 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



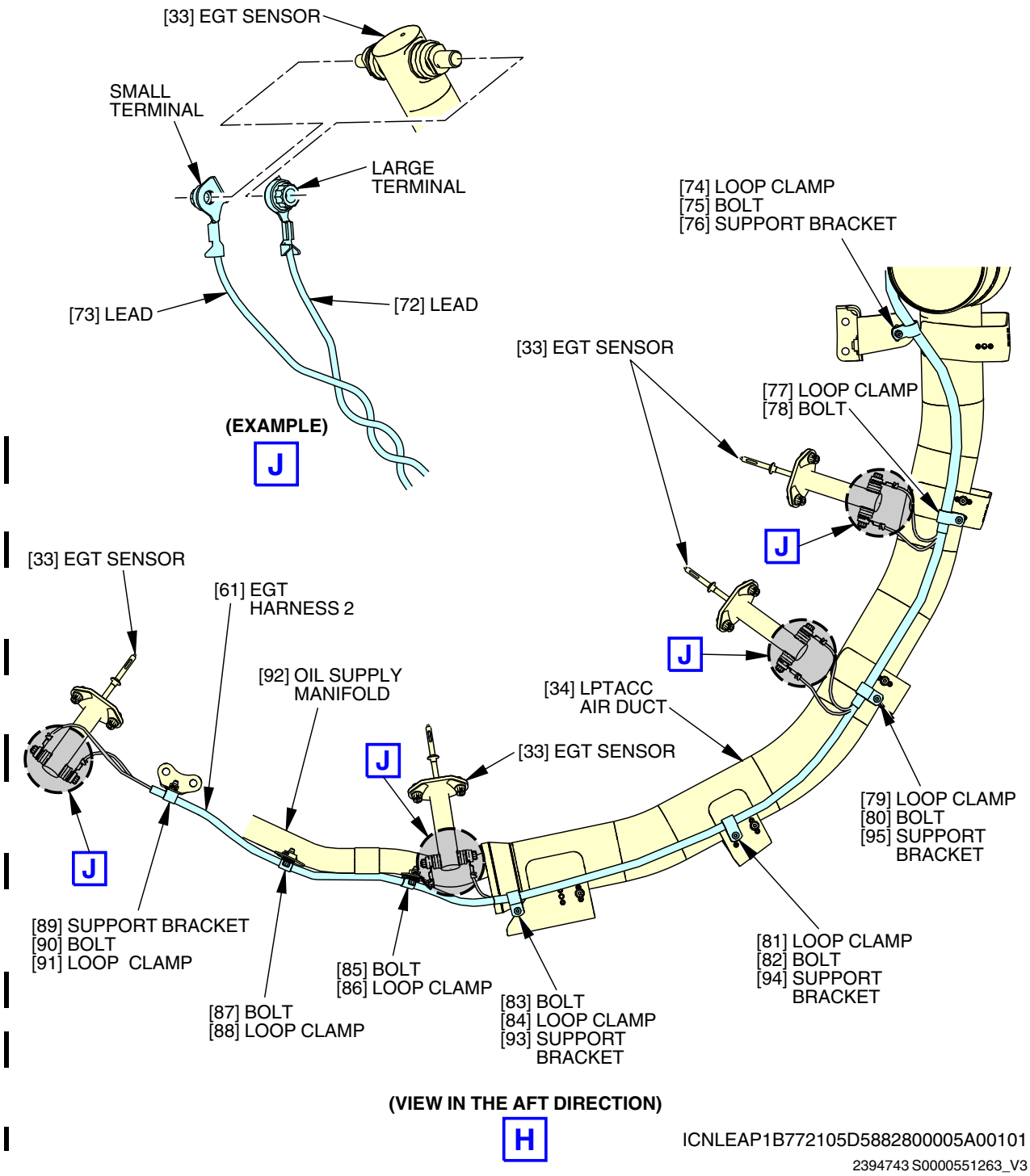
Exhaust Gas Temperature Sensor Harness 2 Installation
Figure 401/77-21-05-990-801-G00 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



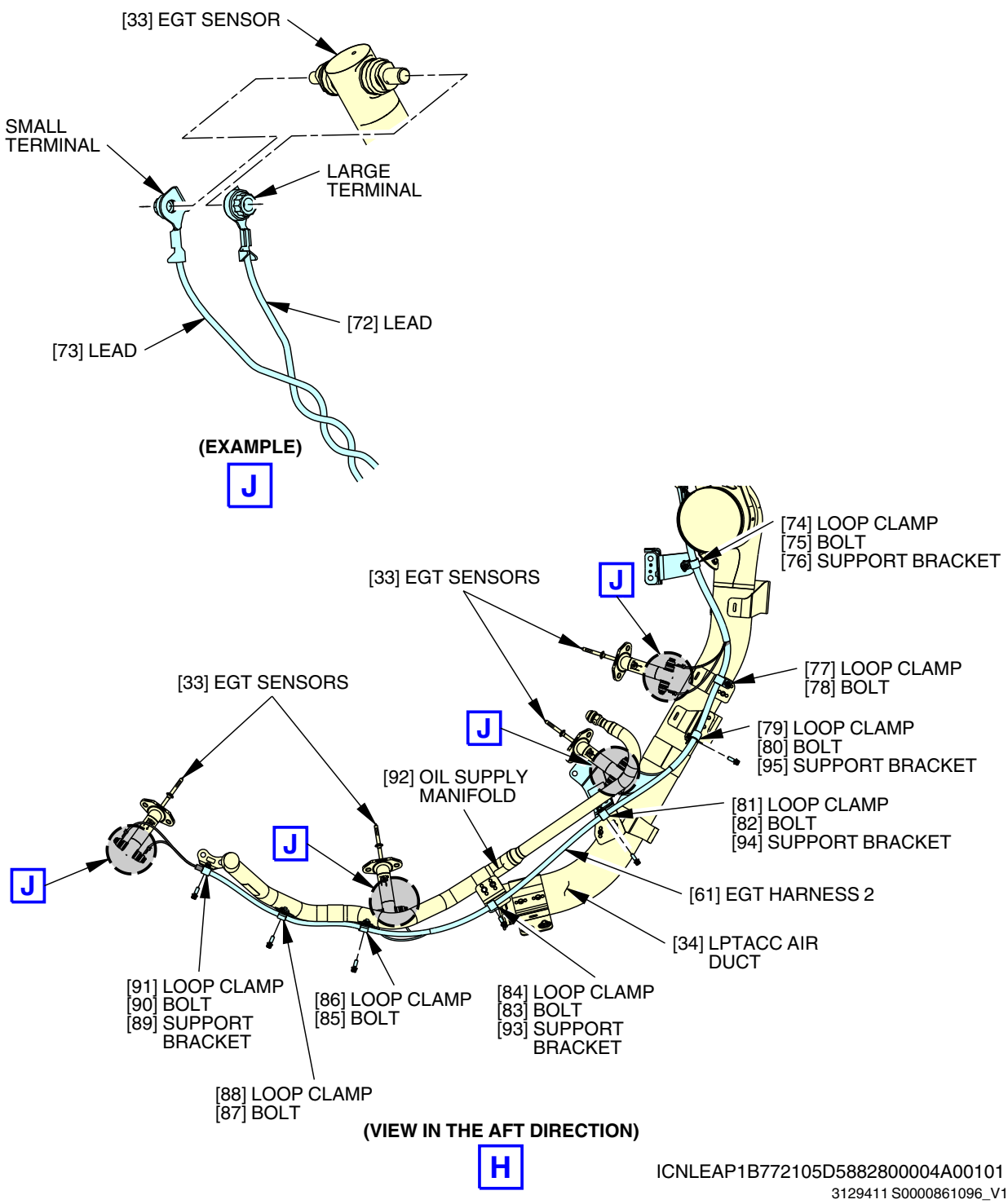
Exhaust Gas Temperature Sensor Harness 2 Installation
Figure 401/77-21-05-990-801-G00 (Sheet 6 of 8)

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Exhaust Gas Temperature Sensor Harness 2 Installation
Figure 401/77-21-05-990-801-G00 (Sheet 7 of 8)

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Exhaust Gas Temperature Sensor Harness 2 Installation
Figure 401/77-21-05-990-801-G00 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 77-21-05-400-801-G00

3. Exhaust Gas Temperature Sensor Harness 2 Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the Exhaust Gas Temperature (EGT) sensor harness 2.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Exhaust Gas Temperature Sensor Harness 2 Installation

SUBTASK 77-21-05-420-001-G00

(1) Install the EGT harness 2 [61] as follows:

(a) Remove the protective covers from the EGT sensors [33] terminals, EGT harness 2 lead [72], and lead [73].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE THAT THE POSITION OF THE HARNESS IS CORRECT. IF YOU DO NOT OBEY, DAMAGE TO THE HARNESS CAN OCCUR WHEN IT RUBS THE ADJACENT COMPONENTS.

(b) Put the EGT harness 2 [61] in position on the engine.

CAUTION MAKE SURE THAT YOU USE THE BOLTS IDENTIFIED IN THE PARTS LIST. THESE BOLTS WILL HAVE THE CORRECT LENGTH. BOLTS THAT ARE TOO LONG CAN CAUSE DAMAGE.

(c) Loosely install the EGT harness 2 loop clamps on the engine brackets, LPTACC air duct [34], and oil supply manifold [92] with the bolts as follows:

1) Lubricate the threads of the bolt [65], bolt [66], bolt [68], and bolt [71] with engine oil, D00599 [CP2442].
2) Lubricate the threads of the bolt [75] and bolt [78] with engine oil, D00599 [CP2442].
3) Lubricate the threads of the bolt [80], bolt [82], and bolt [83] with engine oil, D00599 [CP2442].
4) Lubricate the threads of the bolt [85], bolt [87], and bolt [90] with engine oil, D00599 [CP2442].
5) Install the bolt [90] that attaches the loop clamp [91] and EGT harness 2 [61] to the support bracket [89].
6) Install the bolt [87] that attaches the loop clamp [88] and EGT harness 2 [61] to the oil supply manifold [92].
7) Install the bolt [85] that attaches the loop clamp [86] and EGT harness 2 [61] to the oil supply manifold [92].
8) Install the bolt [83] that attaches the loop clamp [84] and EGT harness 2 [61] to the support bracket [93].
9) Install the bolt [82] that attaches the loop clamp [81] and EGT harness 2 [61] to the support bracket [94].
10) Install the bolt [80] that attaches the loop clamp [79] and EGT harness 2 [61] to the support bracket [95].
11) Install the bolt [78] that attaches the loop clamp [77] and EGT harness 2 [61] to the LPTACC air duct [34].
12) Install the bolt [75] that attaches the loop clamp [74] and EGT harness 2 [61] to the support bracket [76].
13) Install the bolt [71] that attaches the loop clamp [70] and EGT harness 2 [61] to the LPTACC air duct [34].
14) Install the bolt [68] that attaches the loop clamp [69] and EGT harness 2 [61] to the support bracket [16].
15) Install the bolt [66] that attaches the loop clamp [67] and EGT harness 2 [61] to the support bracket [13].
16) Install the bolt [65] that attaches the loop clamp [64] and EGT harness 2 [61] to the support bracket [12].

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CAUTION MAKE SURE THAT THE EGT HARNESS LEADS ARE IN THE CORRECT POSITION BEFORE YOU TIGHTEN THE CAPTIVE NUTS. IF THE LEADS ARE NOT IN THE CORRECT POSITION, YOU CAN CAUSE DAMAGE TO EGT HARNESS LEADS AND THE EGT PROBE.

CAUTION DO NOT TIGHTEN THE CONNECTORS TOO TIGHT. TOO MUCH TORQUE ON THE CONNECTORS CAN CAUSE DAMAGE TO THE EQUIPMENT.

(d) Connect the EGT harness 2 lead [72] and lead [73] to the EGT sensors [33] as follows:

NOTE: The EGT sensor has two terminals and two connector nuts. One terminal has a smaller diameter than the other terminal. The smaller connector nut is installed on the smaller terminal. The larger connector nut is installed on the larger terminal.

1) Install the lead [72] on the larger terminal of the EGT sensor [33].
2) Install the lead [73] on the smaller terminal of the EGT sensor [33].
3) If there is any overlength of the lead [72] and lead [73] connected to the EGT sensor [33], wind the longest branch around the smallest one.
4) Tighten the lead [72] nuts to 29.5 in-lb (3.33 N·m) - 32.5 in-lb (3.67 N·m).
5) Tighten the lead [73] nuts to 16.2 in-lb (1.83 N·m) - 17.8 in-lb (2.01 N·m).

(e) Lubricate the threads of the bolts [63] with engine oil, D00599 [CP2442].

(f) Remove the protective cover from the EGT harness 2 [61] receptacle.

(g) Put the EGT harness 2 [61] in position on the support bracket [9].

(h) Install the two bolts [63] that attach the EGT harness 2 [61] to the support bracket [9].

(i) Tighten the two bolts [63] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(j) Tighten the bolt [65], bolt [66], bolt [68], and bolt [71] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

(k) Tighten the bolt [75] and bolt [78] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

(l) Tighten the bolt [80], bolt [82], and bolt [83] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

(m) Tighten the bolt [85], bolt [87], and bolt [90] to 32.0 in-lb (3.62 N·m) - 38.0 in-lb (4.29 N·m).

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(n) Connect the electrical connectors as follows (TASK 70-00-01-910-803-G00):

1) Remove the protective covers from the electrical connector [7], electrical connector [5], electrical connector [62], and electrical connector [3].

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2) Remove the protective covers from the EGT harness 1 [1], TCF accelerometer cable [2], and from the receptacle on the HPTACC valve [4].

3) Examine the orientation of the harness.

4) Connect the electrical connector [62] to the EGT harness 2 [61].

5) Connect the electrical connector [7] to the EGT harness 1 [1].

6) Connect the electrical connector [5] to the TCF accelerometer cable [2].

7) Connect the electrical connector [3] to the HPTACC valve [4].

8) Use a teflon-jawed pliers, STD-664, to tighten the electrical connectors (TASK 70-00-01-910-803-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 77-21-05-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 77-21-05-860-003-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 77-21-05-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Exhaust Gas Temperature Sensor Harness 2 Installation Test

SUBTASK 77-21-05-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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TURBINE CENTER FRAME ACCELEROMETER - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Turbine center frame accelerometer removal

(2) Turbine center frame accelerometer installation.

TASK 77-31-06-000-801-G00

2. Turbine Center Frame Accelerometer Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the Turbine Center Frame (TCF) accelerometer.

(2) You must open the left thrust reverser to get access to the TCF accelerometer.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE
STD-15636	Socket - Hex, 1-5/16 Inch

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 77-31-06-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

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(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 77-31-06-010-001-G00

(2) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. TCF Accelerometer Removal

SUBTASK 77-31-06-020-001-G00

CAUTION DO NOT BEND THE ACCELEROMETER CABLE TOO TIGHTLY. IF YOU BEND THE ACCELEROMETER CABLE TO A RADIUS THAT IS LESS THAN 2 IN. (51 MM), DAMAGE TO THE CABLE CAN OCCUR.

(1) Remove the TCF accelerometer [1] as follows:

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

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(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(a) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [2] from the TCF accelerometer [1] (TASK 70-00-01-910-803-G00).

1) Install the protective covers on the electrical receptacle on the TCF accelerometer [1] and electrical connector [2].

(b) Remove the bolt [6] that attaches the loop clamp [5] and TCF accelerometer [1] to the support bracket [4].

(c) Remove the bolt [8] that attaches the loop clamp [9] and TCF accelerometer [1] to the support bracket [10].

(d) Remove the bolt [15] that attaches the loop clamp [14] and TCF accelerometer [1] to the support bracket [13].

(e) Remove the two bolts [7] that attach the TCF accelerometer [1] to the support bracket [3].

(f) Use a 1-5/16 inch socket, STD-15636, to remove the three bolts [11] that attach the TCF accelerometer [1] and support bracket [13] to the accelerometer adapter [12].

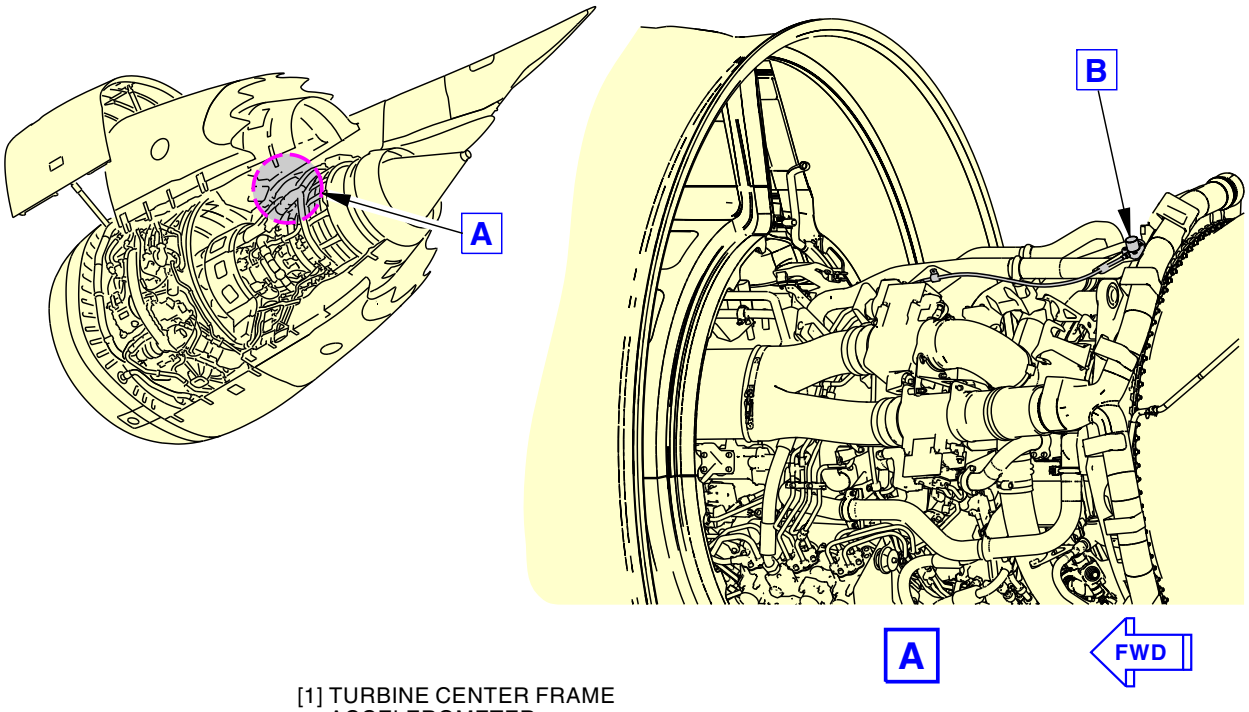
(g) Remove the support bracket [13] from the engine.

(h) Remove the TCF accelerometer [1] from the engine.

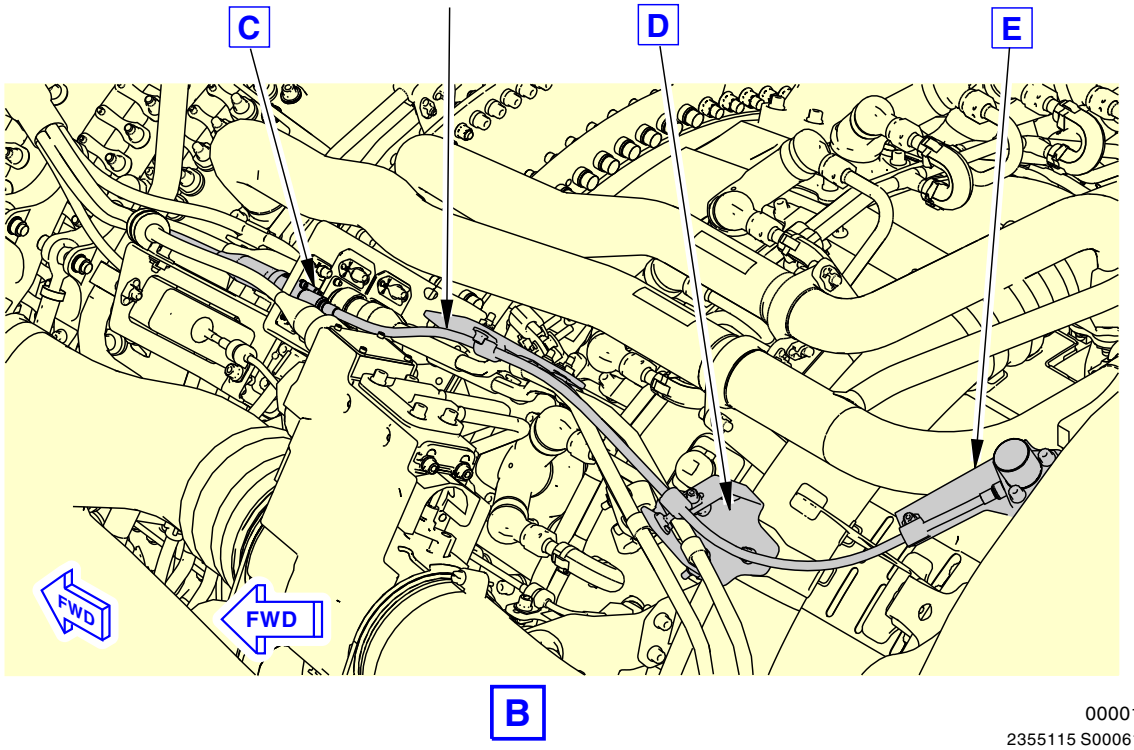
(i) Put the TCF accelerometer [1] in a protective cover.

END OF TASK

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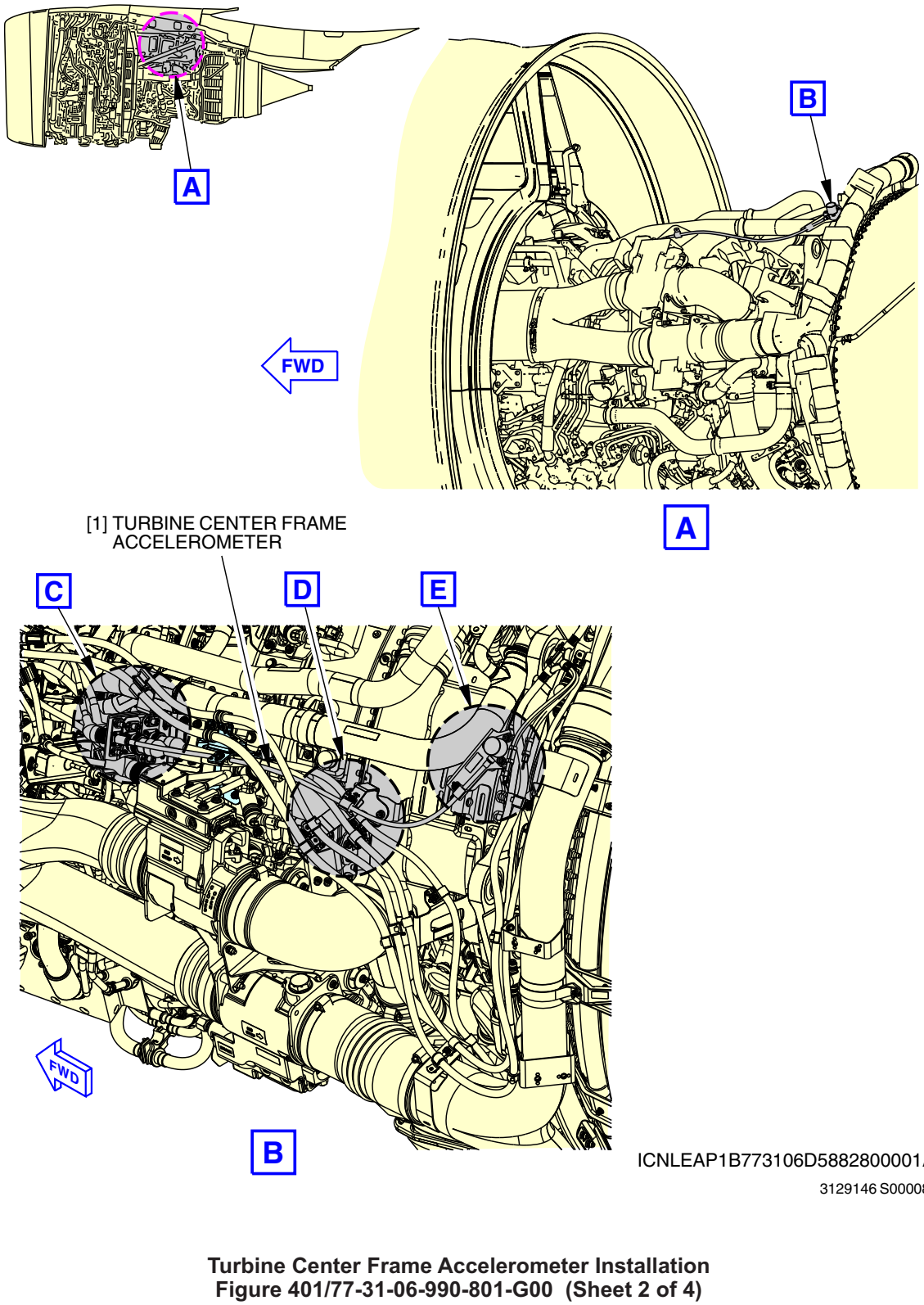
[1] TURBINE CENTER FRAME ACCELEROMETER



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2355115 S00061516568_V2

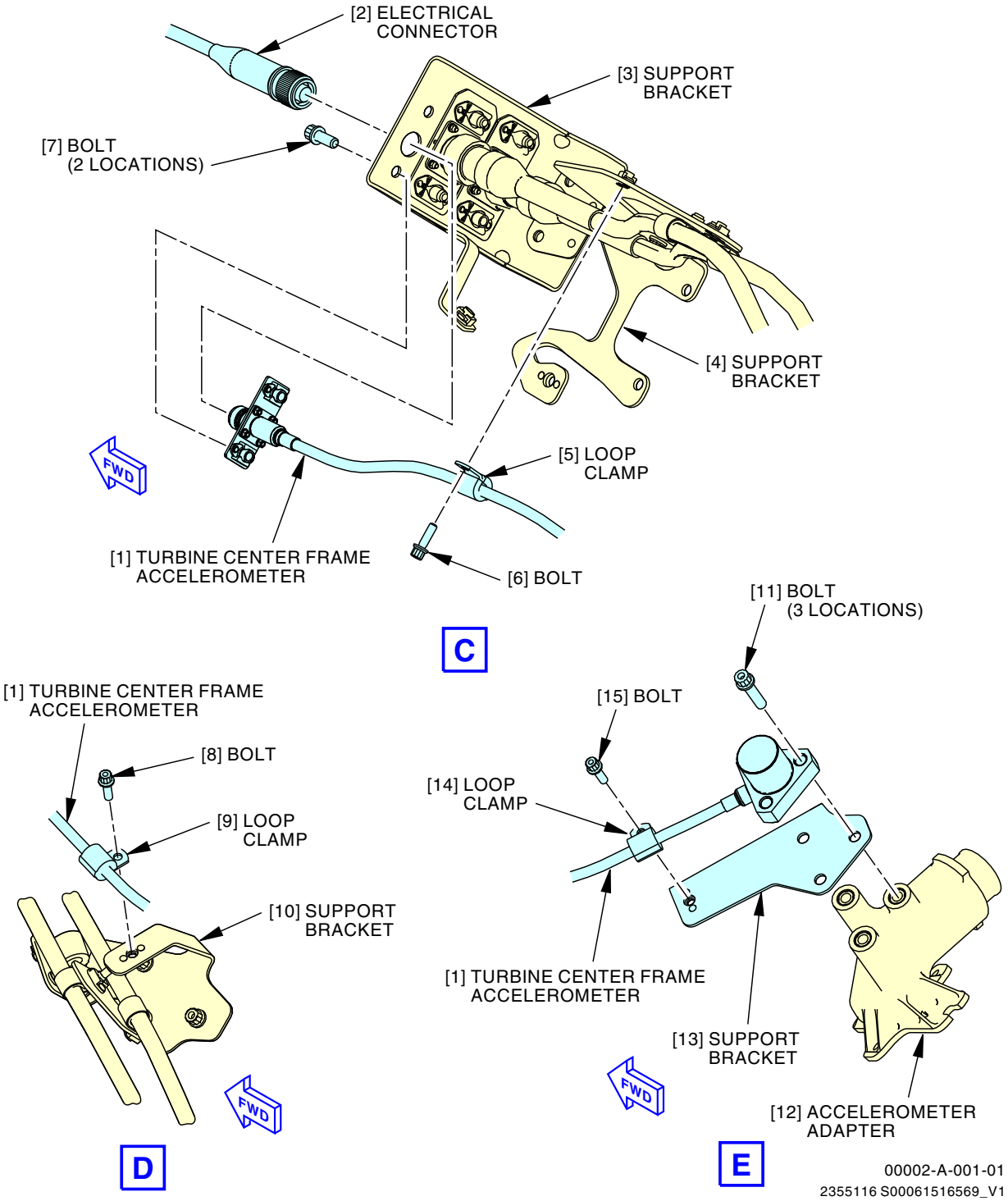
Turbine Center Frame Accelerometer Installation
Figure 401/77-31-06-990-801-G00 (Sheet 1 of 4)

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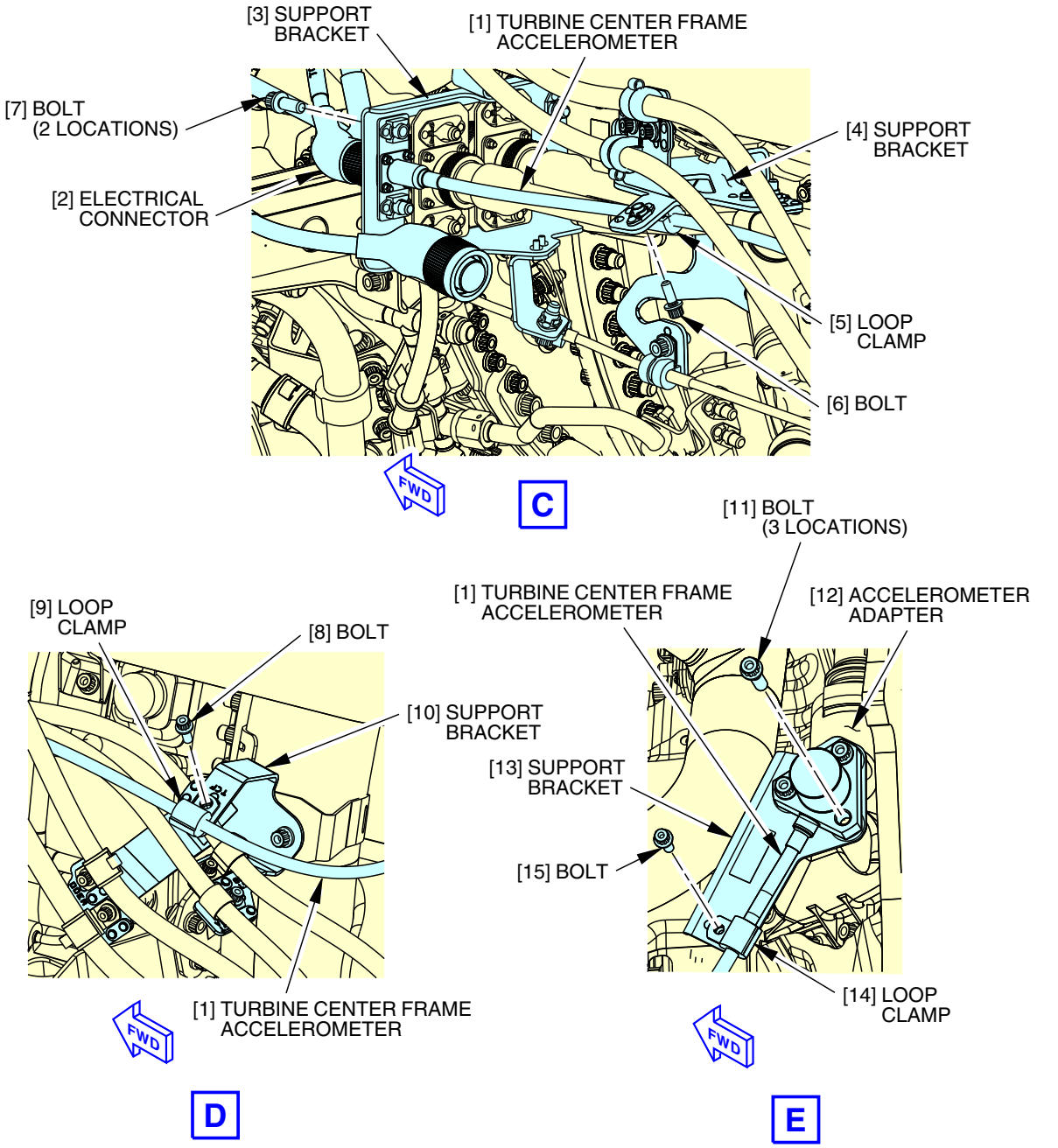
Turbine Center Frame Accelerometer Installation
Figure 401/77-31-06-990-801-G00 (Sheet 2 of 4)

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Turbine Center Frame Accelerometer Installation
Figure 401/77-31-06-990-801-G00 (Sheet 3 of 4)

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Turbine Center Frame Accelerometer Installation
Figure 401/77-31-06-990-801-G00 (Sheet 4 of 4)

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TASK 77-31-06-400-801-G00

3. Turbine Center Frame Accelerometer Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the Turbine Center Frame (TCF) accelerometer.

(2) You must do the tests that are listed in the power plant test reference table after you install TCF accelerometer.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-15636	Socket - Hex, 1-5/16 Inch

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

G. TCF Accelerometer Installation

SUBTASK 77-31-06-420-001-G00

CAUTION

DO NOT BEND THE ACCELEROMETER CABLE TOO TIGHTLY. IF YOU BEND THE ACCELEROMETER CABLE TO A RADIUS THAT IS LESS THAN 2 IN. (51 MM), DAMAGE TO THE CABLE CAN OCCUR.

(1) Install the TCF accelerometer [1] as follows:

(a) Remove the TCF accelerometer [1] from the protective cover.

(b) Lubricate the threads of the bolt [6], bolts [7], bolt [8], bolts [11], and bolt [15] with clean oil, D00599 [CP2442].

(c) Put the TCF accelerometer [1] and support bracket [13] in position on the engine.

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(d) Install the three bolts [11] that attach the TCF accelerometer [1] and support bracket [13] to the accelerometer adapter [12].

1) Use a 1-5/16 inch socket, STD-15636, to tighten the three bolts [11] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

(e) Install the two bolts [7] that attach the TCF accelerometer [1] to the support bracket [3].

1) Tighten the two bolts [7] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(f) Use the Teflon-jawed pliers to connect the electrical connector [2] to the TCF accelerometer [1].

1) Remove the protective covers from the electrical connector [2].

2) Connect the electrical connector [2] to the TCF accelerometer [1].

a) Tighten the electrical connector [2].

(g) Install the bolt [15] that attaches the loop clamp [14] and TCF accelerometer [1] to the support bracket [13].

(h) Install the bolt [8] that attaches the loop clamp [9] and TCF accelerometer [1] to the support bracket [10].

(i) Install the bolt [6] that attaches the loop clamp [5] and TCF accelerometer [1] to the support bracket [4].

(j) Adjust the tension in the cable of the accelerometer until there is the same quantity of cable between the three loop clamps.

(k) Tighten the bolt [6], bolt [8], and bolt [15] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 77-31-06-410-001-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

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1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 77-31-06-860-003-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Turbine Center Frame Accelerometer Installation Test

SUBTASK 77-31-06-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

SUBTASK 77-31-06-710-001-G00

(2) Monitor the vibration data on the subsequent flight or engine operation for a satisfactory indication.

——— END OF TASK ———

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